The Two-World Search for Euler's Constant

Shear invariance, collision worlds, forgeries, and the map of possible worlds

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Abstract

Let γ be Euler's constant, $\mathrm{Ein}(z) = \sum_{n\geq 1}(-1)^{n-1}z^n/(n\,n!)$, and $\delta = eE_1(1)$ the Euler–Gompertz constant, so that $\mathrm{Ein}(1) = \gamma + \delta/e$. We study the alternative A: $\gamma \in \overline{\mathbb{Q}}$ against B: $\gamma \notin \overline{\mathbb{Q}}$ by building inside each world and asking what ordinary mathematics can reject. The outcome is a structure theorem rather than a decision. Every unconditional arithmetic theorem in which γ occurs that we could identify is invariant under a *framing shear* $\gamma \mapsto c$, $E_1(\lambda) \mapsto E_1(\lambda) + \gamma - c$, $\delta \mapsto \delta + e(\gamma - c)$, and the theory generated by shear-invariant statements and finitely many numerical enclosures has models in which the constant standing for γ is algebraic and models in which it is transcendental; more generally no shear-invariant theorem places γ inside or outside any class that is dense with dense complement (algebraic numbers, E-, G-, D-values, periods, logarithms). On the level of functions the shear is the twist $\Gamma \mapsto \Gamma(s)e^{\tau(1-s)}$, which preserves every connection identity and moves the Euler constant to $\gamma + \tau$; on linear forms it forces the tie $|b - ec| \leq 2\eta/\ell$, so shear-invariant families of integer forms in $1, \gamma, \delta$ have exponent at most 1, generically $\frac{1}{2}$, which Rivoal's forms realise ($b_n/c_n - e = 1.4 \cdot 10^{-255}$ at $n = 39$). The sharpest consequences of A are derived (the collision $\delta \in (\mathbf{E} \cap \mathbf{D}) \setminus \overline{\mathbb{Q}}$; the floor $|a + b\gamma + c\delta| \geq H^{-(d3^d-1)-\varepsilon}$; $\mu(\delta) = 2$ if $\gamma \in \mathbb{Q}$; the reduction to "is δ an E-value"; the collision spectrum of the polyexponential tower) and compared with what is provable; a ledger of twelve A-constructions shows that every one expected to be false is a membership statement whose rejection is a known conjecture at one element. Built for their own sake, the worlds acquire shadow systems of honest E-values, phantom Galois conjugates, a Galois norm, certificate loci dual to the A-world's Beukers theory, a gallery of forgeries (explicit members of ten worlds agreeing with γ to 30 and 100 digits at the uniform price $\mathrm{H} \approx 10^{p/(m+1)}$), and a map of the possible worlds on which the expected truth is the wild region where no integer-relation search can succeed. Certified lattice computations accompany every finite claim. Nothing here proves the irrationality or transcendence of γ or δ; what is proved is why the available tools cannot, and what the first tool that could would have to look like.

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1 Introduction

1.1 The question and the programme

Whether Euler's constant $\gamma = \lim_n(H_n - \log n)$ is irrational, let alone transcendental, is open. The cumulative manuscript [1] and its companions [2, 3] developed the arithmetic of γ through the exponential integral: the E-function $\mathrm{Ein}(z)$ satisfies the connection formula $\mathrm{Ein}(z) = \gamma + \log z + E_1(z)$, its value at 1 is $\gamma + \delta/e$ with δ the Euler–Gompertz constant, and the pair (γ, δ) sits at the meeting point of the two rings of Fischler and Rivoal: the ring $\mathbf{E}$ of values of E-functions at algebraic points and the ring $\mathbf{D}$ of Borel sums of arithmetic Gevrey series.

This paper records a programme proposed in the working session that produced it. Instead of attacking γ directly, one splits mathematics into two worlds,

$$\mathrm{A}: \ \gamma \in \overline{\mathbb{Q}} \qquad \text{and} \qquad \mathrm{B}: \ \gamma \notin \overline{\mathbb{Q}},$$

builds inside each—derives consequences, constructs objects, computes—and asks whether the real world C (mathematics as it stands) rejects anything so built. Fermat's Last Theorem is the model: the consequence "a Frey curve exists" was rejected by modularity, a theorem about objects of a different kind. The question is whether γ admits an analogue of the Frey curve. The programme was run in five stages, and the five parts of this paper are those stages, lightly edited and cross-referenced but otherwise as written, so that the reader can see what was tried, in what order, and where each attempt stopped.

1.2 What was found

The answer to the programme's question is structural, and it is the main result.

The shear. Every unconditional arithmetic theorem in which γ occurs that we could identify reaches γ only through connection formulas of the type $\mathrm{Ein}(\lambda) = \gamma + \log\lambda + E_1(\lambda)$, and is therefore invariant under the *framing shear* $\gamma \mapsto c$, $E_1(\lambda) \mapsto E_1(\lambda) + \gamma - c$, $\delta \mapsto \delta + e(\gamma - c)$ (Part I, Section 4). The theory $\mathcal{K}(\mathbb{F};\mathcal{C})$ generated by shear-invariant statements together with finitely many numerical enclosures has models in which the constant standing for γ is algebraic and models in which it is transcendental (Theorem 4.2); no shear-invariant theorem places γ, δ or $E_1(\lambda)$ inside or outside any class that is dense with dense complement (Theorem 15.2, Corollary 15.3); and in every terminating pair of construction and rejection at least one half must use an analytic definition of the constant (Corollary 15.4). At the level of functions the shear is the twist $\Gamma_\tau(s) = \Gamma(s)e^{\tau(1-s)}$, which preserves every connection identity, moves the Euler constant to $\gamma + \tau$, acts on the polyexponential tower as the rescaling $\lambda \mapsto \lambda e^\tau$, and breaks only the normalisations of the classical Gamma identities (Proposition 24.1, Remark 24.2); it extends to the incomplete Gamma function and to the mixed values of Fischler–Rivoal, fixing the E-side and moving the D-side (Propositions 32.1 and 32.2). On the period matrix of Fresán–Jossen it is a $\mathbb{C}$-linear change of de Rham basis fixing the period $\mathrm{Ein}(1)$ of the cycle $\sigma_1 + \sigma_2$ (Remark 24.3).

The quantitative shadow of the shear. A bound $|a+b\Gamma+c\Delta|<\eta$ proved shear-invariantly forces $|b-ec|\le 2\eta/\ell$ (Theorem 18.1); hence shear-invariant families of integer forms in $1,\gamma,\delta$ have exponent at most 1, generically $\frac{1}{2}$, and can never meet Nesterenko's threshold (Corollary 18.2). Rivoal's forms are verified to be tied, $b_n/c_n\to e$ (Proposition 18.4), lattice-optimal shear-invariant forms sit at exponent 0.50–0.54, and the D-side Stieltjes forms are untied but grow (Proposition 18.5). The first Diophantine construction that could matter is thereby specified: integer forms smaller than their own tie defect, equivalently engine ratios $-(a_n+c_n e\,\mathrm{Ein}(1))/(b_n-ec_n)$ converging to γ with a proof that uses a bridge (Specification 18.6).

What A *implies.* Under A, $\delta\in(\mathbf{E}\cap\mathbf{D})\setminus\overline{\mathbb{Q}}$ (Proposition 5.1); integer forms obey the floor $|a+b\gamma+c\delta|\ge c_\varepsilon H^{-(d3^d-1)-\varepsilon}$ (Theorem 5.2); $\mu(\delta)=2$ if $\gamma\in\mathbb{Q}$ (Proposition 5.5); the whole question reduces to "is δ an E-value" (Theorem 5.6); the polyexponential tower has an explicit collision spectrum, $\omega_{\lambda,r}\in\mathbf{E}$ iff $P_r(\lambda)=[t^r](\Gamma(1-t)\lambda^t)\in\mathbf{E}$ (Theorem 11.1), and A forces only the collision at $(1,1)$ (Corollary 11.3); the A-world proves, by Beukers' theorem, that $\{1,e^\lambda,\mathrm{Ein}(\lambda)-\gamma\}_{\lambda\in\Lambda}$ is linearly independent for every finite Λ (Theorem 23.6), a sentence C does not at present prove. The ledger of Part III (Table 6, Proposition 19.1) shows the pattern: every A-construction expected to be false is a membership, dependence or splitting statement whose rejection is a known conjecture—$\mathbf{E}\cap\mathbf{D}=\overline{\mathbb{Q}}$, fullness of the period realisation, $\gamma\notin\mathbf{E}$—evaluated at one element; every Diophantine construction is expected to be true or is a finite certificate. A linear hierarchy $\mathrm{B}\Leftarrow\mathrm{X}(\overline{\mathbb{Q}}+\overline{\mathbb{Q}}e)\Leftarrow\mathrm{X}(W)\Leftarrow\mathrm{X}(\mathbf{E})$ grades the distance between what is known and what is wanted (Theorem 18.7).

The worlds for their own sake. For every algebraic c the *shadow system* $(c,\delta_c=e(\mathrm{Ein}(1)-c),E_1(\lambda)_c)$ satisfies all connection identities and consists of E-values, $\delta_c=\sum_n(H_n-c)/n!$ (Proposition 23.2); the A-world has $d-1$ phantom Galois conjugates and a Galois norm $P_\gamma(\mathrm{Ein}(1))=\prod_\sigma E_1(1)^{(\sigma)}$ (Proposition 23.4). Every certificate is a camouflage read backwards: the lattice vector certifying "no relation of degree ≤ 12 below height $\approx 10^{p/13}$" is the minimal polynomial of an algebraic number within 10^{-p} of γ (Proposition 23.3), and the same holds on every certificate locus, giving the gallery of forgeries of Part V (Proposition 31.1) with the uniform price $\mathrm{H}\approx 10^{p/(m+1)}$. The A-world's theorems dualise into *certificate loci* of the B-world (Proposition 25.1), searched in Proposition 25.2; the expected truth is the *wild world*, where no integer-relation search can succeed (Proposition 25.4). The map of possible worlds (Figure 1) records which theorem certifies against, and which conjecture is expected to remove, each region.

The verdict. No contradiction was found in either world, and by Lemma 15.1 none could be found without proving B. What the programme produced is a classification of the tools by their relation to the shear, three quantified barriers (Section 20), and three sharpened targets: forms below their tie defect; the point conjecture B_δ: $\delta\in\mathbf{E}\Rightarrow\delta\in\overline{\mathbb{Q}}$ (Definition 10.2); and fullness of the period realisation for the Euler–Mascheroni motive.

1.3 Conventions

Status conventions. As in [1]: *auxiliary theorem* (proved here with displayed hypotheses), *published input* (attributed), *preprint input* (attributed to an unpublished but publicly available source and used only in explicitly labelled statements), *finite certificate* (a computation with exact re-verification), *conditional implication*, *open target*. Nothing in this paper proves the irrationality or transcendence of γ or δ. Where a statement coincides with a published result this is said explicitly; the framework of Fischler and Rivoal [8, 21, 10], Beukers' lifting theorem [5], and the exponential motives of Fresán and Jossen [12] are used throughout.

Notation. We write $H_n=\sum_{k\le n}1/k$ and $\mathrm{Ein}_r(z)=\sum_{n\ge1}(-1)^{n-1}z^n/(n^r n!)$, with $\mathrm{Ein}=\mathrm{Ein}_1$; $E_1(z)=\int_z^\infty e^{-t}\,dt/t$ is the exponential integral and $\delta=eE_1(1)=\int_0^\infty e^{-t}(1+t)^{-1}\,dt$ the Euler–Gompertz constant. The connection formula is $\mathrm{Ein}(z)=\gamma+\log z+E_1(z)$ (principal logarithm), with the instance $\mathrm{Ein}(1)=\gamma+\delta/e$. The rings $\mathbf{E}$, $\mathbf{G}$, $\mathbf{D}$, $\mathbf{S}$ and $\mathbf{V}=\mathbf{S}\cdot\mathbf{E}$ are defined in Part II, Section 10; H is the naive height and μ the irrationality exponent. Each part

opens with its original summary and closes with its computations; the scripts are collected in Appendix A. Cross-references between parts are internal.

Reading order. Part I is the framework and can be read alone. Part II re-runs the search on the ring dichotomy and is independent of Part III. Part III is the ledger and uses Parts I–II. Parts IV and V build without aiming at termination and use everything before them; a reader interested only in the shear as a twist of the Gamma function can read Part IV, Section 24, and Part V, Section 32, after Part I, Section 4.

Part I
The framework: symmetry of finite observations, the framing shear, and the two models

Summary of Part I. The cumulative manuscript names the two arithmetic alternatives for Euler's constant A: $\gamma \in \overline{\mathbb{Q}}$ and B: $\gamma \notin \overline{\mathbb{Q}}$, and asks whether consequences derived under A and under B must eventually collide with an established fact. This note runs that search as far as it can currently go and records the outcome. No collision is obtained. Instead we observe that every unconditional arithmetic theorem in which γ occurs that we could identify is invariant under a *framing shear* $\gamma \mapsto c$, $E_1(\lambda) \mapsto E_1(\lambda) + \gamma - c$, which fixes all E-values, exponentials, logarithms and periods, and we prove that the theory generated by shear-invariant statements together with finitely many numerical enclosures has models in which the constant standing for γ is interpreted by an algebraic number and models in which it is interpreted by a transcendental number, so the search cannot terminate on that theory. We then derive the sharpest consequences of A that we can state with published inputs—$\delta \in \mathbf{E} \cap \mathbf{D}$ with $\delta \notin \overline{\mathbb{Q}}$, an explicit lower bound $|a + b\gamma + c\delta| \geq H^{-(d3^d-1)-\varepsilon}$ for integer linear forms when γ has degree d, and $\mu(\delta) = 2$ when γ is rational—and compare them with what the established world can supply (Rivoal-type explicit forms are guaranteed exponent $1/2 - o(1)$ and deliver about $1/4$; continued-fraction data give 2.000). Finally we show that the whole search reduces to the single question "is δ (equivalently γ, or $E_1(1)$) a value of an E-function at an algebraic point?": a positive answer makes the algebraicity of γ a terminating computation (Adamczewski–Rivoal) once an explicit representation is given, and a negative answer proves transcendence. Neither answer is presently available, and the only known certificates of non-membership in $\mathbf{E}$ are Diophantine. Finite certificates are pushed further with exact LLL data (no relation of degree ≤ 30 and height $< 10^{73}$), and a degree-12 algebraic integer agreeing with γ to 30 decimals with full local counts at 2 and 3 is exhibited to show why such certificates cannot close the question.

2 The proposal and the outcome

Throughout,

$$\gamma = \lim_{n\to\infty}(H_n - \log n), \qquad \mathrm{Ein}(z) = \sum_{n\geq 1} \frac{(-1)^{n-1} z^n}{n\, n!},$$
$$E_1(x) = \int_x^\infty \frac{e^{-t}}{t}\, dt, \qquad \delta = \int_0^\infty \frac{e^{-t}}{1+t}\, dt = eE_1(1), \tag{2.1}$$

and on the positive real axis $\mathrm{Ein}(x) = \gamma + \log x + E_1(x)$, so that $\eta := \mathrm{Ein}(1) = \gamma + \delta/e$. Let $\mathcal{T}$ be the body of established mathematics (any fixed consistent effectively axiomatized theory containing the definitions above will do), let A be the sentence $\gamma \in \overline{\mathbb{Q}}$ and B $= \neg$A. The proposal is: derive in $\mathcal{T} + $ A, derive in $\mathcal{T} + $ B, and wait for a contradiction in $\mathcal{T}$.

The bookkeeping is recorded in the cumulative manuscript [1, Part VI]: $\mathcal{T} + \mathrm{A} \vdash \bot$ if and

only if $\mathcal{T} \vdash \mathrm{B}$, and $\mathcal{T} + \mathrm{B} \vdash \bot$ if and only if $\mathcal{T} \vdash \mathrm{A}$, so the parallel search terminates exactly when $\mathcal{T}$ decides A. The practical question is therefore not whether the search terminates in principle but on which *sub-theory* $\mathcal{K} \subset \mathcal{T}$ it can be run, namely on the theorems actually available. The outcome of this note is the following.

1. **The available sub-theory does not decide** A. Section 4 proves that the theory $\mathcal{K}$ generated by (i) all true statements about any field of constants $\mathbb{F}$—in particular the engine field $\mathbb{F}_0$ generated by all E-values and all classical periods, hence by all exponentials, logarithms, G-values, Bessel and Gamma values—, (ii) the connection identities $E_1(\lambda) = \mathrm{Ein}(\lambda) - \gamma - \log\lambda$ and $\delta = eE_1(1)$, and (iii) finitely many true numerical enclosures, has a model in which the constant standing for γ is algebraic and a model in which it is transcendental (Theorem 4.2). Every unconditional arithmetic theorem about γ known to us is provable in $\mathcal{K}$ (Remark 4.3), because each of them reaches γ only through the connection identities. The manuscript's Stokes-framing barrier [3, §"The Stokes-framing barrier"] is the differential-Galois face of the same shear.

2. **The A-world makes sharp, testable-looking predictions, and all of them survive every test.** Section 5 derives them: $\delta \in \mathbf{E} \cap \mathbf{D}$ and $\delta \notin \overline{\mathbb{Q}}$ (Proposition 5.1); a lower bound for integer linear forms in $1, \gamma, \delta$ with explicit exponent $d3^d - 1$ (Theorem 5.2); $\mu(\delta) = 2$ if $\gamma \in \mathbb{Q}$ (Proposition 5.5). Section 8 compares them with the best explicit constructions: Rivoal-type linear forms are guaranteed exponent $1/2 - o(1)$ and deliver about $1/4$ (Table 1), continued-fraction convergents of δ have exponent 2.000 (Table 2).

3. **The search reduces to membership in the ring E.** Theorem 5.6: $\mathrm{A} \Rightarrow \delta \in \mathbf{E}$; if $\delta \in \mathbf{E}$ with an explicit E-function representation, then A is decidable by the Adamczewski–Rivoal algorithm [4]; if $\delta \notin \mathbf{E}$, then B. So whichever way "$\delta \in \mathbf{E}$?" is answered, the search terminates. The expected answer ([16]) is negative, but the only known certificates of non-membership in $\mathbf{E}$ are Diophantine ([9, Cor. 1, Cor. 3]).

4. **Finite work sharpens parameters and nothing else.** Exact LLL certificates (Table 3) exclude every relation of degree ≤ 30 with height $< 10^{73}$, and Proposition 3.2 exhibits an algebraic integer of degree 12 within $5 \cdot 10^{-31}$ of γ with full local root counts at 2 (depth 10) and 3 (depth 5), which is why no finite certificate of this kind can terminate the search.

3 Algebraic inertness of the two alternatives

Under A with primitive minimal polynomial f of degree d, and under B, the manuscript gives complete descriptions of $\mathbb{Q}[\gamma]$: the relation lattice $f\mathbb{Z}[X]$ versus 0, the vanishing versus existence of a $\mathbb{Q}$-derivation with $D\gamma = 1$, the local counts $N_{p,k}(\gamma) \leq dp^s$ versus p^k, and the value counts $W_D(\gamma) \leq 1 + [D(D+1)]^d M(f)^D$ versus $(D+1)^{D+1}$ [1, Part VI]. Each of these statements is *uniform*: it holds for every algebraic number of degree d, respectively for every transcendental number. They are therefore inert against any established fact about γ that also holds for a set of real numbers containing algebraic numbers of every large degree and transcendental numbers. The manuscript proves that finite observations are of this kind; we record the combined statement once, in the form in which it will be used.

Theorem 3.1 (Symmetry of finite observations)**.** *Let $\Phi(x)$ be the conjunction of finitely many strict polynomial inequalities with real algebraic coefficients, finitely many rational interval enclosures, and finitely many local conditions $N_{p,k}(x) = p^k$ ($p \in S$, $1 \leq k \leq L_p$, S finite), and suppose $\Phi(\gamma)$ holds. Put $M = \prod_{p \in S} p^{L_p}$ and $\mu(M) = \min\{m : M \mid m!\}$. Then $\Phi(x)$ holds for some algebraic integer x of degree exactly $\max(1, \mu(M))$, for algebraic numbers of every degree $\geq \max(1, \mu(M))$, and for a dense set of transcendental numbers in a neighbourhood of γ (with the convention $\mu(1) = 0$ when $S = \varnothing$).*

Proof. The inequalities and enclosures define an open set $U \ni \gamma$. The optimal-camouflage theorem [1, Thm. "Optimal finite camouflage"] produces an algebraic integer in any prescribed open interval $J \subset U$ of degree exactly $\mu(M)$ with the prescribed full counts, and its proof (adding $M\ell^2 A(bX-a)$, with an Eisenstein prime $\ell \notin S$, to a monic polynomial congruent to $(X)_m$ modulo M) works verbatim with m replaced by any $m \geq \max(1, \mu(M))$, since $M \mid (a)_m$ for every integer a whenever $M \mid m!$. Transcendental numbers have full counts at every prime and depth [1, eq. "free counts"], and are dense in U. □

By Tarski's quantifier elimination for real closed fields, every property of γ definable without parameters in the ordered field $(\mathbb{R}; +, \cdot, <)$ is a Boolean combination of polynomial equalities and inequalities with integer coefficients; if none of the polynomials occurring vanishes at γ, the property is open at γ and Theorem 3.1 applies. So first-order real-algebraic facts never terminate the search unless one of them *is* a polynomial equation for γ.

The following example makes the theorem concrete at a scale beyond the manuscript's degree-four, depth-three, six-digit example.

Proposition 3.2 (A degree-twelve camouflage of γ)**.** *Let* $a = 577215664901532860606512090082 = \operatorname{round}(10^{30}\gamma)$ *and*

$$\begin{aligned} f_0(X) &= X^{12} + 746430X^{11} + 1925X^{10} + 1211490X^9 + 606255X^8 + 846090X^7 + 897935X^6 \\ &\quad + 38190X^5 + 2140X^4 + 371880X^3 + 1104480X^2 + 1140480X + 3732480, \\ f(X) &= f_0(X) + 6220800 \cdot 10^{45}\,(10^{30}X - a). \end{aligned}$$

Then f is monic, irreducible over $\mathbb{Q}$ (Eisenstein at 5), has a real root ξ with $|\xi - \gamma| = 4.02 \cdot 10^{-31}$, and its root counts satisfy

$$N_{2,k}(\xi) = 2^k \ (1 \leq k \leq 10), \qquad N_{3,k}(\xi) = 3^k \ (1 \leq k \leq 5),$$

while $N_{2,11}(\xi) = 512$. Its degree $12 = \mu(2^{10}3^5)$ is the least possible for these counts.

Proof. $f_0 \equiv X(X-1)\cdots(X-11) \pmod{2^{10}3^5}$ by construction (coefficientwise Chinese remaindering with the conditions "lower coefficients $\equiv 0 \pmod 5$, constant term $\equiv 5 \pmod{25}$"), and $6220800 = 2^{10}3^5 \cdot 25$, so $f \equiv (X)_{12}$ modulo $M = 2^{10}3^5$; since $(a)_{12} = 12!\binom{a}{12}$ is divisible by M for every integer a, every residue class modulo 2^k ($k \leq 10$) and modulo 3^k ($k \leq 5$) is a root. Eisenstein's criterion at 5 gives irreducibility. The real root near $a/10^{30}$ is at distance at most $|f_0(a/10^{30})|/(6220800 \cdot 10^{75}) < 10^{-75}$ from $a/10^{30}$, because f' is dominated by the constant $6220800 \cdot 10^{75}$ there, and $|a/10^{30} - \gamma| = 4.02 \cdot 10^{-31}$ by the choice of a. Minimality of the degree is the manuscript's fixed-divisor bound. All counts were verified by exhaustive residue enumeration (script `exp05_camouflage.py`). □

4 The framing shear and the two models

The manuscript's Part VII observes that the analytic comparison vector, not the differential module, carries γ, and [3] proves that under $\gamma \in \overline{\mathbb{Q}}$ an algebraic change of Frobenius framing removes the Euler entry of the connection matrix. We now show that the same shear acts on the entire body of available arithmetic theorems, and we make the resulting no-termination statement precise. The formal statement is elementary; its content lies in the inventory of Remark 4.3.

Definition 4.1 (Engine theory over a field of constants)**.** Let T_0 be a sound background theory in which the classical theorems are proved (ZFC with its standard model will do), and let $\mathbb{F} \subset \mathbb{C}$ be any subfield. Expand the language of T_0 by a constant symbol for every element of $\mathbb{F}$ and by three further families of constant symbols Γ, Z_λ ($\lambda \in \overline{\mathbb{Q}} \setminus (-\infty, 0]$), Δ. The theory $\mathcal{K}(\mathbb{F}; \mathcal{C})$ is T_0 together with

(K1) every sentence with parameters in $\mathbb{F}$ that is true in the standard model (the complete elementary diagram of the standard model relative to the parameters $\mathbb{F}$);

(K2) the connection axioms $Z_\lambda = \mathrm{Ein}(\lambda) - \Gamma - \log\lambda$ for all λ (principal logarithm), and $\Delta = e\,Z_1$, in which $\mathrm{Ein}(\lambda)$, $\log\lambda$ and e are the T_0-definable numbers;

(K3) a finite set $\mathcal{C}$ of true *strict* rational enclosures $r < \Gamma < s$, $r < Z_\lambda < s$ (for finitely many real λ), $r < \Delta < s$.

Write A_Γ for the sentence $\Gamma \in \overline{\mathbb{Q}}$ and B_Γ for its negation. The intended interpretation is $\Gamma = \gamma$, $Z_\lambda = E_1(\lambda)$, $\Delta = \delta$, under which A_Γ becomes A; but the theory does *not* contain the analytic definitions of these three families (as limits, integrals, Borel sums or asymptotic constants), nor any identification of them with elements of $\mathbb{F}$ or with T_0-definable numbers. The *engine field* is

$$\mathbb{F}_0 = \overline{\mathbb{Q}}(\mathbf{E},\ \mathcal{P}),$$

the field generated by all values of E-functions at algebraic points and by the ring $\mathcal{P}$ of classical periods; it contains all G-values, π, $\log\lambda$ for $\lambda \in \overline{\mathbb{Q}}^{\times}$, the Bessel values $J_\nu(\lambda)$ ($\nu \in \mathbb{Q}$), and the Gamma values $\Gamma(a/b)$ (whose b-th powers are periods). Whether γ, $E_1(1)$ or δ belongs to $\mathbb{F}_0$ is not known; under A all three do (Proposition 5.1).

Theorem 4.2 (Two models)**.** *Let $\mathbb{F} \subset \mathbb{C}$ be a subfield and $\mathcal{C}$ finite. Then $\mathcal{K}(\mathbb{F};\mathcal{C}) + \mathrm{A}_\Gamma$ and $\mathcal{K}(\mathbb{F};\mathcal{C}) + \mathrm{B}_\Gamma$ are both satisfiable, each by an expansion of the standard model of T_0 in which every constant naming an element of $\mathbb{F}$ is interpreted by that element. Consequently no sentence about Γ that is provable in $\mathcal{K}(\mathbb{F};\mathcal{C})$ decides A_Γ.*

Proof. For $c \in \mathbb{R}$ let $\mathfrak{s}_c$ be the expansion of the standard model of T_0 that sends every constant of $\mathbb{F}$ to itself and

$$\Gamma \mapsto c, \qquad Z_\lambda \mapsto \mathrm{Ein}(\lambda) - c - \log\lambda, \qquad \Delta \mapsto e\big(\mathrm{Ein}(1) - c\big).$$

Axioms (K1) do not mention the new constants and hold in the standard model. Axioms (K2) hold by definition. Each enclosure in (K3) is a strict inequality between a rational number and a quantity that is an affine function of c with slope ± 1 or $\pm e$; since it holds at $c = \gamma$, it holds for all c in an open interval $I \ni \gamma$ (finitely many enclosures, finite intersection). Choosing $c \in I \cap \overline{\mathbb{Q}}$ gives a model of $\mathcal{K} + \mathrm{A}_\Gamma$, choosing $c \in I \setminus \overline{\mathbb{Q}}$ a model of $\mathcal{K} + \mathrm{B}_\Gamma$. The last sentence follows. □

The theorem concerns the symbol Γ, which $\mathcal{K}$ ties to the rest of mathematics only through (K2)–(K3). The sentence A itself, about the T_0-definable number $\gamma = \lim(H_n - \log n)$, may of course be decided by T_0; the passage from A_Γ to A is the assessment of Remark 4.3 that the established theorems about γ reach it only through (K2)–(K3), i.e. that they are theorems about Γ.

We call $\mathfrak{s}_c$ the *framing shear*. On the intended constants it reads

$$\mathfrak{s}_c: \quad \gamma \mapsto c, \qquad E_1(\lambda) \mapsto E_1(\lambda) + (\gamma - c), \qquad \delta \mapsto \delta + e(\gamma - c), \tag{4.1}$$

with every element of $\mathbb{F}$ fixed. It is exactly the change of framing U_t, $t = \gamma - c$, of [3, Thm. "Euler Stokes-framing barrier"], now acting on numbers rather than on solution bases. We *define* a theorem φ about γ to be *shear-invariant* (relative to $\mathbb{F}$) if it is the intended interpretation of a sentence $\psi(\Gamma)$ provable in $\mathcal{K}(\mathbb{F};\mathcal{C})$ for some finite $\mathcal{C}$. Equivalently, φ is shear-invariant if the statement obtained by substituting the sheared constants (4.1) holds for every c in some open interval $(r, s) \ni \gamma$ with rational endpoints: indeed "$\forall c \in (r,s)\,\psi(c)$" is then an axiom of type (K1), and with the enclosure $r < \Gamma < s$ in $\mathcal{C}$ one obtains $\mathcal{K} \vdash \psi(\Gamma)$. In practice the condition is checked on the proof: it must use about $\gamma, E_1(\lambda), \delta$ only (K2)–(K3). Theorem 4.2 says that shear-invariant theorems never decide A_Γ, and the reason is transparent: such a theorem holds equally for the constant c, and a neighbourhood of γ contains algebraic and transcendental c.

Remark 4.3 (Inventory). The unconditional theorems in which γ appears fall, as far as we can see, into two classes. The first class consists of shear-invariant theorems relative to $\mathbb{F}_0$; this is an assessment of proofs, not a theorem, and it is checked theorem by theorem:

- Rivoal's theorem that at least one of $\gamma+\log z$, $G_0(z)$ is transcendental for algebraic $z \notin (-\infty, 0]$ [15, Thm. 2], and his quantitative simultaneous-approximation bound $|\gamma+\log z-p/q|+|G_0(z)-r/q| \geq d(\varepsilon, z)H^{-3-\varepsilon}$ for rational $z > 0$ [15, Thm. 1]: the proofs construct integer forms in the E-values $1, e^{\pm z}, E(\pm z)$ and transfer them through $\mathrm{Ein}(z) = (\gamma + \log z) + e^{-z}G_0(z)$, which is (K2). Under $\mathfrak{s}_c$ the sheared pair $(c + \log z,\ G_0(z) + e^z(\gamma - c))$ satisfies the same identity, the same forms are small for the same reason, and the conclusions hold verbatim.

- The multipoint theorem and its corollaries: the six-constant theorem, the conditional ideals, the T_m-detectors and the fixed-degree detector of [2]; Powers' independence of $\{\gamma^{(n)} + \delta^{(n)}/e\}$ [14] (with symbols for the $\delta^{(n)}$ and their connection formulas added in the same way as for Y_0 below); Mahler's theorem that $\frac{\pi}{2}Y_0(2)/J_0(2) - \gamma$ is transcendental, where γ enters only through the connection formula $Y_0(x) = \frac{2}{\pi}(\log\frac{x}{2} + \gamma)J_0(x) + \frac{2}{\pi}\sum_{k\geq 1}(-1)^{k+1}H_k(x/2)^{2k}/k!^2$ and the two remaining constants are E-values. (To make the last statement literally a sentence of $\mathcal{K}$ one adds a symbol for $Y_0(2)$ with this connection formula as a further axiom of type (K2); the sheared model reinterprets it together with Γ.)

- All finite certificates: nonvanishing boxes, Sturm counts, modular escapes, LLL data (Section 8); these are instances of (K3) together with (K1).

- The algebraic descriptions of Section 3: they are consequences of A_Γ or of B_Γ alone.

The second class consists of theorems that *identify* γ with an analytic expression and are therefore not sentences about Γ at all: $\gamma = -\Gamma'(1)$; Rivoal's representations of γ as a convergent twisted primitive of an E-function and as a Mahler-type integral [16]; the period matrix $\left(\begin{smallmatrix} 1 & \gamma \\ 0 & 2\pi i \end{smallmatrix}\right)$ of the Euler–Mascheroni motive [12]; Fischler–Rivoal's containment theorem for the Stokes constants of E-functions in the $\mathbf{G}[\gamma]$-module $\mathbf{S}$ generated by Gamma values at rationals [8, Thm. 2], together with their observation that γ "always appears through $\gamma - \log(1/z)$". These are the natural starting points of a shear-breaking argument, but none of them has so far produced a statement about Γ beyond (K2).

Not shear-invariant are the conditional routes, precisely because each of them ties a constant of the family $(\gamma, E_1(\lambda), \delta)$ to a subset of $\mathbb{F}_0$ by a property other than (K2): the intersection conjecture $\mathbf{E} \cap \mathbf{D} = \overline{\mathbb{Q}}$ [10] together with the analytic fact "δ is the Borel sum of Euler's series at 1" (so $\delta \in \mathbf{D}$); the exponential period conjecture, or the matched-line lifting problem [3, §"The Euler–Tate matched-line target"], together with the motivic non-splitting theorem of Fresán–Jossen [12]; and the expectation $\gamma \notin \mathbf{E} \cup \mathbf{G} \cup \mathbf{M}$ [16]. Each of these uses a hypothesis about Γ, Z_λ or Δ that is not available in $\mathcal{K}$ and fails in the sheared model for suitable c: for instance the route through $\mathbf{E} \cap \mathbf{D} = \overline{\mathbb{Q}}$ needs "$\Delta \in \mathbf{D}$", whereas $\mathfrak{s}_c$ sends δ to $\delta + e(\gamma - c)$ and nothing in $\mathcal{K}$ asserts that this number lies in $\mathbf{D}$.

Corollary 4.4 (What a terminating fact must look like)**.** *Say that a theorem $\varphi \in \mathcal{T}$, read as a sentence $\psi(\Gamma)$,* terminates the search *if $\mathcal{K}(\mathbb{F}_0; \mathcal{C}) + \psi(\Gamma)$ proves A_Γ or B_Γ for some finite $\mathcal{C}$. Such a φ is not shear-invariant relative to $\mathbb{F}_0$: its proof must use the analytic definition of γ, of some $E_1(\lambda)$, or of δ in a way that is not reducible to (K1)–(K3). Concretely, it must either identify one of these constants with an element of $\mathbb{F}_0$ ("$\gamma = F(\alpha)$ for an explicit E-function F", Theorem 5.6), or exclude such an identification ("$\delta \notin \mathbf{E}$"), or relate the constant to a subset of $\mathbb{C}$ defined independently of γ by a property that is not preserved under* (4.1)*.*

The corollary is a triviality once Theorem 4.2 is available; its value is that it turns "look for a contradiction" into "look for a shear-breaking theorem", and Section 7 lists the shear-breaking statements that are currently on the table. One more consequence deserves to be stated: $E_1(1)$,

δ, the second-kind Bessel values $Y_0(\lambda)$, $K_0(\lambda)$, and the derivatives $\Gamma^{(k)}(1)$ are all related to γ and to elements of $\mathbb{F}_0$ (E-values, zeta values) by connection formulas of the form (K2); if symbols for them are added to the language with those formulas as axioms, the sheared model reinterprets all of them simultaneously. No established theorem attaches arithmetic information to any one of these "second-kind" constants individually, and this is the same phenomenon.

5 Consequences of A and their tests

Let $\mathbf{E}$ be the ring of values at algebraic points of E-functions, $\mathbf{G}$ the ring of G-values, and $\mathbf{D}$ the ring of values at algebraic points, in a non-anti-Stokes direction, of arithmetic Gevrey series of order one (Fischler–Rivoal [8, 10]). E-functions are taken with coefficients in $\overline{\mathbb{Q}}$; when a number field K is specified, the coefficients lie in K.

Proposition 5.1 (Basic consequences)**.** *Assume* A*, let* $K = \mathbb{Q}(\gamma)$*,* $d = [K : \mathbb{Q}]$*. Then:*

(i) $\mathrm{Ein}(z)-\gamma$ *and* $e^z(\mathrm{Ein}(z)-\gamma)$ *are* E*-functions with coefficients in* K*, so* $E_1(1) = \mathrm{Ein}(1)-\gamma \in \mathbf{E}$ *and* $\delta = e\,E_1(1) \in \mathbf{E}$*;*

(ii) δ *is transcendental and* $\operatorname{trdeg}_{\overline{\mathbb{Q}}}\,\overline{\mathbb{Q}}(e,\delta) = 2$*;*

(iii) $\delta \in \mathbf{E} \cap \mathbf{D}$ *and* $\delta \notin \overline{\mathbb{Q}}$*;*

(iv) no element of $\mathbf{E}$*, in particular neither* δ *nor* $E_1(1)$*, is a Liouville number or a U-number [9, Cor. 1, Cor. 3];*

(v) for every $\lambda \in \overline{\mathbb{Q}} \cap \mathbb{R}_{>0}$*,* $\log\lambda + E_1(\lambda)$ *is transcendental, and the family* $\{\log\lambda + E_1(\lambda)\}$ *is algebraically independent over* $\overline{\mathbb{Q}}(e^{\mu} : \mu \in \overline{\mathbb{Q}})$*.*

Proof. (i) Sums and products of E-functions are E-functions and constants in K are E-functions with coefficients in K. (ii) $\mathrm{Ein}(1)$ and e are algebraically independent over $\overline{\mathbb{Q}}$ [1, Thm. "Multipoint Ein input"], so $\overline{\mathbb{Q}}(e,\delta) = \overline{\mathbb{Q}}(e, \mathrm{Ein}(1)-\gamma) = \overline{\mathbb{Q}}(e, \mathrm{Ein}(1))$ has transcendence degree two. (iii) $\delta = \int_0^\infty e^{-t}/(1+t)\,dt$ is the Borel sum in direction 0 of Euler's series $\sum(-1)^n n!\, z^{-n-1}$ at $z = 1$, whose Borel transform $1/(1+\xi)$ is a G-function with its singularity on the negative axis [3, §"The mixed-Gevrey aperture"]; hence $\delta \in \mathbf{D}$, and (i), (ii) give the rest. (iv) is the cited theorem applied to (i). (v) $\log\lambda + E_1(\lambda) = \mathrm{Ein}(\lambda) - \gamma$; apply the multipoint theorem. □

None of these statements can be tested against an established fact. Items (ii) and (v) are expected to be true unconditionally and would follow in the B-world from the exponential period conjecture; item (iv) is expected but follows from no standard conjecture. The membership statements in (i) and (iii) are expected to be *false* ($\gamma \notin \mathbf{E}$ is Rivoal's expectation [16], and $\mathbf{E} \cap \mathbf{D} = \overline{\mathbb{Q}}$ is the intersection conjecture of [10]), but they are compatible with every proved property of $\mathbf{E}$ and $\mathbf{D}$: no explicit computable number is known to lie outside $\mathbf{E}$ except through Diophantine approximation (item (iv) read backwards). The next statements are sharper.

Theorem 5.2 (A-world floor for linear forms in $1, \gamma, \delta$)**.** *Assume* γ *is algebraic of degree* d*. For every* $\varepsilon > 0$ *there is* $c_\varepsilon > 0$ *such that for all* $(a,b,c) \in \mathbb{Z}^3$ *with* $a + b\gamma + c\delta \neq 0$*,*

$$|a + b\gamma + c\delta| \;\geq\; c_\varepsilon \,\max(|a|,|b|,|c|)^{-(d\,3^d-1)-\varepsilon}. \tag{5.1}$$

Moreover $a + b\gamma + c\delta = 0$ *with* $(a,b,c) \neq 0$ *forces* $c = 0$ *and* $d = 1$*.*

Proof. Let $K = \mathbb{Q}(\gamma)$ and $D = \mathrm{den}(\gamma)$. The vector $f = (1, e^z, e^z\,\mathrm{Ein}(z))^t$ consists of E-functions with rational coefficients and satisfies $f' = Af$ with

$$A = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ -z^{-1} & z^{-1} & 1 \end{pmatrix} \in M_3(\mathbb{Q}(z)),$$

because $(e^z \operatorname{Ein}(z))' = e^z \operatorname{Ein}(z) + (e^z - 1)/z$. Since $\delta = e \operatorname{Ein}(1) - e\gamma$,

$$D(a + b\gamma + c\delta) = \lambda_1 \cdot 1 + \lambda_2 \cdot e + \lambda_3 \cdot e \operatorname{Ein}(1), \qquad (\lambda_1, \lambda_2, \lambda_3) = \big(D(a + b\gamma),\, -Dc\gamma,\, Dc\big) \in \mathcal{O}_K^3,$$

with $\max_j \operatorname{house}(\lambda_j) \le C_\gamma \max(|a|, |b|, |c|)$. The linear independence measure of Fischler–Rivoal [9, Thm. 1] for E-functions with coefficients in a number field of degree d at a point of K gives, for nonzero values, $|\Lambda| > c\, H^{-d3^d+1-\varepsilon}$ with $H = \max_j \operatorname{house}(\lambda_j)$ and $N = 3$. This is (5.1). If $a + b\gamma + c\delta = 0$ with $c \neq 0$, then $\delta \in K \subset \overline{\mathbb{Q}}$, contradicting Proposition 5.1(ii); if $c = 0$, then $a + b\gamma = 0$ forces $\gamma \in \mathbb{Q}$. □

Corollary 5.3 (A transcendence criterion with an auxiliary constant)**.** *Write $H(a, b, c) = \max(|a|, |b|, |c|)$. If for every $\kappa > 0$ there are integers (a, b, c) with $H(a, b, c) \ge 2$ and $0 < |a + b\gamma + c\delta| < H(a, b, c)^{-\kappa}$, then γ is transcendental. If for some fixed $\kappa > 2$ there are infinitely many triples with $0 < |a + b\gamma + c\delta| < H(a, b, c)^{-\kappa}$, then $\gamma \notin \mathbb{Q}$.*

Proof. Suppose γ algebraic of degree d and apply (5.1) with $\varepsilon = 1$: $|a + b\gamma + c\delta| \ge c_1 H^{-d3^d}$ for nonzero values. A triple with $|a+b\gamma+c\delta| < H^{-\kappa}$ and $\kappa \ge d3^d + 1$ then has $H < 1/c_1$, so it lies in a finite set $\mathcal{F}$ of triples with nonzero values, whose values are bounded below by some $m > 0$; for $\kappa > \max(d3^d+1, \log(1/m)/\log 2)$ no triple with $H \ge 2$ can satisfy $|a+b\gamma+c\delta| < H^{-\kappa} \le 2^{-\kappa} < m$. This contradicts the hypothesis. The second statement is (5.1) with $d = 1$ (exponent $3 - 1 = 2$) and $\varepsilon < \kappa - 2$: infinitely many triples force $H \to \infty$, and $c_\varepsilon H^{-2-\varepsilon} < H^{-\kappa}$ fails for large H. □

The criterion is the linear, three-term instance of the family of detectors in [2, Cor. "Fixed-degree approximation detector"]; its interest is that the auxiliary number δ is allowed to enter the form. It should be compared with the classical route: forms of exponent $\kappa > 1$ with regular growth already give $\dim_{\mathbb{Q}} \operatorname{span}(1, \gamma, \delta) \ge 3$, hence $\gamma \notin \mathbb{Q}$, by Nesterenko's criterion [13, 7]. Roth's theorem gives $\gamma \notin \overline{\mathbb{Q}}$ from rational approximations of exponent > 2 without any auxiliary constant.

Proposition 5.4 (What the established world supplies)**.** *(i) (Dirichlet) For every $H \ge 1$ there is $(a, b, c) \neq 0$ with $\max(|b|, |c|) \le H$, $|a| \le 2H + 1$ and $|a + b\gamma + c\delta| < 1/H^2$; the exponent 2 is exactly the $d = 1$ floor (5.1), so Theorem 5.2 is compatible with Dirichlet for every d.*

(ii) (Rivoal-type forms) The integer forms $L_n \in \mathbb{Z} + \mathbb{Z}\gamma + \mathbb{Z}\delta$ obtained from the construction of [15, §§3–6] at $\alpha = 0$ and Rivoal's parameter $z = -1$, i.e. at the point 1 of Ein *(the determinant elimination below) satisfy $|L_n| \le C^n/n!$ and $H(L_n) \le C'^n (n!)^2$; hence, for every n with $L_n \neq 0$, the exponent $\tau_n := -\log|L_n|/\log H(L_n)$ satisfies $\tau_n \ge \frac{1}{2} - o(1)$. The computed values for $1 \le n \le 39$ (all nonzero) lie between 0.037 and 0.27 (Table 1). Upper bounds for τ_n are not proved; heuristically $\tau_n \to \frac{1}{2}$.*

Proof. (i) Among the $(H+1)^2$ fractional parts of $b\gamma+c\delta$, $0 \le b, c \le H$, two lie in one of $(H+1)^2-1$ equal subintervals of $[0, 1)$, hence at distance less than $1/(H^2 + 2H) < 1/H^2$; subtract the two expressions. (ii) With Rivoal's notation, $R_{n,0}(-1) = A_n(-1)e^{-1} + B_n(-1)E(-1) + C_n(-1)$, $E(-1) = -\operatorname{Ein}(1) = -\gamma - \delta/e$, so $R_{n,0}(-1) = e^{-1}P_n + Q_n$ with $P_n = A_n(-1) - B_n(-1)\delta$, $Q_n = C_n(-1) - B_n(-1)\gamma$, and $P_{n+1}R_{n,0}(-1) - P_nR_{n+1,0}(-1) = P_{n+1}Q_n - P_nQ_{n+1}$ is the form

$$(A_{n+1}C_n - A_nC_{n+1}) + (A_nB_{n+1} - A_{n+1}B_n)\gamma + (B_nC_{n+1} - B_{n+1}C_n)\delta$$

(all polynomials at -1), in which the $\gamma\delta$ terms cancel. Lemma 4 of [15] gives $|R_{n,0}(-1)| \le C^n/(n!)^2$, $\max(|A_n|, |B_n|, |C_n|) \le C^n n!$, and integrality of $d_nA_n(-1)$, $B_n(-1)$, $d_nC_n(-1)$ with $d_n = \operatorname{lcm}(1, \dots, n) \le e^{n(1+o(1))}$; the stated bounds follow, and $\tau_n \ge (\log n! - n\log C)/(2\log n! + n\log C')$. □

So the A-world floor demands exponent above $d3^d - 1 \geq 2$, the nonconstructive Dirichlet forms sit exactly at 2, and the best explicit construction is guaranteed only $1/2 - o(1)$, delivers about $1/4$ in the computable range, and is not expected to exceed $1/2$. In the language of the manuscript, this is the "fast analytic decay disappears after taking primitive heights" gap, measured: a factor of at least four in the exponent, and unboundedly more as d grows.

Proposition 5.5 (The rational case is Diophantinely rigid)**.** (Preprint input.) *Assume $\gamma \in \mathbb{Q}$. Then the irrationality exponents satisfy $\mu(\delta) = \mu(E_1(1)) = 2$. Unconditionally $\mu(\mathrm{Ein}(1)) = 2$.*

Proof. Fischler–Rivoal [11, Thm. 1] prove: if $f \in \mathbb{Q}[[z]]$ is an E-function and $r \in \mathbb{Q}$, then $f(r) \in \mathbb{Q}$ or $\mu(f(r)) = 2$. Apply it to $f = e^z(\mathrm{Ein}(z) - \gamma)$, $f = \mathrm{Ein}(z) - \gamma$ (rational coefficients because $\gamma \in \mathbb{Q}$) and $f = \mathrm{Ein}$ at $r = 1$; the values are δ, $E_1(1)$, $\mathrm{Ein}(1)$, which are irrational because δ is transcendental under A (Proposition 5.1) and $\mathrm{Ein}(1)$ is transcendental. □

Thus the A-world with rational γ predicts that δ behaves like an almost-all real number with respect to rational approximation. Table 2 shows that the first 2,873 partial quotients of δ are consistent with this prediction (convergent exponents 2.0004 on average over the last 500, maximal partial quotient 17871), as they are with B. The refutation route "$\mu(\delta) > 2$" is weaker than Nesterenko's criterion, as noted after Corollary 5.3; we record it because it is the sharpest statement that the rational A-world makes about a single explicitly computable number.

Theorem 5.6 (Reduction to the ring $\mathbf{E}$)**.** *(i) $\gamma \in \mathbf{E} \iff E_1(1) \in \mathbf{E} \iff \delta \in \mathbf{E}$, and* $\mathrm{A} \Rightarrow \gamma \in \mathbf{E}$.

(ii) Suppose an E-function F and $\alpha \in \overline{\mathbb{Q}}$ with $F(\alpha) = \gamma$ are given in the sense of [4, §2.1]: a linear differential operator over $\overline{\mathbb{Q}}(z)$ annihilating F, enough Taylor coefficients to determine F, and the guarantee that F is an E-function. Then A *is settled by a terminating computation of proved correctness: the algorithm of [4, Thm. 2] first decides whether F is transcendental (if not, $F \in \overline{\mathbb{Q}}[z]$ and $\gamma = F(\alpha) \in \overline{\mathbb{Q}}$), and if it is, outputs the finite list of algebraic β with $F(\beta) \in \overline{\mathbb{Q}}$ together with the values;* A *holds if and only if α is on the list.*

(iii) If $\delta \notin \mathbf{E}$ then B.

(iv) Consequently the two-world search terminates as soon as "$\delta \in \mathbf{E}$" is decided in either direction; in the affirmative direction, given an explicit representation, it terminates algorithmically.

Proof. (i) $\mathbf{E}$ is a ring containing $\overline{\mathbb{Q}}$, $e^{\pm 1}$ and $\mathrm{Ein}(1)$ [8, §1, after Def. 2]; $\gamma = \mathrm{Ein}(1) - E_1(1)$ and $\delta = e\,E_1(1)$, $E_1(1) = e^{-1}\delta$. The implication $\mathrm{A} \Rightarrow \gamma \in \mathbf{E}$ is $\overline{\mathbb{Q}} \subset \mathbf{E}$. (ii) is the cited theorem; note that a representation $\delta = G(\beta)$ with G an E-function yields $\gamma = F(1)$ with $F(z) = \mathrm{Ein}(z) - e^{-z}G(\beta z)$, again an E-function given with its operator. (iii) is the contrapositive of (i). (iv) combines (ii) and (iii). □

Remark 5.7 (Three cases, and the position of the conjectures). Exactly one of the following holds: $\gamma \in \overline{\mathbb{Q}}$; $\gamma \in \mathbf{E} \setminus \overline{\mathbb{Q}}$; $\gamma \notin \mathbf{E}$. Rivoal [16] expects the third and shows instead that γ lies in the extended set $\widetilde{\mathbf{E}}$ of convergent twisted primitives $\int_0^z Q(x)F(x)\,dx$ of E-functions, e.g. $\gamma = \int_0^\infty \frac{1-(1+x)e^{-x}}{x(1+x)}\,dx$. In the second case the algebraicity question would be decidable but the E-function is unknown. Under A, Proposition 5.1(iii) says that $\mathbf{E} \cap \mathbf{D}$ contains the transcendental number δ; hence the intersection conjecture $\mathbf{E} \cap \mathbf{D} = \overline{\mathbb{Q}}$ of [10] implies B (this recovers the conditional transcendence of [10] and [3, Thm. "Conditional simultaneous dilation theorem"]), and in the second case it implies $\delta \in \overline{\mathbb{Q}}$. The reduction does not lower the difficulty of the negative direction: the only explicit computable numbers currently known to lie outside $\mathbf{E}$ are Liouville numbers and, more generally, U-numbers [9, Cor. 1, 3], i.e. non-membership in $\mathbf{E}$ is certified only Diophantinely, exactly as transcendence itself.

Remark 5.8 (Why the reduction is not circular). Theorem 5.6(ii) is not the tautology "if γ were known to be a value of a function with computable Galois theory, its nature would be computable". The content is Beukers' lifting theorem: algebraic relations among values of E-functions at a nonsingular algebraic point come from functional relations, and functional relations are computable from the operator [5, 4, 6]. This is a property of the ring $\mathbf{E}$ that no other ring of constants containing γ is known to have. It also explains how a representation $\gamma = F(\alpha)$ interacts with the shear of Section 4: for algebraic t, $F - t$ is again an E-function with $(F - t)(\alpha) = \gamma - t$, so a representation is stable under algebraic shifts and cannot by itself tell γ from $\gamma - t$; what Beukers' theorem adds is that, for the given F and α, whether $F(\alpha)$ is algebraic—whether one of these shifts lands on 0—is decidable.

6 Consequences of B and their tests

Under B the ring $\mathbb{Z}[\gamma]$ is free, $n_p(\gamma) = p$ and $N_{p,k}(\gamma) = p^k$ for all p, k, $W_D(\gamma) = (D+1)^{D+1}$, and there is a discontinuous $\mathbb{Q}$-derivation with $D\gamma = 1$ [1, Part VI]. Each statement is uniform over transcendental numbers, so by Theorem 3.1 none can be refuted by finite observation, and by Theorem 4.2 none can be refuted by the engine theory. There is no established theorem asserting an algebraic relation for γ, so the B-branch of the search is vacuous at present: every consequence of B is either uniform or expected. The refinement offered by Theorem 5.6 is that B splits into $\gamma \in \mathbf{E} \setminus \overline{\mathbb{Q}}$, in which case a proof of B could be algorithmic once a representation is found, and $\gamma \notin \mathbf{E}$, which is the expected case and is currently reachable only through the shear-breaking conjectures of Remark 4.3.

7 What would terminate the search

By Corollary 4.4, a terminating fact must break the shear (4.1). The following are the shear-breaking statements currently on the table, with what each would give and where each stands.

Shear-breaking input	Would give	Present distance
Rational approximations $\lvert\gamma - p/q\rvert < q^{-2-\varepsilon}$ i.o. (Roth)	B	convergent exponents 2.000 (Table 2); no explicit family of exponent > 1 [1, Parts I–III]
Linear forms in $1, \gamma, \delta$ of exponent κ	$\kappa > 1$ with regular growth: $\gamma \notin \mathbb{Q}$ (Nesterenko); $\kappa \to \infty$: B (Cor. 5.3)	best explicit forms: exponent $\geq 1/2 - o(1)$ proved, $\approx 1/4$ computed, $1/2$ expected (Table 1)
Polynomial values with $\mathcal{R}_\gamma(P_n) \to \infty$ [1, Thm. "Equivalent infinite-family target"]	B	finite boxes only; canonical family collapses under A
An E-function representation $\delta = G(\beta)$	decidability of A (Thm. 5.6)	expected not to exist [16]; would contradict $\mathbf{E} \cap \mathbf{D} = \overline{\mathbb{Q}}$ unless $\delta \in \overline{\mathbb{Q}}$
A proof that $\delta \notin \mathbf{E}$	B	no non-Diophantine tool [9]
$\mathbf{E} \cap \mathbf{D} = \overline{\mathbb{Q}}$ [10]	B, all-order Gamma-jet independence [3]	conjecture
Matched-line lifting for $M(\gamma)$ [3], or the exponential period conjecture [12]	$\gamma \notin \overline{\mathbb{Q}} + \pi\overline{\mathbb{Q}}$	conjecture; motivic non-splitting is a theorem, its period-level content is not
A relation-preserving map $\mathbb{Z}[\gamma] \to \mathbb{Z}_p$ with unbounded counts [1, Parts VI–VII]	B	no construction; prescribed simultaneous limits do not preserve relations

Two remarks on the table. First, the entries are not independent: an E-function representation of δ would, by Beukers' theorem, either produce the minimal polynomial of γ or a proof of B,

and would in the latter case also disprove $\mathbf{E} \cap \mathbf{D} = \overline{\mathbb{Q}}$ unless δ is algebraic. Second, nothing in the table is a lemma that current fixed-point E-function theory can supply: the manuscript's Problem "Common-coefficient separation" [3] is equivalent to B, the E-representation question is a Siegel-type existence problem, and the Diophantine entries are the classical obstructions measured in Section 8.

8 Computations for Part I

All scripts are in the accompanying archive and use only `mpmath`, `sympy` and `fpylll`; exact rational arithmetic is used wherever a certificate is claimed. Floating-point evaluations are performed at 400–900 digits, far beyond the sizes involved.

Identities (`exp01`). $\mathrm{Ein}(1) = \gamma + \delta/e$, $\delta = eE_1(1) = \int_0^\infty e^{-t}/(1+t)\,dt$, $\mathrm{Ein}(x) = \gamma + \log x + E_1(x)$ at $x = \frac{1}{2}, 1, 2, 3, \frac{7}{3}$, and Rivoal's normalisation $\gamma + \log z = \mathrm{Ein}(z) - e^{-z}G_0(z)$ at $z = 1, 2, \frac{1}{3}$ were checked to 80 digits (residuals below $3 \cdot 10^{-81}$), together with $e^{-1} < \mathrm{Ein}(1) < 1$.

Rivoal's construction (`exp03`). The polynomials A_n, B_n, C_n of [15, Prop. 4] were obtained exactly from the series $R_{n,0}(z) = z^{3n+1} \sum_{k \ge 0} \frac{z^k}{k!} \left[\frac{(2n+k)!}{(3n+k+1)!}\right]^2$ (Beta integrals) by solving the $3n+3$ leading coefficient equations and verifying six further orders; the integrality statements of [15, Lemma 4(i)] at $z = -1$ were confirmed for $n \le 40$. Table 1 lists the derived forms.

n	$\log_{10} H(L_n)$	$\log_{10} \lvert L_n \rvert$	$\tau_n = -\log \lvert L_n \rvert / \log H(L_n)$
5	18.9	−3.0	0.159
10	43.5	−7.6	0.174
15	69.6	−13.7	0.197
20	96.7	−21.1	0.218
25	128.1	−25.7	0.201
30	157.7	−33.5	0.213
35	185.2	−44.3	0.239
39	207.6	−53.3	0.257

Table 1: Rivoal-type integer linear forms $L_n = \alpha_n + \beta_n\gamma + \kappa_n\delta$ at the point 1 of Ein (Rivoal's $z = -1$), reduced to primitive integer coefficients. Throughout, for $10 \le n \le 39$, $\log \lvert L_n \rvert / \log((n+1)!) \in [-1.12, -0.89]$, and $\log H(L_n)/\log((n+1)!)$ decreases from 10.0 at $n = 1$ to 4.33 at $n = 39$, consistent with $\lvert L_n \rvert \asymp C^n/n!$ and $H(L_n) \asymp C'^n (n!)^2$; the A-world floor of Theorem 5.2 requires $\tau > d3^d - 1 \ge 2$.

Continued fractions (`exp04`). From 3000-digit values (cross-checked against 7000-digit values: all 2873, respectively 2910, partial quotients agree) we obtained the data in Table 2. The Stieltjes–Padé convergents of δ, whose denominators $q_n = \sum_k k!\binom{n}{k}^2$ appear in [2], have approximation exponent decreasing to about 0.25 by $k = 60$: they are exponentially worse than the continued-fraction convergents, in agreement with the manuscript's denominator analysis.

	partial quotients	largest a_k (index)	max exponent, $k > 10$	mean exponent, last 500
δ	2873	17871 (945)	2.271	2.00043
γ	2910	2076 (528)	2.225	2.00042

Table 2: Simple continued fractions computed from 3000 digits; "exponent" is $-\log \lvert x - p_k/q_k \rvert / \log q_k$ along convergents. Empirical Khinchin means 2.732 and 2.686.

Exact LLL certificates (`exp06`). Let $\tilde{x}$ be γ rounded to p decimals, $\varepsilon = 10^{-p}$, $N = \lfloor 10^p/(2D(D+1))\rfloor$, and $L \subset \mathbb{Z}^{D+2}$ the lattice spanned by $b_i = (e_i, \mathrm{round}(N\tilde{x}^{\,i}))$, $0 \le i \le D$. If $\sum_{i\le D} a_i\gamma^i = 0$ with $\max|a_i| \le H$, then $v = \sum a_i b_i$ has $|v_{D+1}| \le NH\varepsilon D(D+1)/2 + (D+1)H/2 \le (D+1)H$, hence $\|v\|_2 \le (D+1)H\sqrt{D+2}$. For a (ϑ,η)-LLL-reduced basis with $\vartheta = 0.99$, $\eta = 0.51$, $\alpha = 1/(\vartheta-\eta^2)$, every nonzero lattice vector has norm $\ge \alpha^{-D/2}\|b_1\|$. The reduction was computed with `fplll` and the reduction conditions and the unimodularity of the transformation were re-verified in exact rational arithmetic. The certificate depends only on the correctness of the p-digit value of γ (`mpmath`, Brent–McMillan).

D	digits p	no relation of degree $\le D$ with height $< 10^h$, $h =$
2	300	98.6
3	300	73.4
4	300	58.3
6	400	55.1
8	500	53.2
10	600	52.0
12	800	58.8
16	1000	55.7
20	1400	63.3
24	1800	68.3
30	2400	73.3

Table 3: Finite certificates: for each row, every nonzero $P \in \mathbb{Z}[X]$ with $\deg P \le D$ and $H(P) < 10^h$ satisfies $P(\gamma) \neq 0$. Compare the manuscript's exhaustive degree-4, height-4 box.

These certificates push the A-world to $\deg f > 30$ or $H(f) \ge 10^{73}$ and nothing more: Proposition 3.2 and Theorem 3.1 produce algebraic numbers of any larger degree, or of small degree and larger height, that pass every finite test performed.

9 Outcome

The search was run on the largest sub-theory of established mathematics that we could identify as relevant, and it did not terminate. This is not for lack of consequences: the A-world predicts $\delta \in \mathbf{E} \cap \mathbf{D}$ with δ transcendental, a Diophantine floor of exponent $d3^d - 1$ for linear forms in $1, \gamma, \delta$, and $\mu(\delta) = 2$ when γ is rational; the B-world predicts free local arithmetic at every prime. It is because every established theorem that says something arithmetic about γ and that we could identify reaches γ only through connection identities and is therefore invariant under the framing shear (4.1), so that it cannot tell γ from $\gamma + t$ for small algebraic t, while the sentence A is precisely the statement that some such shear lands on zero; the theorems that identify γ analytically have not yet been turned into shear-breaking arithmetic statements, and the statements that would break the shear are all conjectures at present. The one genuinely new lever is structural rather than Diophantine: the question "is δ an E-value?" is at least as hard as the original one in its negative direction, but in its positive direction, given an explicit representation, it would make the algebraicity of γ a finite computation.

Part II
The collision world: the search on the ring dichotomy $\mathbf{E} \cap \mathbf{D} = \overline{\mathbb{Q}}$

Summary of Part II. Part I showed that the alternative $\gamma \in \overline{\mathbb{Q}}$ versus $\gamma \notin \overline{\mathbb{Q}}$ is invisible to every established theorem, because those theorems reach γ only through connection identities and are invariant under a framing shear. Here the search is re-run on the dichotomy chosen in its place: $\mathrm{A}_{\mathbf{S}}$, "some value of an E-function at an algebraic point coincides with a transcendental Borel sum of an arithmetic Gevrey series" ($\mathbf{E} \cap \mathbf{D} \supsetneq \overline{\mathbb{Q}}$), against $\mathrm{B}_{\mathbf{S}}$, the intersection conjecture $\mathbf{E} \cap \mathbf{D} = \overline{\mathbb{Q}}$ of Fischler and Rivoal. The framework is theirs: their paper on relations between values of arithmetic Gevrey series already contains the identity $\delta = e\,\mathrm{Ein}(1) - e\gamma$, the implication "$\mathbf{E} \cap \mathbf{D} = \overline{\mathbb{Q}} \Rightarrow \gamma \notin \overline{\mathbb{Q}}$", the mechanism $\gamma \in \mathbf{E} \Leftrightarrow \delta \in \mathbf{E}$, and a conditional lifting theorem for mixed functions. What this note adds is the following. The polyexponential tower has an explicit collision spectrum: the Borel sum $\omega_{\lambda,r}$ of the order-r Euler series at λ is an E-value if and only if $P_r(\lambda) = [t^r]\big(\Gamma(1-t)\lambda^t\big)$, an explicit polynomial in γ, zeta values and $\log\lambda$, is an E-value (an elementary proof through the incomplete Gamma function is given); the $r = 1$ locus is empty or a coset of the group $\{\beta : \log\beta \in \mathbf{E}\}$; and the only collision that $\gamma \in \overline{\mathbb{Q}}$ is known to force is $\omega_{1,1} = \delta$. We then determine the logical geometry of the route: the collision world formally strengthens the original problem (refuting the one-parameter witness family $e^z(\mathrm{Ein}\,z - c)$, $c \in \overline{\mathbb{Q}}$, *is* the original problem), its only possible advantage is structural, linear disjointness of $\mathbf{S}$ and $\mathbf{E}$ would not refute $\gamma \in \overline{\mathbb{Q}}$ while $\mathbf{E} \cap \mathbf{D} = \overline{\mathbb{Q}}$ does, and every available tool—Beukers' lifting, the Fischler–Rivoal containment theorems, Diophantine measures, E-approximations, adelic product formulas, exponential motives—is collision-compatible: each can explain a collision, none can exclude one, and we know of no separation theorem between any two of the rings $\mathbf{E}, \mathbf{G}, \mathbf{D}$. Certified lattice probes in natural finite windows (a 34-element basis of classical E-values, 13 D-values, 8 $\mathbf{S}$-values, and 78 quadratic monomials) find exactly the known $\mathbf{S}$-mediated identities and nothing else, with explicit height bounds. The outcome is a sharper target, not a contradiction: the single statement "$\delta \in \mathbf{E} \Rightarrow \delta \in \overline{\mathbb{Q}}$", which is the intersection conjecture at the one element δ, would end the search.

10 The rings and the dichotomy

Notation from Part I: γ, Ein, E_1, $\delta = eE_1(1)$, $\mathrm{Ein}_r(z) = \sum_{n\ge1}(-1)^{n-1}z^n/(n^r n!)$, and $\mathrm{Ein}(x) = \gamma + \log x + E_1(x)$ for $x > 0$. We use the rings of Fischler and Rivoal:

- $\mathbf{E}$, the values of E-functions at algebraic points; $\mathbf{G}$, the values of G-functions at algebraic points (including analytic continuations); both are rings containing $\overline{\mathbb{Q}}$ [8, 20].
- $\mathbf{S}$, the $\mathbf{G}[\gamma]$-module generated by the values $\Gamma(a/b)$, $a/b \in \mathbb{Q} \setminus \mathbb{Z}_{\le0}$; it is a ring, equal to the $\mathbf{G}$-module generated by all derivatives $\Gamma^{(k)}(a/b)$ [8, §1].
- $\mathbf{D}$, the values $f_\theta(1)$, where f is an arithmetic Gevrey series of order one—a series $\sum_{n\ge0} n!\,c_n z^{-n-1}$ whose Borel transform $\sum c_n\xi^n$ is a G-function—and f_θ its Borel (Ramis 1-)sum in a non-anti-Stokes direction θ; evaluation at 1 costs nothing by rescaling. $\mathbf{D}$ is a ring containing $\overline{\mathbb{Q}}$ and δ [10, §1], [21, §4].
- $\mathbf{V}$, the $\mathbf{S}$-module generated by $\mathbf{E}$; since $\mathbf{E}$ and $\mathbf{S}$ are rings, $\mathbf{V}$ is a ring.

Theorem 10.1 (Structure theorems; Fischler–Rivoal)**.** (Published inputs.)

(i) [21, Thm. 3] The $\mathbf{S}$-module generated by the numbers $e^\rho\chi$, $\rho \in \overline{\mathbb{Q}}$, $\chi \in \mathbf{D}$, equals $\mathbf{V}$. In particular $\mathbf{D} \subset \mathbf{V} = \mathbf{S}\cdot\mathbf{E}$: every element of $\mathbf{D}$ is a finite sum $\sum_i s_i\varepsilon_i$ with $s_i \in \mathbf{S}$, $\varepsilon_i \in \mathbf{E}$.

(ii) [10, Thm. 1] Every element of $\mathbf{G}$ *is a polynomial with algebraic coefficients in elements of* $\mathbf{E}$ *and* $\mathbf{D}$*.*

(iii) [10, eqs. (4.4)–(4.5)] For rational $a > 0$ *and* $s \geq 0$, $\Gamma^{(s)}(a) = (-1)^s s!\, E_{a,s+1}(-1) + e^{-1} f_{a,s+1;0}(1)$ *with* $E_{a,s}(z) = \sum_{n\geq 0} z^n/(n!(n+a)^s)$ *an* E*-function and* $f_{a,s+1;0}(1) \in \mathbf{D}$*; for* $a = 1$, $s = 1$ *this is* $\gamma = \mathrm{Ein}(1) - e^{-1}\delta$*.*

For δ, (i) is the connection formula $\delta = e\,\mathrm{Ein}(1) - e\gamma$, with $\mathbf{S}$-coefficients 1 and $-\gamma$; (iii) is the same identity. Theorem 10.1 says that this is the general shape: a D-value is an E-value *decorated* by elements of $\mathbf{S}$, and conversely G-values are built from E- and D-values. The three rings are entangled; the question is whether they are entangled *only* through such decorations.

Definition 10.2 (The collision dichotomy)**.** $\mathrm{A}_{\mathbf{S}}$: $\mathbf{E} \cap \mathbf{D} \neq \overline{\mathbb{Q}}$, i.e. there exist an E-function F, an arithmetic Gevrey series $\widehat{D}$ of order one, $\alpha \in \overline{\mathbb{Q}}$ and a non-anti-Stokes direction θ with $F(\alpha) = \mathcal{S}_\theta \widehat{D}(\alpha) \notin \overline{\mathbb{Q}}$ (a *collision*; one may take $\alpha = 1$ by rescaling both series). $\mathrm{B}_{\mathbf{S}}$: $\mathbf{E} \cap \mathbf{D} = \overline{\mathbb{Q}}$, Conjecture 1 of [10]. The *point dichotomy* at δ is A_δ: $\delta \in \mathbf{E} \setminus \overline{\mathbb{Q}}$, B_δ: $\delta \notin \mathbf{E} \setminus \overline{\mathbb{Q}}$.

Proposition 10.3 (Position of the original alternative)**.** *With* A*:* $\gamma \in \overline{\mathbb{Q}}$ *and* B*:* $\gamma \notin \overline{\mathbb{Q}}$*,*

$$\mathrm{A} \implies \mathrm{A}_\delta \implies \mathrm{A}_{\mathbf{S}}, \qquad \mathrm{B}_{\mathbf{S}} \implies \mathrm{B}_\delta \implies \mathrm{B}.$$

Moreover $\delta \in \mathbf{E} \iff \gamma \in \mathbf{E} \iff E_1(1) \in \mathbf{E}$*, so that* $\mathrm{A}_\delta \iff (\gamma \in \mathbf{E}) \wedge (\delta \notin \overline{\mathbb{Q}})$ *and* $\mathrm{B}_\delta \iff (\gamma \in \mathbf{E} \Rightarrow \delta \in \overline{\mathbb{Q}})$*.*

Proof. $\delta \in \mathbf{D}$ always. If $\gamma \in \overline{\mathbb{Q}}$ then $F(z) = e^z(\mathrm{Ein}(z) - \gamma)$ is an E-function with $F(1) = \delta$, and $\delta \notin \overline{\mathbb{Q}}$ by Rivoal's theorem that at least one of γ, δ is transcendental [15, Thm. 2]; hence $\delta \in (\mathbf{E} \cap \mathbf{D}) \setminus \overline{\mathbb{Q}}$. The remaining implications are contrapositives, and the equivalences follow from $\delta = eE_1(1)$, $\gamma = \mathrm{Ein}(1) - E_1(1)$, $e^{\pm 1}, \mathrm{Ein}(1) \in \mathbf{E}$ and the ring property. □

Remark 10.4 (Attribution)*.* The implication $\mathrm{B}_{\mathbf{S}} \Rightarrow \mathrm{B}$ is [10, Thm. 2] for $(a, s) = (1, 1)$, proved there by the same argument; the equivalence $\gamma \in \mathbf{E} \Leftrightarrow \delta \in \mathbf{E}$ is the mechanism of [10, Prop. 1], which derives $\gamma \notin \mathbf{E}$ from Conjectures 1 and 2 of that paper; and [10, Cor. 1] derives $\delta \notin \overline{\mathbb{Q}}$ from their Conjecture 2 (zero removal for Gevrey series) alone. Proposition 10.3 only arranges these facts as a dichotomy. Note that Conjecture 2 alone does not refute A: A implies $\delta \notin \overline{\mathbb{Q}}$ anyway.

11 The collision spectrum of the polyexponential tower

For $\lambda \in \overline{\mathbb{Q}}$, $\lambda > 0$, and $r \geq 1$, the companion manuscript [3, §"The mixed-Gevrey aperture"] considers the order-r Euler series

$$\widehat{D}_{\lambda,r}(z) = \sum_{n\geq 0} (-1)^n \begin{bmatrix} n+1 \\ r \end{bmatrix} \lambda^{-n-1} z^{-n-1},$$

with unsigned Stirling numbers of the first kind, whose Borel transform $\frac{(-1)^{r-1}}{\lambda(r-1)!} \frac{\log^{r-1}(1+\xi/\lambda)}{1+\xi/\lambda}$ is a G-function with its singularity on the negative axis; hence direction 0 is admissible and

$$\omega_{\lambda,r} := \mathcal{S}_0\, \widehat{D}_{\lambda,r}(1) = \int_0^\infty e^{-\xi}\, \frac{(-1)^{r-1} \log^{r-1}(1+\xi/\lambda)}{\lambda\,(r-1)!\,(1+\xi/\lambda)}\, d\xi \;\in \mathbf{D}. \tag{11.1}$$

For $\lambda = 1$ these are, up to the factor $(-1)^{r-1} r!$, the values $f_{1,r+1;0}(1)$ of [10, §4]; $\omega_{1,1} = \delta$. Write

$$\Gamma(1-t) = \sum_{j\geq 0} g_j t^j = \exp\Big(\gamma t + \sum_{k\geq 2} \frac{\zeta(k)}{k} t^k\Big), \qquad P_r(\lambda) := [t^r](\Gamma(1-t)\lambda^t) = \sum_{j+k=r} g_j \frac{(\log\lambda)^k}{k!}, \tag{11.2}$$

so that $g_1 = \gamma$, $g_2 = \frac{1}{2}(\gamma^2 + \zeta(2))$, $g_3 = \frac{1}{6}\gamma^3 + \frac{1}{2}\gamma\zeta(2) + \frac{1}{3}\zeta(3)$, and $P_1(\lambda) = \gamma + \log\lambda$, $P_2(\lambda) = \frac{1}{2}((\gamma + \log\lambda)^2 + \zeta(2))$.

Theorem 11.1 (Collision spectrum)**.** *For every $\lambda \in \overline{\mathbb{Q}}_{>0}$ and $r \geq 1$,*

$$\omega_{\lambda,r} = e^{\lambda}\,\mathrm{Ein}_r(\lambda) - e^{\lambda}P_r(\lambda), \qquad \textit{hence} \qquad \omega_{\lambda,r} \in \mathbf{E} \iff P_r(\lambda) \in \mathbf{E}. \tag{11.3}$$

In particular $\delta \in \mathbf{E} \iff \gamma \in \mathbf{E}$; $\omega_{\lambda,1} \in \mathbf{E} \iff \gamma + \log\lambda \in \mathbf{E}$; $\omega_{1,2} \in \mathbf{E} \iff \gamma^2 + \zeta(2) \in \mathbf{E}$; and $\omega_{1,3} \in \mathbf{E} \iff \gamma^3 + 3\gamma\zeta(2) + 2\zeta(3) \in \mathbf{E}$. At $\lambda = 1$ the identity is [10, eq. (4.4)] with $a = 1$, $s = r - 1$.

Proof. Substitute $u = 1 + \xi/\lambda$ in (11.1):

$$\omega_{\lambda,r} = e^{\lambda}\int_1^{\infty} e^{-\lambda u}\frac{(-\log u)^{r-1}}{(r-1)!}\,\frac{du}{u}, \qquad \sum_{r\geq 1}\omega_{\lambda,r}t^r = e^{\lambda}t\int_1^{\infty} e^{-\lambda u}u^{-t-1}\,du = e^{\lambda}\,t\,\lambda^{t}\,\Gamma(-t,\lambda),$$

with the upper incomplete Gamma function, for $|t|$ small. Now $\Gamma(-t,\lambda) = \Gamma(-t) - \sum_{n\geq 0}(-1)^n\lambda^{n-t}/(n!(n-t))$, and $t\Gamma(-t) = -\Gamma(1-t)$, while for $n \geq 1$, $-t/(n-t) = -\sum_{k\geq 1} t^k/n^k$; the $n = 0$ term contributes λ^{-t}. Hence

$$t\,\Gamma(-t,\lambda) = -\Gamma(1-t) + \lambda^{-t} + \lambda^{-t}\sum_{k\geq 1}t^k\sum_{n\geq 1}\frac{(-1)^{n-1}\lambda^n}{n^k\,n!} = -\Gamma(1-t) + \lambda^{-t}\Big(1 + \sum_{k\geq 1}\mathrm{Ein}_k(\lambda)t^k\Big),$$

and therefore $\sum_{r\geq 1}\omega_{\lambda,r}t^r = e^{\lambda}(1 + \sum_k \mathrm{Ein}_k(\lambda)t^k - \Gamma(1-t)\lambda^t)$. Comparing coefficients of t^r, $r \geq 1$, gives (11.3) (the constant terms cancel). The same identity follows from the generating function of [3, Prop. "All-order mixed jet"] at $z = 1$; it was also checked by direct quadrature of (11.1) to 60 digits for eight pairs (λ, r) (script `exp07`). The equivalence follows because $e^{\pm\lambda}$ and $\mathrm{Ein}_r(\lambda)$ lie in the ring $\mathbf{E}$. The special cases are $P_1(1) = \gamma$, $P_1(\lambda) = \gamma + \log\lambda$, $2g_2$ and $6g_3$. □

Corollary 11.2 (The $r = 1$ locus)**.** *Let $L_{\mathbf{E}} = \{\beta \in \overline{\mathbb{Q}}_{>0} : \log\beta \in \mathbf{E}\}$ and $\Lambda_1 = \{\lambda \in \overline{\mathbb{Q}}_{>0} : \omega_{\lambda,1} \in \mathbf{E}\}$. Then $L_{\mathbf{E}}$ is a multiplicative group containing 1, and Λ_1 is either empty or a coset $\lambda_0 L_{\mathbf{E}}$. If $\mathbf{E}\cap\mathbf{G} = \overline{\mathbb{Q}}$, then $L_{\mathbf{E}} = \{1\}$ and Λ_1 has at most one element. Under* A*, $1 \in \Lambda_1$; under* A *and $\mathbf{E}\cap\mathbf{G} = \overline{\mathbb{Q}}$, $\Lambda_1 = \{1\}$.*

Proof. $\log(\beta_1\beta_2) = \log\beta_1 + \log\beta_2$ and $\log(1/\beta) = -\log\beta$, and $\mathbf{E}$ is a ring, so $L_{\mathbf{E}}$ is a group. If $\lambda, \mu \in \Lambda_1$ then $\log(\lambda/\mu) = (\gamma + \log\lambda) - (\gamma + \log\mu) \in \mathbf{E}$, and conversely $\lambda \in \Lambda_1$, $\beta \in L_{\mathbf{E}}$ give $\lambda\beta \in \Lambda_1$. Logarithms of positive algebraic numbers are G-values, so $\mathbf{E}\cap\mathbf{G} = \overline{\mathbb{Q}}$ forces $\log\beta \in \overline{\mathbb{Q}}$ for $\beta \in L_{\mathbf{E}}$, whence $\beta = 1$ by the Hermite–Lindemann theorem. Under A, $\gamma \in \overline{\mathbb{Q}} \subset \mathbf{E}$. □

Corollary 11.3 (What A is known to force in the tower)**.** *Assume* A*. Then $\omega_{1,1} = \delta \in (\mathbf{E}\cap\mathbf{D})\setminus\overline{\mathbb{Q}}$. For $(\lambda, r) \neq (1,1)$ the statement $\omega_{\lambda,r} \in \mathbf{E}$ is equivalent, under* A*, to $P_r(\lambda) \in \mathbf{E}$, where $P_r(\lambda)$ is now a polynomial with algebraic coefficients in $\log\lambda, \zeta(2), \ldots, \zeta(r)$, hence a G-value. Under $\mathbf{E}\cap\mathbf{G} = \overline{\mathbb{Q}}$ such a collision requires $P_r(\lambda) \in \overline{\mathbb{Q}}$, which is excluded for $r = 1$, $\lambda \neq 1$ (Hermite–Lindemann) and for $(\lambda, r) = (1, 2)$ (π is transcendental); for $(\lambda, 2)$ with $\lambda \neq 1$ it would require $(c + \log\lambda)^2 + \pi^2/6 \notin \overline{\mathbb{Q}}$, and for $r \geq 3$ statements about $\zeta(3), \zeta(5), \ldots$, all of which are open. No collision other than $(1,1)$ is known to be forced by* A*, and none other than $(1,1)$ is known not to be.*

Remark 11.4. The theorem is the collision-world form of the Stokes-framing barrier [3, §"The Stokes-framing barrier"]: the **S**-decorations of the tower are exactly the Taylor coefficients of $\Gamma(1-t)\lambda^t$, i.e. the unipotent connection data of the polyexponential E-operators, and a collision means that one of these decorations is itself an E-value. The framing shear of Part I moves γ inside these decorations and fixes $\mathbf{E}$; a collision is a statement that the shear is *not* free at that decoration.

12 Logical geometry of the collision route

Proposition 12.1 (Three facts about the route)**.** *(i)* $\mathrm{A} \Rightarrow \gamma \in \mathbf{E}$ *carries no information* ($\overline{\mathbb{Q}} \subset \mathbf{E}$)*. The content of* $\mathrm{A} \Rightarrow \mathrm{A}_\delta$ *is entirely in* $\delta \notin \overline{\mathbb{Q}}$*.*

(ii) The witness supplied by A *is the one-parameter family* $F_c(z) = e^z(\mathrm{Ein}(z) - c)$*,* $c \in \overline{\mathbb{Q}}$*, at* $z = 1$*. Refuting the collisions of this family alone, "*$\delta \neq F_c(1)$ *for all* $c \in \overline{\mathbb{Q}}$*", is verbatim the statement* B*. Hence* B_δ *and* $\mathrm{B}_{\mathbf{S}}$ *are formally stronger than* B *(they are not known to be equivalent to it), and the only possible advantage of the collision route is structural: a statement about all* E*-functions at once may be provable where its one-parameter special case is not (the Lindemann–Weierstrass phenomenon).*

(iii) Let $\ell(\mathbf{S}, \mathbf{E})$ *denote the hypothesis that* $\mathbf{S}$ *and* $\mathbf{E}$ *are linearly disjoint over* $\overline{\mathbb{Q}}$ *(*$\overline{\mathbb{Q}}$*-linearly independent elements of* $\mathbf{S}$ *remain* $\mathbf{E}$*-linearly independent). Then* $\ell(\mathbf{S}, \mathbf{E})$ *implies* ($\gamma \notin \overline{\mathbb{Q}} \Rightarrow \delta \notin \mathbf{E}$) *but does* not *refute* A*; the hypothesis that refutes* A *through* δ *is* $\mathbf{E} \cap \mathbf{D} = \overline{\mathbb{Q}}$*, and its content at* δ *is exactly* B_δ*.*

Proof. (i) and (ii) are immediate from Proposition 10.3. (iii) Write $\delta = e\,\mathrm{Ein}(1) \cdot 1 + e \cdot (-\gamma)$. If $\gamma \notin \overline{\mathbb{Q}}$, then $1, \gamma$ are $\overline{\mathbb{Q}}$-linearly independent elements of $\mathbf{S}$, and $\delta - \varepsilon_0 = 0$ with $\varepsilon_0 \in \mathbf{E}$ would be the $\mathbf{E}$-linear relation $(e\,\mathrm{Ein}(1) - \varepsilon_0) \cdot 1 + (-e) \cdot \gamma = 0$, so $\ell(\mathbf{S}, \mathbf{E})$ gives $e = 0$, absurd; thus $\delta \notin \mathbf{E}$. If $\gamma \in \overline{\mathbb{Q}}$, the decorations are algebraic and $\ell(\mathbf{S}, \mathbf{E})$ says nothing. Finally, $\mathbf{E} \cap \mathbf{D} = \overline{\mathbb{Q}}$ at the element $\delta \in \mathbf{D}$ reads $\delta \in \mathbf{E} \Rightarrow \delta \in \overline{\mathbb{Q}}$, which is B_δ by Proposition 10.3. □

The routes to B that the manuscripts and [10] identify sit differently with respect to A: $\mathbf{E} \cap \mathbf{D} = \overline{\mathbb{Q}}$ refutes A through the transcendence of δ; the expectation $\gamma \notin \mathbf{E}$ [16] refutes it because $\overline{\mathbb{Q}} \subset \mathbf{E}$; the exponential period conjecture through the non-splitting of the Euler–Mascheroni motive; Conjecture 2 of [10] gives $\delta \notin \overline{\mathbb{Q}}$ but not B. Among the ring-theoretic statements considered, the smallest one that still refutes A is B_δ.

13 The attempts and where each stops

We ran the collision world against every tool we could bring to bear. The ledger records, for each tool, the exact point of failure. The pattern is uniform: every tool is *collision-compatible*, in the sense that it can explain a collision but cannot exclude one.

Beukers' lifting theorem (*E*-side). Suppose $\delta = F(1)$ for an E-function F. Combined with $\delta = e\,\mathrm{Ein}(1) - e\gamma$ this is the relation $F(1) - e\,\mathrm{Ein}(1) + e\gamma = 0$. If $\gamma \in \overline{\mathbb{Q}}$, its coefficients are algebraic, and Beukers' theorem [5, Thm. 1.3] lifts it (at any point where the system is regular; apparent singularities can be removed, [5, Thm. 1.5]) to a functional relation: F lies in the $\overline{\mathbb{Q}}(z)$-span of $e^z\,\mathrm{Ein}(z)$, e^z and the remaining functions of the joint system, with the specialisation prescribed at 1. The witness $F_\gamma = e^z(\mathrm{Ein}\, z - \gamma)$ realises this. Thus Beukers *confirms* the collision that A forces; it cannot refute it. If $\gamma \notin \overline{\mathbb{Q}}$, the number $e\gamma$ is not known to be an E-value, so there is no relation among E-values to lift unless $\gamma \in \mathbf{E}$, in which case the same lifting shows that every δ-witness F is a $\overline{\mathbb{Q}}(z)$-combination of $e^z\,\mathrm{Ein}(z)$, $e^z P(z)$ and system functions, for any γ-witness P: collisions for δ are *explained* by collisions for γ, again with no contradiction. What the route needs is a lifting theorem for relations among E-values *and* D-values; such a theorem exists, but only conditionally: [10, Thm. 4] lifts relations among values of mixed functions under their Conjecture 3, and Conjecture 3 already implies $\mathbf{E} \cap \mathbf{D} = \overline{\mathbb{Q}}$.

Containment theorems (*D*-side). Theorem 10.1 and [8, Thm. 2] (Stokes constants lie in $\mathbf{S}$) are one-directional: they place $\mathbf{D}$ inside $\mathbf{S} \cdot \mathbf{E}$, $\mathbf{G}$ inside $\overline{\mathbb{Q}}[\mathbf{E}, \mathbf{D}]$, and Stokes data inside $\mathbf{S}$.

Nothing places **S** or **D** *outside* **E**. The direction needed is a separation statement, and we are not aware of one for any pair among $\mathbf{E}, \mathbf{G}, \mathbf{D}$: Fischler and Rivoal remark that $\mathbf{E} \cap \mathbf{G} = \overline{\mathbb{Q}}$ and $\mathbf{G} \cap \mathbf{D} = \overline{\mathbb{Q}}$ "seem currently out of reach" [10], and even a proof that $e \notin \mathbf{G}$ is not known to us.

Diophantine measures. A_δ implies that δ is not a U-number [9, Cor. 3] and, if the witness has rational coefficients (the point 1 being rational), $\mu(\delta) = 2$ [11, Thm. 1] (preprint input). The continued fraction of δ (Part I, Table 2 has convergent exponents 2.000 and Khinchin-typical partial quotients; nothing to contradict.

E-approximations. By [8, Thm. 3], every element of **E** has E-approximations (sequences $P_n/Q_n \to \xi$ whose generating series are algebraic functions times E-functions composed with algebraic functions), so $\mathrm{A}_\delta \Rightarrow \gamma$ has E-approximations; Fischler and Rivoal conjecture it has none. The known approximations to γ (Aptekarev, Rivoal) are of the extended type with a $\log(1-z)$ factor, [8, eq. (1.8)], and are not counterexamples. The conjecture is of the same kind as $\gamma \notin \mathbf{E}$ and no tool attacks it.

Adelic product formulas. Matala-aho and Zudilin's global-relation theorem for Euler's factorial series [23] and the p-adic Padé radius transition of [2] constrain relations that hold simultaneously at many places. The real Borel sum δ is not the "value" of the divergent series at the real place in the sense required, and Part VII of [1] shows that prescribed simultaneous limits do not preserve relations. The collision is a statement at one place.

Exponential motives. E-values are integrals of G-functions against $e^{\pm t}$ over bounded chains, D-values over rapid-decay rays; a collision equates periods of two exponential motives paired with different chains. The period conjecture for exponential motives [12] would exclude a transcendental coincidence, but nothing unconditional does; this is the same barrier as the matched-line lifting problem of [3].

Finite windows. A lattice search can only find witnesses of small height. The witness that A supplies, $\delta = e\,\mathrm{Ein}(1) - e\gamma$ with $\gamma = p/q$, is an integer relation of height about q, and $q > 10^{244663}$ if γ is rational (Papanikolaou 1997; later continued fraction computations give far larger bounds, see [22]); for algebraic γ of degree d the relation is not an integer relation at all but one of degree d over $\mathbb{Q}$. So finite windows cannot see the forced collision; they probe for *independent* low-complexity collisions, which the $\mathrm{B}_{\mathbf{S}}$-world predicts do not exist. Section 14 reports them.

14 Probes (computations for Part II)

All numbers were computed with `mpmath` at 1660 (linear, E–D, E–**S**, D–**S** windows) or 1560 (quadratic window) digits, the reductions with `fplll`, and the reduction conditions and unimodularity re-verified in exact rational arithmetic as in Part I. For a vector $x = (x_0, \dots, x_m)$ rounded to p decimals with $N = \lfloor 10^p/2 \rfloor$, an integer relation of height H yields a lattice vector of norm at most $(m+1)H\sqrt{m+2}$, so that no relation exists with $H < \|b_1\|/(\alpha^{m/2}(m+1)\sqrt{m+2})$, $\alpha = 1/(0.99 - 0.51^2)$. When the reduced basis contains a short vector with tiny last coordinate, it is an integer relation among the rounded values, and it is re-checked against the exact numbers.

The E-basis. Thirty-four classical E-values with no known $\mathbb{Q}$-linear relation: 1; $e^{\pm 1}, e^{\pm 2}, e^{\pm 1/2}, e^3, e^{1/3}$; $\mathrm{Ein}(\pm 1), \mathrm{Ein}(\pm 2), \mathrm{Ein}(\pm\frac{1}{2}), \mathrm{Ein}(3)$, $\mathrm{Ein}_2(\pm 1), \mathrm{Ein}_2(2), \mathrm{Ein}_3(1)$; I_0, I_1, J_0, J_1 at 1 and 2; ${}_1F_1(\frac{1}{2}; \frac{3}{2}; \pm 1)$, ${}_1F_1(\frac{1}{3}; \frac{4}{3}; \pm 1)$, ${}_1F_1(\frac{1}{4}; \frac{5}{4}; 1)$, ${}_1F_1(\frac{2}{3}; \frac{5}{3}; 1)$. Kummer-transform partners and contiguous triples were excluded to avoid known linear dependencies.

Validation. With $e\,\mathrm{Ein}(1)$ and $e\gamma$ adjoined, the search returns exactly $\delta - e\,\mathrm{Ein}(1) + e\gamma = 0$ (residual $< 10^{-1659}$, with δ computed independently as $e\,E_1(1)$); with γ adjoined, $E_1(1) - \mathrm{Ein}(1) + \gamma = 0$. These two runs show that the machinery detects the **S**-mediated identities. A third run returns $\omega_{1,2} - e\,\mathrm{Ein}_2(1) + eg_2 = 0$, but since $\omega_{1,2}$ was computed through Theorem 11.1 it only checks the lattice code; the independent check of that identity is the 60-digit quadrature of script `exp07`.

Part III
Building the A-world: an axiom-search ledger

Summary of Part III. Parts I and II showed that the alternative $\gamma \in \overline{\mathbb{Q}}$ (world A) versus $\gamma \notin \overline{\mathbb{Q}}$ (world B) is invisible to every theorem that reaches γ through its connection formulas, and re-ran the search on the ring dichotomy $\mathbf{E} \cap \mathbf{D} = \overline{\mathbb{Q}}$. This note carries out the programme proposed in the session: build freely inside each world, without aiming at the proof, and look for a construction that ordinary mathematics (world C) rejects. We first make the programme precise. A bookkeeping lemma identifies termination with the refutation of a single consequence; a *density obstruction* theorem shows that no shear-invariant theorem places γ (or δ, or any $E_1(\lambda)$) inside or outside *any* class that is dense with dense complement—algebraic numbers, E-, G-, D- and **S**-values, periods, logarithms of algebraic numbers, numbers of irrationality exponent two; hence every terminating pair (construction, rejection) passes through an analytic definition of the constant. We then build the A-world along ten definitions of γ and compute the Diophantine content that A forces through each: the Euler–Maclaurin and Kummer representations force approximations of $\log n$ by algebraic numbers of exponent $0.42 \to 0.17$ and $0.18 \to 0.07$ (at most 0.36 under the most favourable truncation), decaying like $1/\log n$, far below every transcendence measure; Mertens' product forces none. For linear forms we prove a *tie theorem*: a bound $|a + b\Gamma + c\Delta| < \eta$ proved shear-invariantly forces $|b - ec| \le 2\eta/\ell$, so shear-invariant families of integer forms in $1, \gamma, \delta$ have exponent at most 1, generically $1/2$, and can never meet Nesterenko's threshold. Rivoal's forms are verified to be tied ($b_n/c_n - e = 1.4 \cdot 10^{-255}$ at $n = 39$), LLL-reduced shear-invariant forms sit at exponent 0.50–0.54, and the D-side Stieltjes forms are untied but grow to 10^{44}. This gives a precise specification of the first Diophantine construction that could matter: forms in $1, \gamma, \delta$ that are smaller than their own tie defect, equivalently approximations of γ by ratios of engine numbers whose convergence is proved through a bridge. A linear hierarchy $\mathrm{X}(V)$ of statements between "$\gamma \notin \overline{\mathbb{Q}}$" and "$\delta \notin \mathbf{E}$" is established, with the equivalences $\mathrm{X}(\overline{\mathbb{Q}} + \overline{\mathbb{Q}}e) \Leftrightarrow \gamma \notin \overline{\mathbb{Q}} + \overline{\mathbb{Q}}e^{-1}$ and, under the point conjecture of Part II, $\mathrm{X}(W) \Leftrightarrow \delta \notin \overline{\mathbb{Q}}$. The ledger (twelve A-entries, three B-entries) shows a uniform pattern: every entry expected to be false in C is a membership or splitting statement whose rejection is a known conjecture at one element, and every Diophantine entry is expected to be true or is inert. No contradiction is found; three sharpened targets remain.

15 The programme, made precise

15.1 Worlds, constructions, termination

Let T_0 be the background theory of Part I (ZFC with its standard model), in which $\gamma = \lim(H_n - \log n)$, $\mathrm{Ein}(z) = \sum_{n\ge1}(-1)^{n-1}z^n/(n\,n!)$, E_1, and $\delta = eE_1(1)$ are definable numbers. The worlds are the theories

$$\mathrm{W_A} = T_0 + \mathrm{A}, \qquad \mathrm{W_B} = T_0 + \mathrm{B}, \qquad \mathrm{C} = T_0,$$

with A: $\gamma \in \overline{\mathbb{Q}}$ and B: $\gamma \notin \overline{\mathbb{Q}}$. An A-*construction* is a sentence X of T_0 with $\mathrm{W_A} \vdash X$; a B-construction is a sentence Y with $\mathrm{W_B} \vdash Y$. The programme is: build many X and Y, and try to prove $\neg X$ or $\neg Y$ in C.

Lemma 15.1 (Bookkeeping)**.** *The following are equivalent: (i) some* A*-construction X has $T_0 \vdash \neg X$; (ii)* $\mathrm{W_A}$ *is inconsistent; (iii)* $T_0 \vdash \mathrm{B}$*. Symmetrically for* B*-constructions and* $T_0 \vdash \mathrm{A}$*. Since exactly one of* A, B *is true and* T_0 *is sound, the true world has the standard model and is consistent; at most one of the two worlds is inconsistent.*

Proof. (i)⇒(ii): $\mathrm{W_A} \vdash X \wedge \neg X$. (ii)⇒(iii): deduction theorem. (iii)⇒(i): take $X = \mathrm{A}$. □

So the programme is not a new proof strategy in the logical sense—termination is the same event as proving B—but it is a *search* strategy: it replaces the single target B by the set of all consequences of A, any one of which may be easier to refute than A itself. Fermat's Last Theorem is the model: the consequence "a Frey curve exists" was refuted by modularity, a theorem about objects of a different kind. The question is whether γ admits an analogue of the Frey curve: a consequence of A living in a domain where C has rejecting power.

15.2 The density obstruction

Part I introduced the engine theory $\mathcal{K}(\mathbb{F};\mathcal{C})$ (Definition 4.1 there): T_0 with constants for the elements of a field $\mathbb{F} \subset \mathbb{C}$ and their complete elementary diagram (K1), and with constant symbols Γ, Z_λ, Δ tied to the definable numbers only by the connection axioms $Z_\lambda = \mathrm{Ein}(\lambda) - \Gamma - \log\lambda$, $\Delta = eZ_1$ (K2) and finitely many strict rational enclosures $\mathcal{C}$ (K3). Theorem 4.2 there (two models) shows that $\mathcal{K} + \mathrm{A}_\Gamma$ and $\mathcal{K} + \mathrm{B}_\Gamma$ are both satisfiable, by the sheared models $\mathfrak{s}_c$: $\Gamma \mapsto c$, $Z_\lambda \mapsto \mathrm{Ein}(\lambda) - c - \log\lambda$, $\Delta \mapsto e(\mathrm{Ein}(1) - c)$. Write $I = I(\mathcal{C})$ for the set of $c \in \mathbb{R}$ satisfying all enclosures of $\mathcal{C}$ after this substitution; it is an open interval containing γ, of length $\ell(\mathcal{C}) > 0$. The obstruction is much more general than the alternative $\overline{\mathbb{Q}}/\notin \overline{\mathbb{Q}}$.

Theorem 15.2 (Density obstruction)**.** *Let $\mathcal{P} \subset \mathbb{R}$ be a T_0-definable set such that both $\mathcal{P}$ and $\mathbb{R} \setminus \mathcal{P}$ are dense in some neighbourhood of γ. Then, for every subfield $\mathbb{F} \subset \mathbb{C}$ and every finite $\mathcal{C}$, neither $\Gamma \in \mathcal{P}$ nor $\Gamma \notin \mathcal{P}$ is provable in $\mathcal{K}(\mathbb{F};\mathcal{C})$. The same holds for $\Delta \in \mathcal{P}'$ and $Z_\lambda \in \mathcal{P}''$ (λ real) whenever $\mathcal{P}'$, $\mathcal{P}''$ are dense with dense complement near δ, resp. $E_1(\lambda)$.*

Proof. $I(\mathcal{C})$ is an open interval around γ, so it meets both $\mathcal{P}$ and its complement; pick $c_1 \in I \cap \mathcal{P}$, $c_2 \in I \setminus \mathcal{P}$. Both $\mathfrak{s}_{c_1}$ and $\mathfrak{s}_{c_2}$ are models of $\mathcal{K}(\mathbb{F};\mathcal{C})$ (Part I, proof of Theorem 4.2, and since they are expansions of the standard model, $\Gamma \in \mathcal{P}$ holds in $\mathfrak{s}_{c_i}$ iff $c_i \in \mathcal{P}$. For Δ and Z_λ use that $c \mapsto e(\mathrm{Ein}(1) - c)$ and $c \mapsto \mathrm{Ein}(\lambda) - c - \log\lambda$ are affine bijections of $\mathbb{R}$, so density near δ or $E_1(\lambda)$ transfers to density near γ. □

Corollary 15.3 (Every arithmetic dichotomy is shear-invisible)**.** *Each of the following T_0-definable sets is countable and dense, or dense of measure zero (the Liouville numbers), or of full measure with dense complement (exponent exactly* 2*), hence satisfies the hypothesis of Theorem 15.2: $\mathbb{Q}$; $\overline{\mathbb{Q}}$; $\mathbb{R} \cap \mathbf{E}$, $\mathbb{R} \cap \mathbf{G}$, $\mathbb{R} \cap \mathbf{D}$, $\mathbb{R} \cap \mathbf{S}$; the real periods; the real exponential periods; $\{\log\alpha : \alpha \in \overline{\mathbb{Q}} \cap \mathbb{R}_{>0}\}$; $\overline{\mathbb{Q}} + \overline{\mathbb{Q}}\pi$; $\mathbb{R} \cap e^{-1}V$ for every countable definable $\overline{\mathbb{Q}}$-subspace $V \ni e$ of $\mathbb{C}$ (Section 18); the Liouville numbers; the numbers of irrationality exponent exactly* 2*. Consequently no shear-invariant theorem decides whether γ, δ or $E_1(\lambda)$ belongs to any of these classes. In particular the point dichotomy $\mathrm{A}_\delta/\mathrm{B}_\delta$ of Part II (Definition 10.2 there), the sub-dichotomies of the* B*-world ($\gamma \in \mathbf{E}$?, $\delta \in \overline{\mathbb{Q}}$?), and the geometric alternative $\gamma \in \overline{\mathbb{Q}} + \overline{\mathbb{Q}}\pi$ are all in the same position as* A/B*.*

Theorem 15.2 explains, after the fact, why the change of dichotomy in Part II could not by itself help: the obstruction is not a property of the set $\overline{\mathbb{Q}}$ but of the way γ enters the established theorems. Note that for some of the pairs listed the membership is a *theorem* of T_0: $\gamma \in \mathbf{S}$ by definition of $\mathbf{S}$, $\delta \in \mathbf{D}$ by (D6), and γ, δ are exponential periods by (D5). The corollary says that these theorems are reached only through a bridge, which is precisely the point.

15.3 The shear-filter rule

Call an A-construction $X = \psi(\gamma)$ *shear-invariantly derived* if $\mathcal{K}(\mathbb{F}_0; \mathcal{C}) + \mathrm{A}_\Gamma \vdash \psi(\Gamma)$ for some finite $\mathcal{C}$ (the derivation uses about $\gamma, E_1(\lambda), \delta$ only the connection formulas and enclosures), and call a rejection $T_0 \vdash \neg\psi(\gamma)$ shear-invariant if $\mathcal{K}(\mathbb{F}_0; \mathcal{C}') \vdash \neg\psi(\Gamma)$.

Corollary 15.4 (Shear-filter rule)**.** *In a terminating pair (an* A*-construction $X = \psi(\gamma)$ together with a rejection $T_0 \vdash \neg X$), at most one of the two halves is shear-invariant. The same holds for* B*-constructions.*

Proof. If both were, $\mathcal{K} + \mathrm{A}_\Gamma$ would prove $\psi(\Gamma) \wedge \neg\psi(\Gamma)$ (take $\mathcal{C} \cup \mathcal{C}'$), contradicting its satisfiability. □

The rule turns the programme into a bookkeeping discipline. For each entry of the ledger we record which *bridge*—which analytic definition of γ—the derivation uses, and which the rejection would have to use. The bridges are the theorems of T_0 that identify γ with an analytic expression; none of them is available in $\mathcal{K}$:

(D1) Euler–Maclaurin: $\gamma = H_n - \log n - \frac{1}{2n} + \sum_{k=1}^{K} \frac{B_{2k}}{2k\, n^{2k}} + R_{n,K}$, the remainder being bounded by the first omitted term.

(D2) Kummer/Ein: $\gamma = \mathrm{Ein}(n) - \log n - E_1(n)$, $0 < E_1(n) < e^{-n}/n$.

(D3) Mertens: $e^{-\gamma} = \lim_{x\to\infty} \log x \prod_{p \le x}(1 - 1/p)$, with error $O(1/\log x)$ elementarily.

(D4) Laurent: $\gamma = \lim_{s\to 1}\big(\zeta(s) - \frac{1}{s-1}\big)$, the zeroth Stieltjes constant γ_0.

(D5) Period: $\gamma = -\Gamma'(1) = -\int_0^\infty e^{-t} \log t\, dt$, an exponential period; period matrix $\left(\begin{smallmatrix} 1 & \gamma \\ 0 & 2\pi i \end{smallmatrix}\right)$ of the Euler–Mascheroni motive [12].

(D6) Connection at 1 and Borel sum: $\gamma = \mathrm{Ein}(1) - E_1(1)$, $\delta = eE_1(1) = \int_0^\infty e^{-t}(1+t)^{-1} dt$, the Borel sum of Euler's series $\sum (-1)^n n!$; so $\delta \in \mathbf{D}$.

(D7) Vacca: $\gamma = \sum_{n \ge 1} (-1)^n \lfloor \log_2 n \rfloor / n$; Mahler-type integral representations [16].

(D8) Sondow: $\gamma = \iint_{[0,1]^2} \frac{x-1}{(1-xy)\log(xy)}\, dx\, dy$ [28] and the rationality criteria of [27].

(D9) Expansions: the decimal and continued-fraction expansions of γ (the latter computed to 2910 partial quotients in Part I).

(D10) Exponentiation: e^γ as the Mertens constant, $\lim (\log x)^{-1} \prod (1 - 1/p)^{-1}$.

Each bridge, combined with A, yields a statement of T_0 about a definable object of a different kind (a sequence of rationals, a product over primes, a Laurent coefficient, a period matrix, a Borel sum, a Mahler integral, an expansion), and it is in these other domains that C might have rejecting power. Section 16 runs through them.

16 The A-world along the definitions

Throughout this section assume A, write $c = \gamma \in \overline{\mathbb{Q}}$, $d = [\mathbb{Q}(c) : \mathbb{Q}]$, and $\mathrm{H}(\rho) = \max(|p|, |q|)$ for $\rho = p/q$ in lowest terms.

16.1 Bridges (D1), (D2): forced approximations of $\log n$

Under A, every representation $\gamma = X_n - \rho_n + \varepsilon_n$ with X_n a transcendental "engine number" $(\log n, \mathrm{Ein}(n))$, ρ_n rational and ε_n an explicit error becomes an approximation $X_n \approx \rho_n - c$ of an engine number by an algebraic number of degree $\le d$. This is the place where the analytic definitions produce a statement of a Diophantine kind for which C has testers: transcendence measures of $\log n$ (Baker, Feldman) and irrationality measures ($\mu(\log 2) \le 3.575$ [25]). Define the *forced exponent*

$$\kappa_{n,K} := -\frac{\log|\varepsilon_{n,K}|}{\log \mathrm{H}(\rho_{n,K})}.$$

Proposition 16.1 (Forced approximations are trivial)**.** (Finite certificate; script `exp10`.) *For the Euler–Maclaurin approximants $\rho_{n,K} = H_n - \frac{1}{2n} + \sum_{k\le K} B_{2k}/(2kn^{2k})$, optimised over K, and for the Kummer approximants $S_{n,K} = \sum_{k\le K}(-1)^{k-1}n^k/(k\,k!)$ (for which* A *forces $|\log n + c - S_{n,K}| = |E_1(n) - \mathit{tail}_K|$), truncated by the declared rule $K^* = \min\{K : |\mathrm{Ein}(n) - S_{n,K}| \le E_1(n)/10\}$, the exponents are those of Table 4: they decrease from 0.42 to 0.17 and from 0.18 to 0.07; the most favourable truncation, which can profit from an accidental cancellation between the tail and $E_1(n)$, gives 0.36 at $n = 5$ and $0.17 \to 0.07$ for $n \ge 10$. Under* A *of degree d the algebraic number $\alpha = \rho_{n,K} - c$ has degree $\le d$ and height $\mathrm{H}(\alpha) \le C_c\,\mathrm{H}(\rho_{n,K})^d$, so*

$$C_1\,\mathrm{H}(\alpha)^{-\kappa_{n,K}/d} \;\le\; |\log n - \alpha| \;\le\; C_2\,\mathrm{H}(\alpha)^{-\kappa_{n,K}/d}, \qquad \kappa_{n,K}/d \le 0.43 :$$

the exponent of the forced approximation relative to the height of α is exactly $\kappa_{n,K}/d$. Approximations of this quality exist for every real number—already the rational approximations of Dirichlet's theorem have exponent 2—so no transcendence measure of $\log n$ is contradicted, for any d; in the rational case $d = 1$ one would need $\kappa > \mu(\log n) \ge 2$.

Proof. The first statement is the computation. For the height of α: if P_c is the minimal polynomial of c, then α is a root of $q^d P_c(p/q - x)$, an integer polynomial of degree d and height $\le 2^{d+1}\,\mathrm{H}(P_c)\,\mathrm{H}(\rho)^d$; conversely, with h the absolute logarithmic height, $\log \mathrm{H}(\rho) = h(\rho) = h(\alpha + c) \le h(\alpha) + h(c) + \log 2$ and $h(\alpha) \le d^{-1}\log(\sqrt{d+1}\,\mathrm{H}(\alpha))$, so $\mathrm{H}(\rho) \le C\,\mathrm{H}(\alpha)^{1/d}$. Substituting $\mathrm{H}(\rho) = |\varepsilon|^{-1/\kappa}$ gives the two displayed inequalities. The last sentence is Dirichlet's theorem. □

Remark 16.2 (Why the exponents decay). For (D1) the optimal truncation is at $K \approx \pi n$ with error $\approx e^{-2\pi n}$ ($\log_{10}|\varepsilon| = -437.7$ at $n = 160$ against $-2\pi\cdot 160/\log 10 = -436.6$), while the height is dominated by the denominators n^{2K} and $\mathrm{lcm}(1,\dots,n)$, $\log \mathrm{H} \approx 2\pi n \log n$; hence $\kappa_n \approx 1/\log n$ (heuristic; the table gives 0.166 at $n = 160$ against $1/\log 160 = 0.197$). For (D2) the error forced by A is $E_1(n) \approx e^{-n}/n$ whatever the truncation (apart from accidental cancellations such as the one at $n = 5$, $K = 14$), while the height of the partial sum grows like $K!\,n^K$. This is the manuscript's "fast analytic decay disappears after taking primitive heights" [1], now measured along the definitions themselves. Note also that the difference of two instances, $\log(m/n) \approx \rho_{m,K} - \rho_{n,K'}$, does not involve c at all: the c-free content of (D1) is a theorem of T_0, and the only A-specific content is that a *single* algebraic c makes all the residues $\log n + c - \rho_{n,K}$ small simultaneously. We know of no tester in C that uses more than one n at a time.

16.2 Bridges (D3), (D10): Mertens and the exponential

The rational product $P_x = \prod_{p\le x}(1 - 1/p)$ satisfies $P_x = e^{-\gamma}(\log x)^{-1}(1 + O(1/\log x))$ elementarily, and $1 + O(\exp(-c\sqrt{\log x}))$ by the prime number theorem; in the computed range the error $|\log x \cdot P_x - e^{-\gamma}|$ decays like $x^{-0.55}$ (Table 4) while the height of P_x grows like $e^{0.6x}$–$e^{0.83x}$. Under A this is a rational approximation of the number $e^{-c}/\log x$ with exponent tending to zero; there is nothing to test. Under A the exponential bridge gives, by the Hermite–Lindemann theorem,

Table 4: Forced approximations under A (script `exp10`, 1500 digits). (EM): $\log n + c \approx \rho_{n,K}$ at the optimal K; (Kummer): $\log n + c \approx S_{n,K}$ at the declared truncation K^*, where the forced error equals $E_1(n)$ up to 10%; in parentheses, the largest exponent over $1 \le K \le K^* + n$ (most favourable truncation); (Mertens): $e^{-c}/\log x \approx \prod_{p \le x}(1 - 1/p)$, the height being that of the product.

route	n (or x)	K	$\log_{10} \mathrm{H}$	$\log_{10} \lvert\varepsilon\rvert$	κ
EM	5	15	33.0	−14.0	0.4237
EM	10	31	87.4	−27.8	0.3179
EM	20	63	215.1	−55.2	0.2568
EM	40	125	506.5	−110.0	0.2171
EM	80	251	1174.6	−219.3	0.1867
EM	160	502	2642.2	−437.7	0.1657
Kummer	5	16	16.2	−2.98	0.1834 (0.3580)
Kummer	10	34	35.9	−5.40	0.1505 (0.1736)
Kummer	20	70	93.2	−10.02	0.1076 (0.1142)
Kummer	40	141	231.9	−18.96	0.0817 (0.0829)
Kummer	80	284	559.6	−36.70	0.0656 (0.0676)
Mertens	10^2	–	27.4	−2.13	0.078
Mertens	10^3	–	321.4	−2.66	0.008
Mertens	10^4	–	3503.0	−3.16	0.0009
Mertens	10^5	–	36171.3	−3.77	0.0001

that $e^\gamma = e^c$ is transcendental. This statement is expected to be true in C; its rejection, $e^\gamma \in \overline{\mathbb{Q}}$, would itself prove B (again by Hermite–Lindemann, since $\gamma \neq 0$). So bridge (D10) produces an A-construction that can only be rejected by proving B in another way: inert.

16.3 Bridges (D4), (D7), (D8), (D9): no tester

(D4) says under A that the constant term of the Laurent expansion of ζ at 1 is algebraic; no theorem constrains the arithmetic of Stieltjes constants. (D7) says that a Mahler-type integral has an algebraic value; Mahler's method applies to values of Mahler functions at algebraic points, not to their integrals, and the expectation $\gamma \notin \mathbf{E} \cup \mathbf{G} \cup \mathbf{M}$ recorded in [16] is exactly the statement that this bridge is not covered by a theorem. (D8) gives, for $d = 1$, equivalent reformulations of rationality [27]; they are tests, not testers. (D9) gives finite certificates only: under A with $d \le 30$ the minimal polynomial of γ has height above the bounds of Part I, Table 3 (above 10^{98} for $d = 2$), and under $d = 2$ the continued fraction is eventually periodic with a period that the 2910 computed partial quotients do not display; neither is a rejection.

16.4 Bridges (D5), (D6): the two non-Diophantine constructions

These are the two bridges through which A produces statements that are *expected to be false*. (D6): A gives $\delta = e(\mathrm{Ein}(1) - c) \in \mathbf{E}$, and (D6) gives $\delta \in \mathbf{D}$; so $\delta \in (\mathbf{E} \cap \mathbf{D}) \setminus \overline{\mathbb{Q}}$, a collision (Part II, Proposition 10.3. The rejecting statement is the intersection conjecture of Fischler–Rivoal [10] at the single element δ, i.e. B_δ: $\delta \in \mathbf{E} \Rightarrow \delta \in \overline{\mathbb{Q}}$ (Part II, Definition 10.2 and Proposition 10.3. (D5): A makes the period matrix of the Euler–Mascheroni motive $\overline{\mathbb{Q}}$-split, whereas Fresán–Jossen prove that the motive itself is non-split [12]; the rejecting statement is the fullness of the period realisation for this one motive (a case of the exponential period conjecture), equivalently the impossibility of removing the Euler entry by an algebraic change of Stokes framing [3]. In both derivations the bridge is what ties Δ to $\mathbf{D}$, resp. Γ to the motive; nothing in $\mathcal{K}$ does. Note that the point statement B_δ itself is never shear-invariant: in $\mathfrak{s}_c$ with c algebraic, $\Delta \in \mathbf{E}$ and $\Delta \notin \overline{\mathbb{Q}}$ (Theorem 18.7(i)), so $\mathrm{B}_\delta(\Gamma)$ fails there. A shear-invariant rejection of L2 would have to be the

universal conjecture $\mathbf{E}\cap\mathbf{D}=\overline{\mathbb{Q}}$, a sentence of type (K1), which then rejects $\Delta\in(\mathbf{E}\cap\mathbf{D})\setminus\overline{\mathbb{Q}}$ in every model. Either way the two rejecting statements are precisely the two conjectures already on the table, and neither has a proof. Everything else in the A-world is expected to be true.

17 The B-world

Proposition 17.1 (One-sidedness). *(i) Every* B*-construction obtainable from the tools at hand is a universal negative statement ("no algebraic relation of type T holds for γ", "no algebraic change of framing removes the Euler entry", "the continued fraction is not eventually periodic"), and its rejection is a certificate of a specific relation, hence a proof of* A *or of a stronger relation.*

(ii) The certified lattice searches (Part I, Table 3; Part II, Section 14 exclude every such relation below explicit heights (10^{98} in degree 2, 10^{41} against the 34-element E-basis, $10^{11.5}$ against 78 quadratic monomials). Hence no B*-construction can currently be rejected, and the programme can terminate only on the* A*-side.*

(iii) The B*-world is itself layered by shear-invisible sub-dichotomies (Corollary 15.3): $\gamma\in\mathbf{E}$ or not (equivalently $\delta\in\mathbf{E}$ or not), $\delta\in\overline{\mathbb{Q}}$ or not, $\gamma\in\overline{\mathbb{Q}}+\overline{\mathbb{Q}}e^{-1}$ or not (Section 18). Rivoal's theorem [15, Thm. 2] excludes only the corner γ,δ both algebraic.*

Proof. (i) is an inventory: B is $\forall P\in\overline{\mathbb{Q}}[x]\setminus 0:\ P(\gamma)\neq 0$, and every consequence derived without a bridge is an instance of it; the framing statement is [3, Thm. "Euler Stokes-framing barrier"] read in the B-world; the continued-fraction statement is Lagrange's theorem. (ii) is the record of the certificates. (iii) is Theorem 15.2 with $\mathcal{P}=\mathbb{R}\cap\mathbf{E}$, $\overline{\mathbb{Q}}$ (at δ), $\overline{\mathbb{Q}}+\overline{\mathbb{Q}}e^{-1}$. □

The asymmetry is structural: A is existential (it asserts a polynomial) and its consequences are rich; B is universal and its consequences are its instances. The B-world's usefulness is indirect: it is where the *expected* truth values of the A-entries are read off (Section 19).

18 Linear forms: the tie theorem and the linear hierarchy

18.1 The tie theorem

The Diophantine A-constructions of Part I (the floor Theorem 5.2 and the criterion Corollary 5.3 would be rejected by integer forms $a_n+b_n\gamma+c_n\delta$ of large exponent. We show that shear-invariance forces the coefficients of γ and δ to be tied, and that this caps the exponent.

Theorem 18.1 (Tie theorem). *Let a,b,c be T_0-definable complex numbers (or constants of $\mathbb{F}$), $\eta>0$, and suppose that the sentence $|a+b\Gamma+c\Delta|<\eta$ is provable in $\mathcal{K}(\mathbb{F};\mathcal{C})$, where $\mathcal{C}$ contains at least one enclosure, so that $I(\mathcal{C})$ is bounded, of length ℓ and with endpoints of absolute value $\le M$. Then*

$$|b-ec|\le\frac{2\eta}{\ell},\qquad |a+c\,e\,\mathrm{Ein}(1)|\le\eta\Big(1+\frac{2M}{\ell}\Big). \tag{18.1}$$

Proof. In the model $\mathfrak{s}_{c'}$, $a+b\Gamma+c\Delta$ is interpreted as $a+bc'+ce(\mathrm{Ein}(1)-c')=(a+c\,e\,\mathrm{Ein}(1))+(b-ec)\,c'$, an affine function $f(c')=\alpha+\beta c'$ of c', and the sentence holds in $\mathfrak{s}_{c'}$ for every $c'\in I(\mathcal{C})$. An affine function bounded by η in modulus on an interval of length ℓ has $|\beta|\ell\le 2\eta$, and then $|\alpha|\le\eta+|\beta|M$. □

(If $\mathcal{C}$ is empty, $I=\mathbb{R}$ and the conclusion reads $b=ec$, $|a+ce\,\mathrm{Ein}(1)|<\eta$.) The theorem is elementary; its content is that it is the exact quantitative shadow of Part I's two-model theorem on linear forms. The "shear derivative" of a form is $\partial_\Gamma-e\,\partial_\Delta$, and a shear-invariantly small form

must have small shear derivative, i.e. $b \approx ec$: the form sees γ and δ only through $e\gamma+\delta = e\,\mathrm{Ein}(1)$, which is an E-value.

Corollary 18.2 (Exponent ceiling for shear-invariant families)**.** *Let $(a_n, b_n, c_n) \in \mathbb{Z}^3$, $H_n = \max(|a_n|, |b_n|, |c_n|) \to \infty$, $\eta_n > 0$, and suppose the single sentence $\forall n\ |a_n + b_n\Gamma + c_n\Delta| < \eta_n$ is provable in some $\mathcal{K}(\mathbb{F}; \mathcal{C})$. Then for every $\varepsilon > 0$ there is $C_\varepsilon > 0$ with $\eta_n \geq C_\varepsilon H_n^{-1-\varepsilon}$ for all n with $(b_n, c_n) \neq (0, 0)$; hence $\limsup_n(-\log \eta_n/\log H_n) \leq 1$. Consequently:*

(i) no shear-invariant family meets the hypothesis of Nesterenko's criterion [13] for $\dim_{\mathbb{Q}} \mathrm{span}(1, \gamma, \delta) = 3$, which needs exponent > 1 (with its regularity conditions), nor the hypothesis of Part I, Corollary 5.3 (exponent $\to \infty$), nor a Roth-type exponent > 2;

(ii) a shear-invariant family in $1, \Gamma$ alone ($c_n = 0$) is trivial for $\eta_n < \min(1, \ell/2)$;

(iii) exponent $\frac{1}{2} - o(1)$ is always attainable shear-invariantly, by Dirichlet's simultaneous approximation of the pair $(e, e\,\mathrm{Ein}(1))$, and $\frac{1}{2}$ is the exponent of almost every pair (Khintchine); the ceiling 1 would require b_n/c_n to be a convergent-quality approximation of e and, simultaneously, $-a_n/c_n$ one of $e\,\mathrm{Ein}(1)$ with the same denominator.

Proof. By Theorem 18.1, $|b_n - ec_n| \leq 2\eta_n/\ell$. If $c_n = 0$ then $b_n \neq 0$ forces $\eta_n \geq \ell/2$, which is $\geq (\ell/2)H_n^{-1}$; if $b_n = 0$ too, the form is a_n and $|a_n| < \eta_n$ gives $\eta_n > H_n$. If $c_n \neq 0$, the irrationality exponent of e equals 2 (from its continued fraction; [24]), so $|b_n - ec_n| \geq C'_\varepsilon |c_n|^{-1-\varepsilon} \geq C'_\varepsilon H_n^{-1-\varepsilon}$ and $\eta_n \geq (\ell/2)C'_\varepsilon H_n^{-1-\varepsilon}$. (i) follows from Nesterenko's inequality $\dim \geq (\tau_1 + 1)/(1 + \tau_1 - \tau_2)$ with $\tau_1 = \tau_2 = \tau$, which exceeds 2 only for $\tau > 1$. (ii) was shown. (iii): Dirichlet gives, for every Q, integers $1 \leq c \leq Q$, a, b with $|ce - b| \leq Q^{-1/2}$ and $|c\,e\,\mathrm{Ein}(1) + a| \leq Q^{-1/2}$; then $|a + bc' + c(\delta + e(\gamma - c'))| \leq Q^{-1/2}(1 + |c'|)$ for all c', a shear-invariant bound with $H \leq (1+e\,\mathrm{Ein}(1))Q$; the metric statement is Khintchine's theorem on simultaneous approximation [26]; the last clause restates (18.1). □

Remark 18.3 (Named forms, provable bounds). Two precisions. (a) The tie theorem concerns *named* forms: the triple (a, b, c) is fixed (T_0-definable or in $\mathbb{F}$) and does not depend on Γ. The existential Dirichlet statement of Part I, Proposition 5.4(i) ("for every H there is a triple with $|a + b\Gamma + c\Delta| < H^{-2}$") *is* provable in $\mathcal{K}$, since the box principle applies to any reals; but the triple it produces depends on Γ, and no fixed sequence of triples is exhibited. (b) The ceiling bounds the *provable* η_n, not the actual size $|L_n|$, which may be smaller by cancellation at the true value of γ: Rivoal's forms have $|L_{39}| = 10^{-53.3}$ below their tie defect $10^{-47.7}$ (Table 5). Such a cancellation is itself a statement about γ and not shear-invariant (Specification 18.6).

Proposition 18.4 (Rivoal's forms are tied; LLL-reduced shear-invariant forms)**.** (Finite certificate; script `exp11`.) *Let $L_n = a_n + b_n\gamma + c_n\delta$ be the integer forms of Part I, Proposition 5.4(ii), obtained by eliminating e^{-1} between $R_{n,0}(-1)$ and $R_{n+1,0}(-1)$. Then, up to the denominator-clearing factor,*

$$b_n - ec_n = e(B_{n+1}(-1)R_{n,0}(-1) - B_n(-1)R_{n+1,0}(-1)), \tag{18.2}$$

so the tie defect obeys the same bound $C^n/n!$ as $|L_n|$, and the family is shear-invariantly small (its proof uses $|R_{n,0}(-1)|$, an enclosure of Δ, and the connection formula only). Numerically (Table 5): at $n = 39$, $|L_n| = 10^{-53.3}$, $|b_n - ec_n| = 10^{-47.7}$, $H_n = 10^{207.6}$, $b_n/c_n - e = 1.39 \cdot 10^{-255}$; the tie exponent $-\log|b_n - ec_n|/\log H_n$ rises from 0.06 to 0.23 as τ_n rises from 0.16 to 0.26. LLL-reduced shear-invariant forms with $|c| \leq 10^k$, $k = 10, \ldots, 100$, chosen to minimise the provable bound $\eta = |a + ce\,\mathrm{Ein}(1)| + 0.6\,|b - ec|$ (enclosure $0.5 < \Gamma < 0.6$), have $-\log\eta/\log H$ between 0.499 and 0.537, and actual exponent $-\log|L|/\log H$ between 0.499 and 0.537 as well.

Proof. With Rivoal's notation $R_n := R_{n,0}(-1) = e^{-1}A_n + B_nE(-1) + C_n$ (polynomials at -1, $E(-1) = -\mathrm{Ein}(1)$), L_n has $b_n = A_nB_{n+1} - A_{n+1}B_n$ and $c_n = B_nC_{n+1} - B_{n+1}C_n$ (Part I, proof of Proposition 5.4, so $b_n - ec_n = B_{n+1}(A_n + eC_n) - B_n(A_{n+1} + eC_{n+1})$, and $A_n + eC_n =$

Table 5: Tie defect of Rivoal's forms (script `exp11`, 1000 digits) and LLL-reduced shear-invariant forms. $\tau_n = -\log|L_n|/\log H_n$; tie exponent $= -\log|b_n - ec_n|/\log H_n$; η is the provable bound of Proposition 18.4.

n	$\log_{10}\lvert L_n\rvert$	$\log_{10}\lvert b_n - ec_n\rvert$	$\log_{10} H_n$	τ_n	tie exp.
5	−3.01	−1.12	18.89	0.159	0.059
10	−7.55	−4.85	43.46	0.174	0.112
20	−21.12	−17.19	96.68	0.218	0.178
30	−33.51	−28.62	157.66	0.213	0.182
39	−53.34	−47.71	207.58	0.257	0.230
$\log_{10} Q$	$\log_{10}\lvert L\rvert$	$\log_{10}\lvert b - ec\rvert$	$\log_{10} H$	$-\log\lvert L\rvert/\log H$	$-\log\eta/\log H$
20	−10.29	−10.06	20.15	0.511	0.510
40	−20.68	−20.42	40.26	0.514	0.512
60	−30.34	−30.31	59.05	0.514	0.514
80	−40.10	−40.04	80.30	0.499	0.499
100	−51.21	−50.73	100.06	0.512	0.508

$e(R_n - B_nE(-1))$ gives (18.2) after the $B_nB_{n+1}E(-1)$ terms cancel. Rivoal's Lemma 4 bounds $|B_n| \le C^n n!$ and $|R_n| \le C^n/(n!)^2$ [15]. The numbers are the computation; the lattice forms are the best of the three LLL-reduced basis vectors of the lattice generated by $(1, Ne, Ne\,\mathrm{Ein}(1))$, $(0, -N, 0)$, $(0, 0, N)$ with $N = Q^{3/2}$ (not a proven optimum), and the reported sizes are recomputed from the recovered integers. □

Proposition 18.5 (D-side forms are untied but large)**.** (Finite certificate; script `exp10`.) *The convergents s_k/q_k of Stieltjes' continued fraction $\delta = 1/(1+1/(1+1/(1+2/(1+2/(1+3/(1+3/(1+\cdots)))))))$, i.e. the Padé approximants at infinity of Euler's series evaluated at 1, give the integer forms $q_k\delta - s_k$ in $1, \delta$ alone. They are maximally untied ($b - ec = -eq_k$), but $|q_k\delta - s_k|$ grows: $10^{-0.72}$ at $k = 2$, $10^{0.88}$ at $k = 12$, $10^{15.8}$ at $k = 42$, $10^{44.0}$ at $k = 82$ (where $q_{82} = 10^{54.0}$, so $|\delta - s_k/q_k| = 10^{-10}$). The exponent $-\log|\delta - s_k/q_k|/\log q_k$ of these rational approximations decreases from 0.79 ($k = 12$) through 0.30 ($k = 42$) to 0.19 ($k = 82$), against $\mu(\delta) = 2$ expected, and proved in the case $\gamma \in \mathbb{Q}$ (Part I, Proposition 5.5.*

This is the denominator explosion of [2] in its simplest instance: the D-side approximants converge subexponentially while their denominators grow factorially.

Specification 18.6 (What a rejecting Diophantine construction must look like)**.** Write an integer form as $L_n = a_n + b_n\gamma + c_n\delta = \alpha_n + \beta_n\gamma$ with the *engine coefficients* $\alpha_n = a_n + c_n e\,\mathrm{Ein}(1)$, $\beta_n = b_n - ec_n \in \mathbb{Z} + \mathbb{Z}e + \mathbb{Z}e\,\mathrm{Ein}(1)$. A sequence of such forms rejects an A-entry only if (i) $|L_n| \le H_n^{-1-\varepsilon}$ with Nesterenko regularity (giving $\dim_{\mathbb{Q}} \mathrm{span}(1, \gamma, \delta) = 3$, hence $\gamma \notin \mathbb{Q}$), or (i′) $-\log|L_n|/\log H_n \to \infty$ (giving $\gamma \notin \overline{\mathbb{Q}}$ by Part I, Corollary 5.3. In either case $|\beta_n| \ge C_\varepsilon H_n^{-1-\varepsilon/2}$ by $\mu(e) = 2$ (proof of Corollary 18.2), so $|L_n|/|\beta_n| \to 0$: the forms are *smaller than their own tie defect*, and the engine ratios $-\alpha_n/\beta_n \in \mathbb{Q}(e, e\,\mathrm{Ein}(1))$ converge to γ, with $|\gamma + \alpha_n/\beta_n| = |L_n|/|\beta_n|$. By Corollary 18.2 the bound (i) or (i′) cannot be proved shear-invariantly; in particular its proof must use a bridge, i.e. some information about γ beyond the connection formula. The E-side produces forms whose size is comparable to their tie defect (Proposition 18.4: $10^{-53.3}$ against $10^{-47.7}$ at $n = 39$), the D-side produces untied forms that are large (Proposition 18.5); the specification is a sequence of engine ratios converging to γ with $|L_n| \le H_n^{-1-\varepsilon}$, and no example is known.

18.2 The linear hierarchy

For a $\overline{\mathbb{Q}}$-subspace $V \subset \mathbb{C}$ with $e \in V$ define the statement

$$\mathrm{X}(V): \qquad \delta \notin e\,\mathrm{Ein}(1) + V.$$

Since $\delta = e\,\mathrm{Ein}(1) - e\gamma$, $\mathrm{X}(V)$ says $e\gamma \notin V$, i.e. $\gamma \notin e^{-1}V$. Put $W = \mathrm{span}_{\overline{\mathbb{Q}}}(1, e, e\,\mathrm{Ein}(1))$.

Theorem 18.7 (Linear hierarchy)**.** *(i)* (Published input.) $1, e^{-1}, \mathrm{Ein}(1)$ *are linearly independent over* $\overline{\mathbb{Q}}$*; equivalently* $1, e, e\,\mathrm{Ein}(1)$ *are.*

(ii) $\mathrm{X}(\overline{\mathbb{Q}}e) \iff \mathrm{B}$.

(iii) $\mathrm{X}(\overline{\mathbb{Q}} + \overline{\mathbb{Q}}e) \iff 1, e, e\gamma$ *are* $\overline{\mathbb{Q}}$*-linearly independent* $\iff \gamma \notin \overline{\mathbb{Q}} + \overline{\mathbb{Q}}e^{-1}$.

(iv) $\mathrm{X}(W) \iff 1, \gamma, e^{-1}, \mathrm{Ein}(1)$ *are* $\overline{\mathbb{Q}}$*-linearly independent* $\iff \gamma \notin \overline{\mathbb{Q}} + \overline{\mathbb{Q}}e^{-1} + \overline{\mathbb{Q}}\,\mathrm{Ein}(1)$.

(v) $\mathrm{X}(\mathbf{E}) \iff \delta \notin \mathbf{E} \iff \gamma \notin \mathbf{E}$.

(vi) $\mathrm{X}(\mathbf{E}) \Rightarrow \mathrm{X}(W) \Rightarrow \mathrm{X}(\overline{\mathbb{Q}} + \overline{\mathbb{Q}}e) \Rightarrow \mathrm{B}$*; and under* B_δ *of Part II (*$\delta \in \mathbf{E} \Rightarrow \delta \in \overline{\mathbb{Q}}$*) one has* $\mathrm{X}(W) \iff \delta \notin \overline{\mathbb{Q}}$.

(vii) Each $\mathrm{X}(V)$ *fails in the* A*-world, is expected to hold in* C*, and is shear-invisible (Theorem 15.2 with* $\mathcal{P} = e^{-1}V$*).*

Proof. (i) The functions $1, e^{-z}, \mathrm{Ein}(z)$ are E-functions with rational coefficients, solutions of $y' = Ay$ with A regular outside $z = 0$ (since $\mathrm{Ein}'(z) = (1 - e^{-z})/z$), and they are linearly independent over $\mathbb{C}(z)$: a relation $p + qe^{-z} + r\,\mathrm{Ein}(z) = 0$ with polynomial coefficients, restricted to $z \to +\infty$ where $\mathrm{Ein}(z) = \log z + \gamma + o(1)$, forces $r = 0$ and then $p = q = 0$. Beukers' refinement of the Siegel–Shidlovskii theorem [5, Thm. 1.3, Cor. 1.4] lifts any $\overline{\mathbb{Q}}$-linear relation among the values at the non-singular point 1 to one among the functions; hence there is none. (The same three functions underlie Theorem 2 of [15].) Multiplying by e gives the second form. (ii) $e^{-1}(\overline{\mathbb{Q}}e) = \overline{\mathbb{Q}}$. (iii) $e^{-1}(\overline{\mathbb{Q}} + \overline{\mathbb{Q}}e) = \overline{\mathbb{Q}}e^{-1} + \overline{\mathbb{Q}}$; and a relation $\alpha + \beta e + \xi e\gamma = 0$ with $\xi = 0$ would make e algebraic, so $1, e, e\gamma$ are dependent iff $e\gamma \in \overline{\mathbb{Q}} + \overline{\mathbb{Q}}e$. (iv) $e^{-1}W = \mathrm{span}(e^{-1}, 1, \mathrm{Ein}(1))$; a relation $\alpha + \beta\gamma + \xi e^{-1} + \zeta\,\mathrm{Ein}(1) = 0$ with $\beta = 0$ contradicts (i), so dependence means $\gamma \in \mathrm{span}(1, e^{-1}, \mathrm{Ein}(1))$. (v) $e^{-1}\mathbf{E} = \mathbf{E}$ since $\mathbf{E}$ is a ring containing $e^{\pm 1}$; and $\gamma \in \mathbf{E} \iff \delta \in \mathbf{E}$ (Part II, Proposition 10.3. (vi) The implications are the inclusions $\overline{\mathbb{Q}}e \subset \overline{\mathbb{Q}} + \overline{\mathbb{Q}}e \subset W \subset \mathbf{E}$. Assume B_δ. If $\mathrm{X}(W)$ fails, $\gamma = \alpha + \beta e^{-1} + \xi\,\mathrm{Ein}(1)$ with $\alpha, \beta, \xi \in \overline{\mathbb{Q}}$, so $\delta = e(\mathrm{Ein}(1) - \gamma) = e(1 - \xi)\,\mathrm{Ein}(1) - e\alpha - \beta \in \mathbf{E}$, hence $\delta \in \overline{\mathbb{Q}}$ by B_δ; then $e(1 - \xi)\,\mathrm{Ein}(1) - e\alpha \in \overline{\mathbb{Q}}$ and (i) forces $\xi = 1$, $\alpha = 0$, $\delta = -\beta$. Conversely $\delta \in \overline{\mathbb{Q}}$ gives $\gamma = \mathrm{Ein}(1) - \delta e^{-1} \in e^{-1}W$. So under B_δ, $\neg\mathrm{X}(W) \iff \delta \in \overline{\mathbb{Q}}$. (vii) Under A, $\gamma \in \overline{\mathbb{Q}} = e^{-1}(\overline{\mathbb{Q}}e) \subset e^{-1}V$. Expectation: the B-world reading of Rivoal's expectation $\gamma \notin \mathbf{E}$ [16] is $\mathrm{X}(\mathbf{E})$, the strongest. Shear-invisibility: $e^{-1}V$ is a countable $\overline{\mathbb{Q}}$-space containing $\overline{\mathbb{Q}}$, hence dense with dense complement. □

The hierarchy grades the distance between what is known and what is wanted. The weakest statement strictly above B is (iii): $1, e, e\gamma$ *are linearly independent over* $\overline{\mathbb{Q}}$. It is the first place where γ would be separated from e; no tool reaches it, for the same reason as for B. Statement (iv) is where γ would be separated from the E-function Ein at the single point 1; under the point conjecture B_δ it is the same as $\delta \notin \overline{\mathbb{Q}}$, the conclusion of Fischler–Rivoal's Corollary 1 in [10] (there derived from their Conjecture 2). Statement (v) is the point conjecture's E-side.

19 The ledger

Table 6 collects the constructions. "Bridge" is the definition used in the derivation (none: the derivation is shear-invariant, using only the connection formula and enclosures). "Expected" is

the truth value in C predicted by the B-world together with the standard conjectures (intersection conjecture, exponential period conjecture, Rivoal's expectation); it is a prediction, not a theorem. "Rejecting statement" is what C would have to prove to refute the entry.

Table 6: The ledger. A-entries L1–L12 (consequences of A), B-entries M1–M3. "Bridge" is the bridge used in the derivation.

	entry	bridge	expected	rejecting statement (tool)	status
L1	$\delta \in \mathbf{E}$ ($\Leftrightarrow \gamma \in \mathbf{E} \Leftrightarrow E_1(1) \in \mathbf{E}$)	none	false	$\delta \notin \mathbf{E}$; Diophantine route only (E-values are not Liouville [9]), but $\mu(\delta) = 2$ expected	open target
L2	$\delta \in (\mathbf{E} \cap \mathbf{D}) \setminus \overline{\mathbb{Q}}$ (collision)	D6	false	$\mathbf{E} \cap \mathbf{D} = \overline{\mathbb{Q}}$ at δ, i.e. B_δ (Part II, Definition 10.2	open target
L3	period matrix of $M(\gamma)$ is $\overline{\mathbb{Q}}$-split; Euler entry removable by algebraic framing	D5	false	fullness of the period realisation for $M(\gamma)$ [12, 3]	open target (conditional)
L4	floor $\lvert a + b\gamma + c\delta \rvert \geq c_\varepsilon H^{-(d3^d-1)-\varepsilon}$ (Part I, Theorem 5.2	none	true	forms of exponent $\to \infty$: impossible shear-invariantly (Cor. 18.2); needs untied small forms (Spec. 18.6)	inert
L5	$\mu(\delta) = \mu(E_1(1)) = 2$ if $\gamma \in \mathbb{Q}$ (Part I, Proposition 5.5	none	true	–	inert
L6	$\log n$ has degree-d algebraic approximations of forced exponent κ_n/d, $\kappa_n \leq 0.42$, $\kappa_n \approx 1/\log n$	D1, D2	true (trivially)	a measure below κ_n/d: none exists, Dirichlet already gives 2	inert (Table 4)
L7	$e^{-\gamma}$ is the Mertens limit; no exponent	D3	true	–	inert
L8	$e^\gamma \notin \overline{\mathbb{Q}}$ (Hermite–Lindemann; D10 only names the object)	none	true	$e^\gamma \in \overline{\mathbb{Q}}$, which proves B directly	inert
L9	$1, e, e\gamma$ are $\overline{\mathbb{Q}}$-dependent	none	false	X($\overline{\mathbb{Q}} + \overline{\mathbb{Q}}e$): $\gamma \notin \overline{\mathbb{Q}} + \overline{\mathbb{Q}}e^{-1}$; no tool	open target (weakest)
L10	$1, \gamma, e^{-1}, \mathrm{Ein}(1)$ are $\overline{\mathbb{Q}}$-dependent	none	false	X(W); under B_δ this is $\delta \notin \overline{\mathbb{Q}}$	open target
L11	minimal polynomial of γ has height $> 10^{h(d)}$, $d \leq 30$ (10^{98} for $d = 2$); for $d = 2$ the continued fraction is eventually periodic	none (rejection needs D9)	true	exhibiting the polynomial or the period (proves A)	finite certificate
L11′	the Laurent constant γ_0 and the Vacca value are algebraic	D4, D7	no expectation / false ([16])	no theory for Stieltjes constants or for Mahler-type integrals	inert
L12	the second-kind constants $E_1(\lambda)$, δ, $Y_0(\lambda)$, $K_0(\lambda)$, $\Gamma^{(k)}(1)$ all lie in the engine field $\mathbb{F}_0$	none	false	separating one second-kind constant from $\mathbb{F}_0$; no tool and no standard conjecture beyond L1	inert
M1	no algebraic relation for γ; continued fraction not eventually periodic	none	true	exhibiting a relation (proves A); excluded below 10^{98} in degree 2	certificates only
M2	no algebraic change of Stokes framing removes the Euler entry [3]	D5	true	exhibiting a framing (proves A)	–
M3	the sub-dichotomies $\gamma \in \mathbf{E}$?, $\delta \in \overline{\mathbb{Q}}$?, $\gamma \in \overline{\mathbb{Q}} + \overline{\mathbb{Q}}e^{-1}$? are open and shear-invisible	none	–	–	Cor. 15.3

Proposition 19.1 (Pattern of the ledger)**.** *(i) Every* A*-entry expected to be false (L1, L2, L3, L9, L10, L11′, L12) is a membership, dependence or splitting statement. For L1, L2, L3,*

L9, L10 the rejecting statement is a known conjecture, or a consequence of one, evaluated at a single element: $\gamma \notin \mathbf{E}$ (Rivoal), $\mathbf{E} \cap \mathbf{D} = \overline{\mathbb{Q}}$ at δ (Fischler–Rivoal), fullness for $M(\gamma)$ (period conjecture), and the graded linear statements $\mathrm{X}(V)$; for L11′ and L12 not even a conjectural framework is available.

(ii) Every A*-entry of Diophantine type (L4–L8, L11) is expected to be true, or is a finite certificate; none can be rejected, and the only one with a conceivable rejection (L4) requires forms that Corollary 18.2 excludes for every shear-invariant construction (Specification 18.6).*

(iii) The two entries derived through a bridge and expected to be false (L2, L3) are two of the three conditional routes of Part I, Remark 4.3 (the third, Rivoal's expectation, is the rejection of L1); the entries derived without a bridge and expected to be false (L1, L9, L10, L12) can be rejected only through a bridge (Corollary 15.4), and no bridge-based rejection is available.

Proof. Inspection of the table, with Theorem 18.7 for L9–L10 and Corollary 18.2 for L4. □

20 Outcome

20.1 The three barriers, with numbers

The programme "build in A, reject in C" has been run through the ten bridges and every tool at hand. It stops at three barriers, each now with a quantitative face.

The shear barrier (Part I, Theorem 4.2; here Theorem 15.2 and Corollary 15.4). Every dichotomy about γ or δ that has been proposed—$\overline{\mathbb{Q}}$, $\mathbf{E}$, $\mathbf{D}$, periods, the linear spaces $e^{-1}V$—is dense with dense complement, hence invisible to the connection-formula theorems. On linear forms this becomes the tie (18.1): Rivoal's forms have $|b_n/c_n - e| = 1.4 \cdot 10^{-255}$ at height 10^{208}, and every shear-invariant family has exponent ≤ 1, generically $\frac{1}{2}$ (computed 0.50–0.54).

The denominator barrier ([2]; here Propositions 16.1 and 18.5). Along the definitions, the height outruns the error: forced exponents $0.42 \to 0.17$ (Euler–Maclaurin), $0.18 \to 0.07$ (Kummer), ≈ 0 (Mertens), and on the D-side $|q_k\delta - s_k| \to \infty$ (10^{44} at $k = 82$). The untied forms are large, the small forms are tied.

The realisation barrier ([12, 3]; here L3). The one construction that produces a categorical object—the split period matrix—is rejected by the motive's non-splitting only if the period realisation is full for $M(\gamma)$, which is the exponential period conjecture at this motive.

20.2 Verdict

No contradiction was found in either world, and none was expected: by Lemma 15.1 a contradiction is a proof of B. What the programme produced is a classification. The A-world is large and, on its Diophantine side, entirely compatible with C: everything A forces about approximations, measures and heights is either true anyway or below every threshold. Its non-Diophantine side consists of membership statements, each of which is expected to be false, and each of whose rejections is a standing conjecture at one element. The B-world is small and cannot be rejected by anything short of a relation. The loop does not close because C currently has no theorem that reaches γ or δ through a bridge and separates them from an engine class; the shear-filter rule says that any closing theorem must be of this kind.

The sharpened targets, in increasing strength, are the three that survive the ledger:

(T1) *Forms below their tie defect*: integer forms in $1, \gamma, \delta$ with $|L_n| \leq H_n^{-1-\varepsilon}$, equivalently engine ratios $-(a_n + c_n e\,\mathrm{Ein}(1))/(b_n - ec_n)$ converging to γ, with a proof that uses a

bridge (Specification 18.6); the first Diophantine construction that would matter, and the quantitative meaning of "break the E–D tie".

(T2) *The point conjecture* B_δ: $\delta \in \mathbf{E} \Rightarrow \delta \in \overline{\mathbb{Q}}$ (Part II, Definition 10.2, which rejects L2 and, with $\delta \notin \overline{\mathbb{Q}}$, gives $\mathrm{X}(W)$ (Theorem 18.7(vi)).

(T3) *Fullness for* $M(\gamma)$, which rejects L3.

Below all three sits the weakest unproved statement above B, the linear independence of $1, e, e\gamma$ over $\overline{\mathbb{Q}}$ (L9), for which the ledger records no route at all.

21 Computations for Part III

Scripts (Python 3.11, `mpmath` 1.3, `sympy` 1.14, `fpylll` 0.6.4), archived with their outputs.

- `exp10_aworld_profiles.py`: forced approximations along (D1), (D2), (D3) (exact rational heights, errors at 1500 digits; for (D1) K optimised over $1 \le K \le 4n$, for (D2) the declared rule K^* and the most favourable K on $1 \le K \le K^* + n$); Stieltjes convergents of δ and the forms $q_k\delta - s_k$ (Table 4, Proposition 18.5).
- `exp11_tie_defect.py`: tie defect of Rivoal's forms from the exact coefficients of Part I (script `exp03`), with the identity $L_n = (a_n + c_n e\,\mathrm{Ein}(1)) + (b_n - ec_n)\gamma$ checked to 10^{-700}; LLL-reduced shear-invariant forms by `fplll` with exact recovery of (a, b, c) and the provable bound η (Table 5).

The only non-exact inputs are the values γ, e, $\log n$, $E_1(n)$, $\mathrm{Ein}(1)$ and $\delta = eE_1(1)$ supplied by `mpmath` at 1000–1500 digits; all heights and coefficients are exact integers. Run times: `exp10` about 5 minutes, `exp11` under one minute.

Part IV
Two worlds for their own sake: shadows, twists, certificates, and the map of possible worlds

Summary of Part IV. Parts I–III asked what would terminate the two-world search. This part asks nothing of the kind: it builds inside the world A ($\gamma \in \overline{\mathbb{Q}}$) and the world B ($\gamma \notin \overline{\mathbb{Q}}$) for their own sake and records what is found. In A: for every algebraic c near γ the *shadow system* $(c,\ \delta_c = e(\mathrm{Ein}(1) - c),\ E_1(\lambda)_c)$ satisfies every connection identity of the true system and consists of E-values ($\delta_c = \sum_n (H_n - c)/n!$) and of E-values minus logarithms; Rivoal's forms and every certified lattice search below 1050 digits cannot tell the shadow world of the degree-12 camouflage of Part I from the real world, and the lattice search that certifies "no relation of degree ≤ 12 and height $< 10^{p/13}$" at p digits returns as its first vector the minimal polynomial of a degree-12 number within 10^{-p} of γ: every certificate is a camouflage read backwards. The A-world has $d - 1$ *phantom conjugates* $\sigma(\gamma)$, a Galois norm $P_\gamma(\mathrm{Ein}(1)) = \prod_\sigma E_1(1)^{(\sigma)}$, and a Beukers theory proving, for every finite set Λ of algebraic points, that $\{1, e^\lambda, \mathrm{Ein}(\lambda) - \gamma\}_{\lambda \in \Lambda}$ is linearly independent—sentences that C does not at present prove. The shear of Part I is identified at the level of functions: it is the twist $\Gamma \mapsto \Gamma_\tau(s) = \Gamma(s)e^{\tau(1-s)}$, which preserves every connection identity, moves the Euler constant to $\gamma + \tau$, acts on the polyexponential tower as the rescaling $\lambda \mapsto \lambda e^\tau$, and breaks the normalisations of the functional, reflection and multiplication formulas by constant factors; on the period matrix of Fresán–Jossen it acts as a $\mathbb{C}$-linear (not $\overline{\mathbb{Q}}$-linear) change of de Rham basis that fixes the period $\mathrm{Ein}(1)$ of the cycle $\sigma_1 + \sigma_2$. In B: the A-world's theorems dualise

into *certificate loci*—explicit countable sets of transcendental numbers on which a proved identity would make γ visibly transcendental (exponential, logarithmic, π-, e-, Gamma-value loci and the Ein-hyperplanes)—and certified searches show that γ lies on none of them below heights 10^{28} to 10^{74}; the expected truth is the *wild* world, where no integer-relation search can ever succeed. A map of the possible worlds records which theorem or conjecture prunes each region. A curiosity closes the note: the engine ratios $-(a_n + c_n e\,\mathrm{Ein}(1))/(b_n - ec_n)$ of Rivoal's forms approach γ numerically, 10^8 times more slowly than Stieltjes's convergents of comparable height.

22 Building without termination

Part III, Lemma 15.1, records that a contradiction inside a world is the same event as a proof of the other world, so a search for contradictions is a search for a proof under another name. This part drops the search. The instruction was: go where the interest is, keep the worlds separate, and let them grow. What follows is what grew. Notation and the engine theory $\mathcal{K}(\mathbb{F};\mathcal{C})$, the sheared models $\mathfrak{s}_c$, the bridges (D1)–(D10) and the rings $\mathbf{E},\mathbf{G},\mathbf{D},\mathbf{S}$ are those of Parts I–III.

23 The A-world

23.1 Shadow systems

Definition 23.1 (Shadow system)**.** For $c \in \mathbb{C}$ the *shadow system* at c is the family

$$\gamma_c := c, \qquad E_1(\lambda)_c := \mathrm{Ein}(\lambda) - c - \log\lambda \ (\lambda \in \overline{\mathbb{Q}} \setminus (-\infty, 0]), \qquad \delta_c := eE_1(1)_c = e(\mathrm{Ein}(1) - c),$$

(principal logarithm), together with $\omega_{\lambda,r}(c) := e^{\lambda}\,\mathrm{Ein}_r(\lambda) - e^{\lambda}P_r(\lambda;c)$, where $P_r(\lambda;c) = [t^r](e^{ct}\lambda^t \exp\sum_{k\geq 2}\zeta(k)t^k/k)$ is the tower polynomial of Part II with γ replaced by c. At $c=\gamma$ the shadow system is the true system. The sheared model $\mathfrak{s}_c$ of Part I interprets $\Gamma, Z_\lambda, \Delta$ by the shadow system at c.

Proposition 23.2 (The shadow system is a system of E-values)**.** *Let $c \in \overline{\mathbb{Q}}$ and $K = \mathbb{Q}(c)$. Then*

$$F_c(z) := e^z\big(\mathrm{Ein}(z) - c\big) = \sum_{n\geq 0} \frac{H_n - c}{n!}\, z^n \qquad (H_0 = 0) \tag{23.1}$$

is an E-function with coefficients in K, $\delta_c = F_c(1)$, and for every $\lambda \in \overline{\mathbb{Q}} \setminus (-\infty, 0]$ the number $e^{\lambda}E_1(\lambda)_c = F_c(\lambda) - e^{\lambda}\log\lambda$ is an E-value minus $e^{\lambda}\log\lambda$. Every connection identity of the true system holds verbatim in the shadow system: $\mathrm{Ein}(\lambda) = \gamma_c + \log\lambda + E_1(\lambda)_c$, $\delta_c = eZ_1$, $\omega_{\lambda,r}(c) = e^{\lambda}\,\mathrm{Ein}_r(\lambda) - e^{\lambda}P_r(\lambda;c)$, $\delta_c = \sum_n (H_n - c)/n!$.

Proof. $F = e^z\,\mathrm{Ein}(z)$ satisfies $F' = F + (e^z - 1)/z$ (Part I, proof of Theorem 5.2, so its coefficients f_n in $\sum f_n z^n/n!$ obey $f_{n+1} = f_n + 1/(n+1)$, $f_0 = 0$, i.e. $f_n = H_n$; and $ce^z = \sum_n cz^n/n!$. The difference is (23.1), an E-function since E-functions with coefficients in K form a ring containing K. The identities are definitions or the true identities with γ renamed. □

So for every algebraic c there is an E-value δ_c satisfying all of δ's connection identities (not, of course, its Borel-sum or Stieltjes definitions). The intersection conjecture of Fischler–Rivoal at δ implies that δ is not one of them: $\delta \neq \delta_c$ for all $c \in \overline{\mathbb{Q}}$, i.e. $\gamma \notin \overline{\mathbb{Q}}$.

A day in the shadow world. Let c be the degree-12 camouflage of Part I (Proposition 3.2 there): $c - \gamma = -4.02 \cdot 10^{-31}$, $\mathrm{H}(P_c) = 10^{81.8}$. Then (script `exp12`): $\delta_c - \delta = 1.09 \cdot 10^{-30}$; the series $\sum_n (H_n - c)/n!$ truncated at 400 terms agrees with δ_c to 868 digits (the truncation error); Rivoal's forms of Part I evaluated on the shadow pair (c, δ_c) differ from their true values by

$|b_n - ec_n|\,|\gamma - c|$, which is $10^{-31.5}$, $10^{-47.6}$, $10^{-78.1}$ at $n = 5, 20, 39$—the tie of Part III makes the forgery *better* than the enclosure, by the factor $|b_n - ec_n|$; and the certified lattice searches for an algebraic relation of degree ≤ 12 return indistinguishable certificates for c and for γ at 300, 600, 900, 1000 and 1050 digits (heights $10^{20.6}, 10^{43.5}, 10^{66.8}, 10^{74.4}, 10^{78.3}$, equal to within 0.1 in the exponent). At 1080 digits the search on c returns P_c itself (residual 10^{-3260}), while the search on γ returns a Dirichlet near-relation (residual $10^{-997} \approx H^{-12}$) and the bound $10^{80.6}$; at 1200 digits, $10^{89.8}$. Below 1050 digits the shadow world is the real world for every instrument we own.

23.2 Certificates are camouflages read backwards

Proposition 23.3 (Camouflage–certificate duality)**.** (Finite certificate; script `exp12c`, with exact re-verification.) *Let b_1 be the first vector of the LLL-reduced lattice with rows $(e_i, \lfloor N\gamma^i \rceil)$, $i = 0, \dots, 12$, $N = 10^p/2$, and $Q_p(x) = \sum_{i\le 12} b_{1,i}x^i$. Then Q_p certifies that γ satisfies no integer relation of degree ≤ 12 and height below $\|b_1\|/(\alpha^6 \cdot 13\sqrt{14})$ (the normalisation of Part II, Section 14, and at the same time Q_p is irreducible of degree 12 (checked for the four values of p below) with a real root c_p such that*

$$\mathrm{H}(c_p) = \mathrm{H}(Q_p) \approx 10^{p/13}, \qquad |\gamma - c_p| \approx 10^{-p}:$$

$p = 100$: $\mathrm{H} = 10^{7.5}$, $|\gamma - c_p| = 10^{-99.0}$; $p = 300$: $10^{22.9}$, $10^{-299.5}$; $p = 600$: $10^{45.8}$, $10^{-600.3}$; $p = 1200$: $10^{92.1}$, $10^{-1200.5}$.

Proof. The certificate is that of Parts I–II. For the root: the last coordinate of b_1 is $NQ_p(\gamma)$ up to the rounding term, so $|Q_p(\gamma)| \le (\|b_1\| + 13\,\mathrm{H}(Q_p)/2)/N \approx 10^{-p+p/13}$, while $|Q_p'(\gamma)| \approx \mathrm{H}(Q_p)$ (computed: $10^{6.5}, 10^{22.1}, 10^{45.8}, 10^{92.2}$); Newton's method from γ converges to a root at distance $\approx |Q_p(\gamma)|/|Q_p'(\gamma)|$. The numbers are the computation, with the root located at $2p + 300$ digits and irreducibility checked with `sympy`. □

This is Part I's symmetry of finite observations (Theorem 3.1 there) made operational: the very vector that proves "no algebraic γ of small height" is the minimal polynomial of an algebraic number indistinguishable from γ to the precision used. The degree-12 camouflage of Part I is expensive ($\mathrm{H} = 10^{82}$ for 30 digits) only because it also carries full local root counts; the Dirichlet-optimal camouflage of 30 digits in degree 12 has height about $10^{2.3}$.

23.3 Galois theory of the A-world

Assume A with $d = [\mathbb{Q}(\gamma) : \mathbb{Q}]$ and let σ run over the embeddings of $K = \mathbb{Q}(\gamma)$.

Proposition 23.4 (Phantom conjugates and the Galois norm)**.** *(i) The shadow systems at the conjugates $\sigma(\gamma)$ are the Galois conjugates of the true system: $F_\gamma^\sigma(z) = F_{\sigma(\gamma)}(z)$, $\delta^{(\sigma)} := \delta_{\sigma(\gamma)} = \delta + e(\gamma - \sigma(\gamma))$, $E_1(1)^{(\sigma)} = E_1(1) + \gamma - \sigma(\gamma)$; all are E-values.*

(ii) (Galois norm.) *With P_γ the monic minimal polynomial of γ,*

$$P_\gamma(\mathrm{Ein}(1)) = \prod_\sigma (\mathrm{Ein}(1) - \sigma(\gamma)) = \prod_\sigma E_1(1)^{(\sigma)} = e^{-d}\prod_\sigma \delta^{(\sigma)}, \tag{23.2}$$

a nonzero element of $\mathbb{Q}[\mathrm{Ein}(1)]$; in particular $E_1(1)$ divides $P_\gamma(\mathrm{Ein}(1))$ in $\overline{\mathbb{Q}}[\mathrm{Ein}(1)]$, and the number $P_\gamma(\mathrm{Ein}(1))$ has irrationality exponent 2 unconditionally (preprint input [11], as in Part I, Proposition 5.5.

(iii) (Conjugate towers.) *$P_r(\lambda; \sigma(\gamma)) = P_r(\lambda e^{\sigma(\gamma)-\gamma}; \gamma)$, with the branch $\log(\lambda e^\tau) := \log\lambda + \tau$: the conjugate tower is the true tower at the transcendental points $\lambda e^{\sigma(\gamma)-\gamma}$.*

(iv) (Phantoms.) *For $\sigma \neq \mathrm{id}$ the conjugate $\sigma(\gamma)$ is the Euler constant of the twisted function $\Gamma_\tau(s) = \Gamma(s)e^{\tau(1-s)}$, $\tau = \sigma(\gamma) - \gamma$, which satisfies $\Gamma_\tau(s+1) = se^{-\tau}\Gamma_\tau(s)$ and takes the transcendental values $n!\,e^{-\tau n}$ at the integers $n+1 \geq 2$; by Bohr–Mollerup no log-convex f with $f(1) = 1$ and $f(s+1) = sf(s)$ has $-f'(1) = \sigma(\gamma)$.*

Proof. (i) Apply σ to the coefficients $(H_n - \gamma)/n!$ of (23.1). (ii) The product over the roots of P_γ; $P_\gamma(\mathrm{Ein}(1)) \neq 0$ because $\mathrm{Ein}(1)$ is transcendental; the exponent statement is [11, Thm. 1] applied to the E-function $P_\gamma(\mathrm{Ein}(z)) \in \mathbb{Q}[[z]]$ at $z = 1$. (iii) $e^{\sigma(\gamma)t}\lambda^t = e^{\gamma t}(\lambda e^{\sigma(\gamma)-\gamma})^t$ and Definition 23.1. (iv) Proposition 24.1 below with $\tau = \sigma(\gamma) - \gamma$; $e^{-\tau n}$ is transcendental for algebraic $\tau \neq 0$ (Hermite–Lindemann). □

The phantoms are the A-world's most peculiar feature. Algebraic special values of analytic functions usually come with their conjugates as special values of the same function at conjugate points ($\cos(2\pi/7)$ and $\cos(4\pi/7)$); here the conjugates $\sigma(\gamma)$ would be Euler constants of functions that fail the functional equation by an algebraic exponential twist, and would have no analytic definition independent of γ (they are $-\Gamma_\tau'(1)$ with $\tau = \sigma(\gamma) - \gamma$). The rational world $d = 1$ is the only A-world without phantoms, which may be why "$\gamma \in \mathbb{Q}$?" is the traditional form of the question; Part II recorded that it is also the form pushed farthest by computation ($q > 10^{244663}$).

23.4 Sentences of the A-world that C does not prove

Lemma 23.5 (Multipoint independence over $\mathbb{C}(z)$)**.** *For distinct nonzero $\lambda_1, \dots, \lambda_m \in \mathbb{C}$ and any finite set $M \subset \mathbb{C} \setminus \{0\}$, the functions 1, $e^{\mu z}$ $(\mu \in M)$, $\mathrm{Ein}(\lambda_i z)$ $(1 \leq i \leq m)$ are linearly independent over $\mathbb{C}(z)$.*

Proof. A relation among 1 and the $e^{\mu z}$ alone with polynomial coefficients is trivial (distinct exponentials are linearly independent over $\mathbb{C}(z)$). Suppose $\sum_i q_i(z)\,\mathrm{Ein}(\lambda_i z) + \sum_\mu p_\mu(z)e^{\mu z} + r(z) = 0$ with polynomial coefficients, not all q_i zero, and choose such a relation (over all finite exponent sets M) with $D = \max_i \deg q_i$ minimal. Since $\frac{d}{dz}\mathrm{Ein}(\lambda z) = (1 - e^{-\lambda z})/z$, the combination $D \cdot (\text{relation}) - z \cdot \frac{d}{dz}(\text{relation})$ is a relation of the same shape (exponent set enlarged by $-\lambda_i$) whose Ein-coefficients $Dq_i - zq_i'$ have degree $< D$; by minimality they all vanish, so $q_i = \kappa_i z^D$. Now z times the derivative of the relation minus D times the relation is a relation among 1 and the $e^{\mu z}$, $\mu \in M \cup (-\Lambda)$, in which $e^{-\lambda_i z}$ has coefficient $-\kappa_i z^D + (zp_\mu' + \mu z p_\mu - Dp_\mu)$ for $\mu = -\lambda_i$ if $-\lambda_i \in M$, and $-\kappa_i z^D$ otherwise. This coefficient vanishes. In the second case $\kappa_i = 0$. In the first, if $p_\mu \neq 0$ has degree k and leading coefficient π, the term $\mu z p_\mu$ has degree $k+1$ and leading coefficient $\mu\pi \neq 0$, which only $-\kappa_i z^D$ can cancel, so $D = k + 1$; the coefficient of z^0 gives $p_\mu(0) = 0$, and ascending through $z^1, \dots, z^k$ the recursion $(m - D)\pi_m = -\mu\pi_{m-1}$ (with $m - D \neq 0$) gives $p_\mu = 0$, hence $\kappa_i = 0$. So all $q_i = 0$, contradicting the choice. □

Theorem 23.6 (A-world Beukers theory)**.** *Assume* A. *For every finite set Λ of nonzero algebraic numbers the numbers*

$$1, \qquad e^\lambda\ (\lambda \in \Lambda), \qquad \mathrm{Ein}(\lambda) - \gamma = \log\lambda + E_1(\lambda)\ (\lambda \in \Lambda)$$

are linearly independent over $\overline{\mathbb{Q}}$. In particular $\mathrm{Ein}(\lambda) - \gamma$ is transcendental for every algebraic $\lambda \neq 0$ (for $\lambda \notin (-\infty, 0]$ this is $\log\lambda + E_1(\lambda)$), and $\mathrm{Ein}(\lambda) - \gamma \notin \overline{\mathbb{Q}} + \overline{\mathbb{Q}}e^\lambda$.

Proof. Under A, $\mathrm{span}\{1, \mathrm{Ein}(\lambda z) - \gamma\} = \mathrm{span}\{1, \mathrm{Ein}(\lambda z)\}$, so the statement is the unconditional multipoint theorem for the functions 1, $e^{\mu z}$ $(\mu \in \Lambda \cup (-\Lambda))$, $\mathrm{Ein}(\lambda z)$ $(\lambda \in \Lambda)$ plus A. These are E-functions with coefficients in the number field $\mathbb{Q}(\Lambda)$, they satisfy a first-order system with coefficients in $\overline{\mathbb{Q}}(z)$ (entries $0, \pm\lambda, \pm 1/z$) singular only at 0, because $\frac{d}{dz}\mathrm{Ein}(\lambda z) = (1 - e^{-\lambda z})/z$, and they are $\mathbb{C}(z)$-linearly independent by Lemma 23.5 with $M = \Lambda \cup (-\Lambda)$. Beukers' theorem [5, Cor. 1.4] at the non-singular point $z = 1$ gives the linear independence of the values, hence of the subfamily $1, e^\lambda, \mathrm{Ein}(\lambda)$; and $\gamma \in \overline{\mathbb{Q}}$ lets one replace $\mathrm{Ein}(\lambda)$ by $\mathrm{Ein}(\lambda) - \gamma$. □

So the A-world theorem is the unconditional multipoint theorem of [1] with one algebraic constant renamed; its content lies entirely in the fact that the same sentence, read in C without A, is the open statement $\gamma \notin \mathcal{A}(\Lambda)$ of Proposition 25.1. In C the argument proves only the shear-invariant part (with $\mathrm{Ein}(\lambda_i) - \mathrm{Ein}(\lambda_1)$ in place of $\mathrm{Ein}(\lambda_i) - \gamma$); the part concerning γ itself is what Section 25 reads backwards.

24 The shear is a twist of the Gamma function

Part I defined the shear on numbers, [3] on Stokes framings. Here it is on functions.

Proposition 24.1 (The twist Γ_τ)**.** *For $\tau \in \mathbb{C}$ put $\Gamma_\tau(s) := \Gamma(s)\, e^{\tau(1-s)}$. Then:*

(i) $\Gamma_\tau(1) = 1$, $\log \Gamma_\tau$ *is convex on $\mathbb{R}_{>0}$ for real τ, and $\Gamma_\tau(s+1) = s\, e^{-\tau}\, \Gamma_\tau(s)$; $\Gamma_\tau(n+1) = n!\, e^{-\tau n}$.*

(ii) $-\Gamma_\tau'(1) = \gamma + \tau$, *and $1/\Gamma_\tau(s) = e^{-\tau}\, s\, e^{(\gamma+\tau)s} \prod_{n\geq 1}(1 + s/n)e^{-s/n}$: the Weierstrass product of Γ_τ is that of Γ with Euler constant $\gamma + \tau$, up to the constant $e^{-\tau}$ that restores $\Gamma_\tau(1) = 1$.*

(iii) $\Gamma_\tau(1-t)\lambda^t = \Gamma(1-t)(\lambda e^\tau)^t$, *so the tower polynomials satisfy $P_r(\lambda; \gamma + \tau) = P_r(\lambda e^\tau; \gamma)$ (branch $\log(\lambda e^\tau) = \log \lambda + \tau$), and the shadow system at $c = \gamma + \tau$ is obtained from the true system by replacing Γ with Γ_τ in the definitions $\gamma = -\Gamma'(1)$ and $\Gamma(1-t)\lambda^t$ (not in $E_1 = \Gamma(0, \cdot)$), and λ with λe^τ in every tower polynomial.*

(iv) The shear $\mathfrak{s}_c$ of Part I is the twist with $\tau = c - \gamma$.

Proof. (i)–(iii) are one-line computations from $\Gamma(s+1) = s\Gamma(s)$, $-\Gamma'(1) = \gamma$, the Weierstrass product, and $\Gamma(1-t) = \exp(\gamma t + \sum_{k\geq 2} \zeta(k) t^k/k)$. (iv) compares (ii) with the definition of $\mathfrak{s}_c$. □

Remark 24.2 (What pins $\tau = 0$)*.* By Bohr–Mollerup, Γ is the unique log-convex function with $f(1) = 1$ and $f(s+1) = sf(s)$. The twists Γ_τ satisfy the first two conditions and the functional equation with the constant $e^{-\tau}$ in place of 1. So one property of Γ that singles out γ from the line $\gamma + \tau$ is the *normalisation of the functional equation*, equivalently the single value $\Gamma(2) = 1$ (since $\Gamma_\tau(2) = e^{-\tau}$; rationality at the integers is not enough, as $\Gamma_{\log 2}(n+1) = n!/2^n$ shows). The reflection and multiplication formulas change by the constant factors e^τ and $e^{\tau/2}$, Stirling's formula changes by $e^{-\tau s}$, and Euler's integral becomes $e^\tau \int_0^\infty u^{s-1} e^{-e^\tau u}\, du$ (the twist is the rescaling $u \mapsto e^\tau u$); each of these normalisations also pins $\tau = 0$. Every connection identity, every E-function, every Stokes structure and every tower identity is insensitive to the twist: this is Part I's Theorem 4.2 in function-theoretic form, and it is also the Fischler–Rivoal observation that γ "always appears through $\gamma - \log(1/z)$" [8], since by (iii) the twist is a rescaling of z.

Remark 24.3 (The twist on the period matrix)*.* Fresán and Jossen realise γ as a period of $H^2(\mathbb{A}^2, \{xy(x-1)(y-1) = 0\}, xy)$, a three-dimensional exponential motive whose period matrix, in suitable bases which we call $\sigma_1, \sigma_2, \sigma_3$ (Betti) and $\omega_1, \omega_2, \omega_3$ (de Rham), is [12, 30]

$$\Pi = \begin{pmatrix} 1 & 0 & \gamma \\ 0 & e^{-1} & E_1(1) \\ 0 & 0 & 2\pi i \end{pmatrix}, \qquad E_1(1) = \int_1^\infty e^{-t}\, \frac{dt}{t},$$

and the quotient $M(\gamma)$, an extension of $\mathbb{Q}(-1)$ by $\mathbb{Q}(0)$ with matrix $\left(\begin{smallmatrix} 1 & \gamma \\ 0 & 2\pi i \end{smallmatrix}\right)$, is a non-trivial extension not isomorphic to a classical motive. Three readings. (a) The connection formula is additivity of periods: the cycle $\sigma_1 + \sigma_2$ has periods $(1, e^{-1}, \mathrm{Ein}(1))$, so $\mathrm{Ein}(1) = \gamma + E_1(1)$ is "E-value = period of σ_1 + period of σ_2"; the collision of Part II asks whether the period of σ_2 (times e) can also be the period of an E-function. (b) The twist by τ acts on Π by right multiplication with $D_\tau = I + \tau(E_{13} - eE_{23})$ (E_{ij} the matrix units), i.e. the change of de Rham basis $\omega_3 \mapsto \omega_3 + \tau\omega_1 - e\tau\,\omega_2$: it sends the third column to $(\gamma + \tau,\ E_1(1) - \tau,\ 2\pi i)^t$ and fixes the

periods of $\sigma_1 + \sigma_2$ and σ_3. This is a $\mathbb{C}$-linear, not a $\overline{\mathbb{Q}}$-linear, change of basis: the coefficient $-e\tau$ on ω_2 is forced, because the shear moves $\delta = eE_1(1)$ by $e\tau$. The legitimate $\overline{\mathbb{Q}}$-changes are $\omega_3 \mapsto \omega_3 + \tau\omega_1$, $\tau \in \overline{\mathbb{Q}}$, which move γ alone—this is why γ is defined as a period only modulo $\overline{\mathbb{Q}}$—and the A-world is the statement that the choice $\tau = -\gamma$ is among them, in which case the third column becomes $(0, E_1(1), 2\pi i)^t$ and $\mathbb{Q}(0)$ splits off at the level of periods while, by Fresán–Jossen, it does not split motivically. (c) The matrices D_τ form a one-parameter unipotent subgroup of $\mathrm{GL}_3(\mathbb{C})$, and the two-model theorem of Part I says that the established theorems are invariant under it for *every real* τ. If the unipotent radical of the motivic Galois group of H^2 contains both shifts E_{13} and E_{23} (the expected situation if the two extension classes are independent), then D_τ is a $\mathbb{C}$-point of the Galois group for every τ, the twist is the action of a one-parameter subgroup of the Galois group on the period torsor, shear-invariance is invariance under that subgroup, and the ledger of Part III reads: every established theorem about γ is a theorem about the Galois torsor, not about the point on it. We record this as a reading, not a claim.

25 The B-world

25.1 Duality: A-world theorems are B-certificate loci

For a finite set Λ of nonzero algebraic numbers let $V(\Lambda) = \mathrm{span}_{\overline{\mathbb{Q}}}\{1, e^\lambda, \mathrm{Ein}(\lambda) : \lambda \in \Lambda\}$; by Lemma 23.5 (with $M = \Lambda \cup (-\Lambda)$) and Beukers' theorem the spanning set is a basis, so the Ein-*weight* $w(v) = \sum_\lambda a_\lambda$ of $v = a_0 + \sum b_\lambda e^\lambda + \sum a_\lambda \mathrm{Ein}(\lambda)$ is well defined. Put

$$\mathcal{A}(\Lambda) := \{v \in V(\Lambda) : \ w(v) = 1\},$$

an affine hyperplane of $V(\Lambda)$; it is countable, and $\mathcal{A}(\Lambda) \cap \mathbb{R}$ is dense in $\mathbb{R}$ with dense complement as soon as Λ contains a real point λ_0 (it contains $\mathrm{Ein}(\lambda_0) + (\overline{\mathbb{Q}} \cap \mathbb{R})$).

Proposition 25.1 (Duality)**.** *(i) The conclusion of Theorem 23.6 for* Λ *is equivalent, in* C*, to* $\gamma \notin \mathcal{A}(\Lambda)$*.*

(ii) Consequently $\gamma \in \mathcal{A}(\Lambda) \Rightarrow$ B*: each hyperplane is a* certificate locus*, a countable set of explicit numbers* v *such that a proved identity* $\gamma = v$ *would yield* B *by Beukers' theorem. A lattice search can only supply the candidate* v*; the identity itself would still have to be proved.*

(iii) The statement $\gamma \notin \mathcal{A}(\Lambda)$ *is shear-invisible (Part III, Theorem 15.2, true in* A *(Theorem 23.6), and expected in* B *(period conjecture).*

Proof. (i) A relation $a_0 + \sum b_\lambda e^\lambda + \sum a_\lambda(\mathrm{Ein}(\lambda) - \gamma) = 0$ has Ein-weight $s = \sum a_\lambda$. If $s = 0$ it is a relation in $V(\Lambda)$, impossible. If $s \neq 0$ it says $\gamma = s^{-1}(a_0 + \sum b_\lambda e^\lambda + \sum a_\lambda \mathrm{Ein}(\lambda)) \in \mathcal{A}(\Lambda)$. (ii) If $\gamma = v \in \mathcal{A}(\Lambda)$ and γ were algebraic, $v - \gamma = 0$ would be a relation in $V(\Lambda)$ with Ein-weight 1. (iii) $\mathcal{A}(\Lambda) \cap \mathbb{R} \supset \mathrm{Ein}(\lambda_0) + (\overline{\mathbb{Q}} \cap \mathbb{R})$ is dense; the rest is Part III. □

So the A-world's strongest theorems, read from the B-side, are a list of places where γ would be *visibly* transcendental. The same reading applies to every transcendence theorem for an explicit countable class: Hermite–Lindemann gives the exponential locus $\{\log \alpha : \alpha \in \overline{\mathbb{Q}}^\times, \ \alpha \neq 1\}$ (on which $e^\gamma \in \overline{\mathbb{Q}}$) and the e-locus $\overline{\mathbb{Q}}(e) \setminus \overline{\mathbb{Q}}$; Lindemann gives the π-locus $\overline{\mathbb{Q}}(\pi) \setminus \overline{\mathbb{Q}}$ (the searches below probe the larger sets "γ algebraic of degree ≤ 3 over $\mathbb{Q}(\pi)$", resp. over $\mathbb{Q}(e)$, which the decision argument of Proposition 25.2 also covers); Chudnovsky gives the Gamma loci $\overline{\mathbb{Q}} + \overline{\mathbb{Q}}\pi + \overline{\mathbb{Q}}\,\Gamma(\frac{1}{4})$ and $\overline{\mathbb{Q}} + \overline{\mathbb{Q}}\pi + \overline{\mathbb{Q}}\,\Gamma(\frac{1}{3})$ (minus $\overline{\mathbb{Q}}$); Lindemann–Weierstrass and Beukers give the Ein-hyperplanes and the exponential spans. The B-world is stratified by these loci.

25.2 The certificate zoo

Proposition 25.2 (Certified searches on the loci)**.** (Finite certificate; script `exp12`, 300 digits, exact re-verification as in Parts I–II.) *On each of the following loci, any integer relation would decide the alternative (*B *by the named theorem, or* A *in the degenerate case where all transcendental coefficients vanish); none exists below the stated height:*

locus	*numbers searched*	*theorem*	*no relation below*
exponential	$1, e^{\gamma}, \dots, e^{6\gamma}$	*Hermite–Lindemann*	$10^{41.2}$
logarithmic	$1, \gamma, \log 2, \log 3, \log 5, \log 7$	*Hermite–Lindemann*	$10^{48.4}$
π*-locus*	$\gamma^i \pi^j$, $i + j \le 3$	*Lindemann*	$10^{28.0}$
e-locus	$\gamma^i e^j$, $i + j \le 3$	*Hermite*	$10^{28.0}$
Ein-*hyperplane*	$1, \gamma, e^{\pm 1}, e^{\pm 2}, \mathrm{Ein}(1), \mathrm{Ein}(-1), \mathrm{Ein}(2)$	*Beukers, Lindemann–W.*	$10^{31.2}$
$\Gamma(1/4)$*-locus*	$1, \gamma, \pi, \Gamma(1/4)$	*Chudnovsky*	$10^{73.5}$
$\Gamma(1/3)$*-locus*	$1, \gamma, \pi, \Gamma(1/3)$	*Chudnovsky*	$10^{73.8}$

Together with Part II's linear window (γ *against* 34 *E-values,* 10^{41}*) and quadratic window (*78 *monomials,* $10^{11.5}$*), these are the* B*-certificate searches performed so far.*

Proof. The certificates are the computation. For the decisions: a relation $\sum k_i e^{i\gamma} = 0$ makes e^{γ} algebraic, so $\gamma \neq 0$ is a logarithm of an algebraic number, transcendental. A relation $k_0 + k_1\gamma + \sum k_p \log p = 0$ with $k_1 \neq 0$ and some $k_p \neq 0$ makes γ algebraic only if $\log \prod p^{k_p}$ is, which it is not; $k_1 = 0$ is impossible (then $e^{k_0} = \prod p^{-k_p}$ is rational, so $k_0 = 0$ by Hermite–Lindemann and all $k_p = 0$ by unique factorisation), and $k_1 \neq 0$ with all $k_p = 0$ is $\gamma \in \mathbb{Q}$. A polynomial relation $P(\gamma, \pi) = 0$ with γ algebraic forces $P(\gamma, y) \equiv 0$, i.e. P divisible by the minimal polynomial of γ (of degree ≤ 3, excluded below 10^{28} by the same certificate); otherwise $\gamma \notin \overline{\mathbb{Q}}$. Same for e. On the Ein-hyperplane, a relation with $k_1 \neq 0$ and some Ein-coefficient nonzero contradicts Beukers if γ is algebraic; with only exponential coefficients it contradicts Lindemann–Weierstrass; with neither it is $\gamma \in \mathbb{Q}$; and $k_1 = 0$ is impossible. On the Gamma loci, π and $\Gamma(1/4)$ (resp. $\Gamma(1/3)$) are algebraically independent [29], so a relation with $k_1 \neq 0$ and $(k_2, k_3) \neq 0$ forces $\gamma \notin \overline{\mathbb{Q}}$, and $k_1 = 0$ is impossible. □

25.3 The wild world

Definition 25.3 (Certifiable classes; the wild world)**.** A *certifiable class* is a T_0-definable countable set $\mathcal{Z} \subset \mathbb{R}$ of transcendental numbers, given by explicit closed forms in classical constants and E-values, whose definition and transcendence proof do not involve γ, δ or the $E_1(\lambda)$ (the loci above; more generally the transcendental values of E-functions at algebraic points given explicitly, whose algebraicity is decided by [4]). The *wild world* is the sub-world of B in which γ belongs to no certifiable class.

Proposition 25.4 (No certificate in the wild world)**.** *In the wild world no integer-relation search on a certifiable class can succeed, and no proof of* B *can consist of exhibiting* γ *as a member of a certifiable class. Hence in the wild world* B *can be proved only by a theorem that applies to* γ *without exhibiting it as such a member: a Diophantine construction of the type specified in Part III (engine ratios converging to* γ*, proved through a bridge), or a structure theorem such as the fullness of the period realisation for* $M(\gamma)$*.*

Proof. Immediate from the definition. □

The wild world is where every standard expectation places γ: Rivoal's $\gamma \notin \mathbf{E} \cup \mathbf{G} \cup \mathbf{M}$ [16], Fischler–Rivoal's $\mathbf{E} \cap \mathbf{D} = \overline{\mathbb{Q}}$ [10], and the period conjecture through [30] ("γ is transcendental over the field generated by all classical periods"). This is the honest shape of the situation: the world we believe we live in is precisely the one in which no search can ever succeed, and each

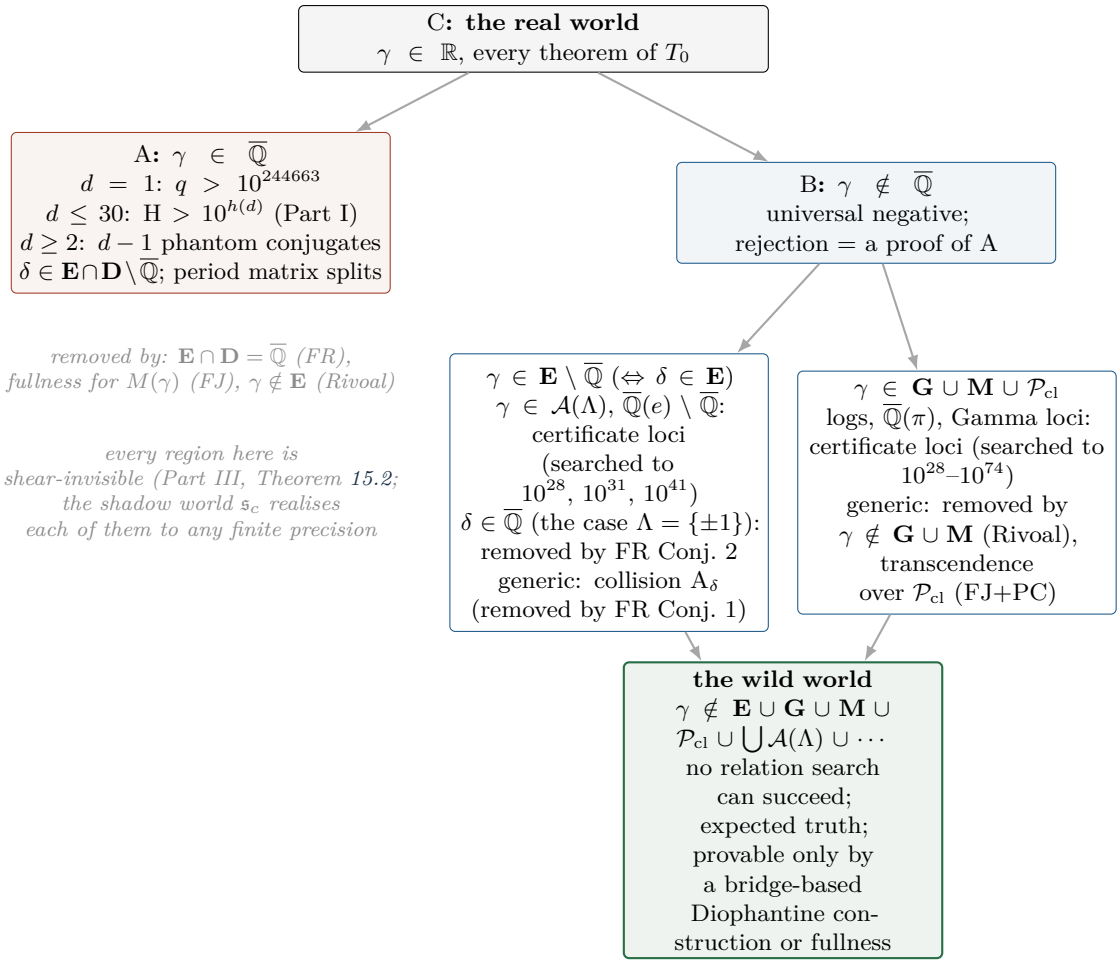


Figure 1: The map of possible worlds for γ. Boxes are regions consistent with every proved theorem; "certificate" regions are those in which a proved identity would make γ visibly transcendental; "removed by" names the conjecture expected to empty a region. $\mathcal{P}_{\mathrm{cl}}$: classical periods; $\mathbf{M}$: Mahler values; $\mathcal{A}(\Lambda)$: the Ein-hyperplanes of Proposition 25.1. The regions overlap (a logarithm is a G-value); the map is a stratification by certificate type, not a partition.

certificate search is a search of a region we expect to be empty. The searches are still worth running—a hit would end the matter—but their negative outcome is the expected outcome, and the map below records it as such.

26 The map of possible worlds

Figure 1 arranges the worlds. Every region is consistent with every proved theorem (the density obstruction of Part III guarantees that no shear-invariant theorem can remove any of them); the labels say which theorem gives a finite certificate against a region, which conjecture is expected to remove it, and where the expected truth lies.

Three features of the map deserve a sentence each. First, the A-world is a *single* region, but it is realised to every finite precision by a shadow world (Section 23); it is small in the sense of measure and large in the sense of evidence. Second, every certificate region is countable, explicit, and searchable, and all searches to date are negative; the regions are the "visible" transcendence of γ, and their emptiness below 10^{28}–10^{74} is the only positive information the B-side has produced. Third, the wild world is the complement of everything explicit; it is where the conjectures put γ, and it is where no search can reach. The two-world programme, run for its own sake, ends by

drawing this map rather than by closing it.

27 A curiosity: the engine ratios of Rivoal's forms

Part III wrote Rivoal's forms as $L_n = \alpha_n + \beta_n\gamma$ with engine coefficients $\alpha_n = a_n + c_n e\,\mathrm{Ein}(1)$, $\beta_n = b_n - ec_n$ and asked for sequences of engine ratios $-\alpha_n/\beta_n$ converging to γ with $|L_n| \le H_n^{-1-\varepsilon}$. Rivoal's own forms provide an empirical sequence, far from that specification.

Proposition 27.1 (Engine ratios)**.** (Finite certificate; script `exp12`.) *The ratios $r_n := -\alpha_n/\beta_n \in \mathbb{Q}(e, \mathrm{Ein}(1))$ approach γ numerically for $n \le 39$, equivalently $t_n := e(\mathrm{Ein}(1) - r_n)$ approaches δ, with*

$$-\log_{10}|\gamma - r_n| = 1.89,\ 2.70,\ 3.92,\ 4.89,\ 5.64 \quad (n = 5, 10, 20, 30, 39),$$

and $-\log_{10}|\gamma - r_n|/\sqrt{n}$ increases slowly from 0.85 to 0.90. Explicitly, with Rivoal's polynomials at -1 and $\rho_n = R_{n+1,0}(-1)/R_{n,0}(-1)$,

$$r_n = \frac{C_{n+1} - \rho_n C_n}{B_{n+1} - \rho_n B_n}, \qquad \gamma - r_n = e^{-1}\,\frac{(A_{n+1} - \rho_n A_n) - \delta(B_{n+1} - \rho_n B_n)}{B_{n+1} - \rho_n B_n}. \tag{27.1}$$

Taking $|B_n(-1)| \approx H_n^{1/2}$ as the height of t_n, the Stieltjes convergents of δ of the same height are far better: at height 10^{103}, $|\delta - t_{39}| = 10^{-5.2}$ against $|\delta - s_{137}/q_{137}| = 10^{-13.2}$ (at the full height $H_{39} = 10^{208}$ the gap is larger still).

Proof. With $R_n = e^{-1}A_n + B_nE(-1) + C_n$, $E(-1) = -\mathrm{Ein}(1)$, $P_n = A_n - B_n\delta$, $Q_n = C_n - B_n\gamma$ one has $R_n = e^{-1}P_n + Q_n$, $L_n = P_{n+1}Q_n - P_nQ_{n+1} = e(R_{n+1}Q_n - R_nQ_{n+1})$ and $\beta_n = e(B_{n+1}R_n - B_nR_{n+1})$ (Part III, Proposition 18.4, eq. (18.2)); dividing gives (27.1) after substituting $Q_n = C_n - B_n\gamma$ and $Q_{n+1} - \rho_nQ_n = -e^{-1}(P_{n+1} - \rho_nP_n)$. The numbers are the computation at 1300 digits. □

The observed approach is slow, of the subexponential type $\exp(-c\sqrt{n})$ that Part III found on the D-side (nothing is proved about it: no bound on $|L_n|/|\beta_n|$ is available), and it is dominated by Stieltjes's convergents by eight orders of magnitude at comparable height. The E-side, once forced to speak about γ rather than about $e\gamma + \delta$, speaks in the D-side's voice and less clearly. This is the tie theorem seen from the other end.

28 Questions each world asks next

Questions, not targets. They are what the worlds, left to themselves, want to know.

The A-world asks.

(A1) What is the degree? Every certificate pushes the height of P_γ up ($10^{89.8}$ for $d \le 12$ at 1200 digits, Section 23, script `exp12b`); the A-world with $d = 1$ has no phantoms and the largest bound ($q > 10^{244663}$).

(A2) Are the phantom conjugates real or complex? Nothing constrains them; the camouflage of Part I has one real and ten complex phantoms, all of modulus about $2.7 \cdot 10^7$ (the eleven other roots of $x^{12} + 6.2 \cdot 10^{81}x - 3.6 \cdot 10^{81} \approx 0$ lie near a circle of radius $\mathrm{H}(P_c)^{1/11}$).

(A3) Where does the period conjecture fail? In the A-world it fails at $M(\gamma)$ (Remark 24.3(b)), but in A the number $\gamma + \log\alpha$ is transcendental for every algebraic $\alpha \ne 1$ (Hermite–Lindemann applied to $\log\alpha$), so along the family of Euler constants $\gamma + \log\alpha$—which is the twist family, by Proposition 24.1(iii)—the splitting at the level of periods happens at $\alpha = 1$ only. Is there a motivic reason why the split point of the family should be its base point?

(A4) Does the Galois norm $\mathrm{N}(E_1(1)) = P_\gamma(\mathrm{Ein}(1))$ have a meaning? It is an E-value with rational coefficients whose irrationality exponent is 2; the A-world knows its degree but not its value.

(A5) The p-adic cousin: Diamond's p-adic Euler constant γ_p has the same unknown arithmetic nature, and the p-adic theory has a Baker-type statement (for a prime q, at most one of the numbers γ_p, $\gamma_p(r,q)$, $1 \le r < q$, is algebraic [31]) in place of Rivoal's disjunction. Does a twist $\Gamma_{p,\tau}$ of Morita's Γ_p play the role of Proposition 24.1 there, and is the p-adic two-world problem shear-invisible in the same sense?

The B-world asks.

(B1) Is δ algebraic? This is the one certificate world inside $\gamma \in \mathbf{E}$ not yet removed by a theorem (only by Conjecture 2 of [10]); a relation would give $\gamma = \mathrm{Ein}(1) - \delta/e$ explicitly.

(B2) Which certificate locus is most plausible? None is expected to contain γ; but the Ein-hyperplanes are the only loci that the A-world's own theorems point to.

(B3) In the wild world, what is the irrationality exponent of γ? Expected 2, consistent with the 2910 computed partial quotients (Part I, Table 2; no theorem gives any bound at all.

(B4) Does the wild world have an internal structure? Part III's hierarchy $\mathrm{X}(V)$ grades it linearly; the twist of Section 24 suggests grading it by which normalisations of Γ (functional equation, reflection, multiplication) a statement uses.

Both worlds ask. Whether the unipotent matrices D_τ of Remark 24.3(c) are $\mathbb{C}$-points of the motivic Galois group of the polyexponential tower. If they are, the two-model theorem is a theorem about the Galois torsor, and the whole series of notes has been describing a single orbit.

29 Computations for Part IV

Scripts (Python 3.11, `mpmath` 1.3, `sympy` 1.14, `fpylll` 0.6.4), archived with outputs.

- `exp12_two_worlds.py`: engine ratios from the exact coefficients of Part I's Rivoal forms and Stieltjes convergents of comparable height (Proposition 27.1); the shadow world of the degree-12 camouflage (root by Newton at 1300 digits, shadow constants, $F_c(1)$, blindness of the forms, Galois norm and conjugate shadows, certified lattice searches on c and γ); the certificate zoo (Proposition 25.2).
- `exp12b_find_camouflage.py`: certified searches on c and γ at 300–1200 digits, with a candidate relation accepted only if it vanishes to $3p$ digits and compared exactly with P_c.
- `exp12c_camouflage_duality.py`: the roots c_p of the certificate vectors, with exact re-verification of the reductions and an irreducibility check of Q_p (Proposition 23.3). The detector of `exp12` (inherited from Part II) reports only relations with coefficients below 10^6, which is why its 1100-digit run on c prints no relation although b_1 is P_c; `exp12b` uses the uncapped test.

Lattice reductions use `fplll` and are re-verified exactly (unimodularity by Bareiss, the $(0.99, 0.51)$ reduction conditions in rational arithmetic) as in Parts I–II. The only non-exact inputs are the values of γ, e, π, $\log p$, $\Gamma(1/3)$, $\Gamma(1/4)$, $\mathrm{Ein}(\lambda)$, $E_1(1)$ supplied by `mpmath`. Run times: under ten minutes in total.

Part V
Forgeries, twists and the δ-world

Summary of Part V. Three things from Part IV are grown here. (1) The camouflage–certificate duality is extended from the algebraic world to the certificate worlds of the map: at precision p, the lattice vector that certifies "γ is not on this locus below height $\approx 10^{p/(m+1)}$" is, read backwards, an explicit member of the locus within 10^{-p} of γ. We exhibit the resulting *gallery of forgeries*: rational, quadratic and cubic algebraic numbers, a logarithm of an algebraic number, an element of $\mathbb{Q}+\mathbb{Q}\log 2+\cdots+\mathbb{Q}\log 7$, numbers cubic over $\mathbb{Q}(\pi)$ and over $\mathbb{Q}(e)$, an explicit E-value on an Ein-hyperplane, and an element of $\mathbb{Q}+\mathbb{Q}\pi+\mathbb{Q}\Gamma(\frac{1}{4})$, all agreeing with γ to 30 digits with heights of 3–14 digits; the same at 100 digits, with a degree-12 algebraic number added. Their heights obey $\mathrm{H}\approx 10^{p/(m+1)}$ uniformly across the worlds, and the approximation exponents of γ by algebraic numbers of degree $d\le 12$ are $d+1$ to within 1.1% at 300 digits: no finite observation prefers any world, quantitatively. (2) The twist Γ_τ is extended to the incomplete Gamma function and to the mixed values of Fischler–Rivoal: the residue-normalised twist $\widetilde{\Gamma}_\tau(a)=e^{-\tau a}\Gamma(a)$ gives $\widetilde{\Gamma}_\tau(a,z)=e^{-\tau a}\Gamma(a)-\gamma(a,z)$ with $\widetilde{\Gamma}_\tau(0,z)=E_1(z)-\tau$, regular at $a=0$ but, for real $\tau\neq 0$, at no other non-positive integer; in $\Gamma^{(s)}(a)=(-1)^s s!\,E_{a,s+1}(-1)+e^{-1}f_{a,s+1;0}(1)$ the E-side is fixed and the D-side moves by $e\,(\Gamma_\tau^{(s)}(a)-\Gamma^{(s)}(a))$, which is e times a $\mathbb{Q}[\tau]$-combination of Gamma derivatives at $a=1$ and involves $e^{\tau(1-a)}$ elsewhere; for algebraic τ the twisted mixed values stay in the ring $\mathbf{V}=\mathbf{S}\cdot\mathbf{E}$ of Fischler–Rivoal. On the τ-line, the worlds A, "$\delta\in\overline{\mathbb{Q}}$" and "$e^\gamma\in\overline{\mathbb{Q}}$" are the statements that a coordinate of the twisted system vanishes at some τ in $\overline{\mathbb{Q}}$, $e^{-1}\overline{\mathbb{Q}}$ or $\log\overline{\mathbb{Q}}^\times$ respectively, and $\gamma+\log\alpha\in\overline{\mathbb{Q}}$ for at most one α, unconditionally. (3) The δ-world ($\delta\in\overline{\mathbb{Q}}$), the one certificate world inside $\gamma\in\mathbf{E}$ that no theorem removes, is given its certificate (δ satisfies no relation of degree ≤ 12 below $10^{20.6}$), its forgery, and its phantoms: shears by $\tau\in e^{-1}\overline{\mathbb{Q}}$.

30 What grows from what

Part IV, Proposition 23.3, observed that the lattice vector certifying "no algebraic relation of degree ≤ 12 for γ below height $\approx 10^{p/13}$" is the minimal polynomial of an algebraic number within 10^{-p} of γ. Part IV, Section 25, listed the certificate loci of the B-world. Section 31 puts the two together. Part IV, Section 24, identified the shear with the twist Γ_τ; Section 32 lets the twist act on everything else that carries a Gamma function. Part IV, question (B1), asked about the δ-world; Section 33 answers what can be answered.

31 The gallery of forgeries

31.1 Construction

Fix a *world* W given by a finite list $x_0,\dots,x_m$ of real numbers containing γ, or containing the powers of γ (or of e^γ), such that a proved relation $\sum k_i x_i=0$ with the coefficient of the γ-part nonzero would exhibit γ as an explicit member of W. The certified lattice search of Parts I–II at p digits returns a first vector $b_1=(k_0,\dots,k_m)$ with $|\sum k_i x_i|\lesssim\|b_1\|/N$, $N=10^p/2$, and $\|b_1\|\approx 10^{p/(m+1)}$. Solving the near-relation for the γ-part produces a number v_W of the world W with

$$|\gamma-v_W|\approx 10^{-p},\qquad \mathrm{H}(v_W)\approx 10^{p/(m+1)},\tag{31.1}$$

which we call the *forgery of γ in W at precision p*. The same vector b_1 certifies that γ satisfies no relation in W below height $\|b_1\|/(\alpha^{m/2}(m+1)\sqrt{m+2})$ (Part II, Section 14, which is $10^{1.5}$ to $10^{2.5}$ below $\mathrm{H}(v_W)$. The forgery is verified to be different from γ at $3p+100$ digits, and it is a transcendental number whenever the theorem attached to W applies to it (checked case by case in Table 7). For the Ein-hyperplane $\mathcal{A}(\Lambda)$ of Part IV the search is run on the weight-one basis

Table 7: Forgeries of γ at 30 digits (scripts `exp13b`, `exp13c`). v is the number defined in the second column; $|\gamma - v|$ is computed at 190 digits. In every row $v \neq \gamma$.

world	forgery v	$\log_{10}\lvert\gamma - v\rvert$	$v \notin \overline{\mathbb{Q}}$ by
rational	$12668145165383/21946987817013$	-28.4	(algebraic)
quadratic	root near 0.577 of $1825589204x^2 - 2377304883x + 763971598$	-28.4	(algebraic)
cubic	root near 0.577 of $10806397x^3 - 12006300x^2 + 5013002x - 971587$	-28.7	(algebraic)
exponential	$\log\alpha$, α the real root near 1.781 of $3899x^6 - 1521x^5 + 501x^4 - 6072x^3 - 15134x^2 - 13627x + 4341$ (irreducible)	-31.0	Hermite–Lindemann
logarithmic	$(19275 - 19985\log 2 + 50615\log 3 + 11986\log 5 - 27071\log 7)/47888$	-29.8	Hermite–Lindemann
π-locus	root near 0.577 of $170x^3 + (334\pi - 436)x^2 + (116 - 627\pi - 75\pi^2)x + (173 - 305\pi + 113\pi^2 + 30\pi^3)$ (irreducible in $\mathbb{Q}[x, y]$)	-30.3	Lindemann
e-locus	root near 0.577 of $-312x^3 - (36 + 76e)x^2 - (383 + 157e + 325e^2)x + (210 + 742e + 292e^2 - 119e^3)$ (irreducible in $\mathbb{Q}[x, y]$)	-31.5	Hermite
Ein-hyperplane (weight 1)	$\big(-2521 + 721e - 1376e^{-1} + 106e^2 + 2322e^{-2} + 3798\,\mathrm{Ein}(1) - 315\,\mathrm{Ein}(-1) - 1968\,\mathrm{Ein}(2)\big)/1515$	-29.6	Beukers
Ein-span (weight $-593/720$)	$\big(281 + 976e + 338e^{-1} - 260e^2 + 1407e^{-2} + 778\,\mathrm{Ein}(1) - 105\,\mathrm{Ein}(-1) - 1266\,\mathrm{Ein}(2)\big)/720$	-29.7	Beukers
$\Gamma(\frac{1}{4})$-locus	$\big(-25383337 + 17880293\,\pi - 7430501\,\Gamma(\frac{1}{4})\big)/6668499$	-29.7	Chudnovsky

$\gamma - \mathrm{Ein}(1)$, 1, $e^{\pm 1}$, $e^{\pm 2}$, $\mathrm{Ein}(-1) - \mathrm{Ein}(1)$, $\mathrm{Ein}(2) - \mathrm{Ein}(1)$, so that the forgery has Ein-weight exactly 1; the search on the full span $V(\Lambda)$ (the window of Part IV's zoo) gives a forgery of weight $\neq 1$, also listed.

Proposition 31.1 (The gallery)**.** (Finite certificate; scripts `exp13`, `exp13b`, `exp13c`.) *Table 7 lists forgeries of γ at 30 digits in ten worlds; Table 8 the heights at 100 digits. In each transcendental row the forgery is a transcendental number, by the theorem named, and in each row $v \neq \gamma$. The bivariate polynomials of the π- and e-rows are irreducible in $\mathbb{Q}[x, y]$, so those forgeries have degree exactly 3 over $\mathbb{Q}(\pi)$, resp. $\mathbb{Q}(e)$. The heights follow (31.1): at $p = 60, 100, 200, 300$ the ratio $p/\log_{10} \mathrm{H}(v_W)$ equals $m + 1$ to within 3% for $p \geq 100$ (Table 9).*

Proof. The construction and the verification at $3p$ digits are the computation. Transcendence: $\log\alpha$ with $\alpha \neq 1$ algebraic is transcendental (Hermite–Lindemann); so is $q_0 + q_1 \log r$ for rational q_0, $q_1 \neq 0$ and rational $r \neq 1$ (here $q_1 = 1/47888$ and $r = 2^{-19985}3^{50615}5^{11986}7^{-27071}$); a root v of $P(x, \pi) = 0$ is algebraic only if $P(v, y) \equiv 0$, i.e. only if the minimal polynomial of v divides every coefficient of P as a polynomial in y, which is excluded by irreducibility (same for e); the two Ein forgeries are $\mathbb{Q}$-linear combinations of $1, e^{\pm 1}, e^{\pm 2}, \mathrm{Ein}(\pm 1), \mathrm{Ein}(2)$ with nonzero Ein coefficients, hence not algebraic, because these eight numbers are linearly independent over $\overline{\mathbb{Q}}$ (Part IV, Lemma 23.5 with $\Lambda = \{1, -1, 2\}$, $M = \{\pm 1, \pm 2\}$, and Beukers' theorem [5]); the Gamma forgery is algebraic only if $17880293\pi - 7430501\Gamma(\frac{1}{4})$ is, contradicting the algebraic independence of π and $\Gamma(\frac{1}{4})$ [29]. □

31.2 What the gallery says

Part I's symmetry of finite observations (Theorem 3.1 there) was a statement about the algebraic world. The gallery is its quantitative form for every world at once: at precision p, γ is a member of each world W at the cost $10^{p/(m+1)}$, the cost depends on W only through the dimension $m + 1$ of its basis, and no world is cheaper than its dimension predicts. A world in which γ *really* lived

Table 8: Heights of the forgeries at 100 digits (script `exp13`); $m+1$ is the number of basis elements, $p/(m+1)$ the Dirichlet prediction for $\log_{10}\mathrm{H}$.

world	$m+1$	$\log_{10}\mathrm{H}(v)$	$p/(m+1)$	$\log_{10}\lvert\gamma - v\rvert$
rational	2	49.6	50.0	−100.3
quadratic	3	33.1	33.3	−100.3
degree 12	13	7.5	7.7	−99.0
exponential (degree ≤ 6)	7	14.2	14.3	−100.6
logarithmic	6	16.6	16.7	−99.4
π-locus (degree ≤ 3)	10	9.9	10.0	−100.4
e-locus (degree ≤ 3)	10	9.9	10.0	−100.3
Ein-hyperplane (weight 1)	8	12.4	12.5	−99.3
Ein-span	9	10.8	11.1	−99.7
$\Gamma(1/4)$-locus	4	24.6	25.0	−98.9

Table 9: The law $\mathrm{H}(v_W) \approx 10^{p/(m+1)}$: the ratio $p/\log_{10}\mathrm{H}(v_W)$ at four precisions (scripts `exp13`, `exp13c`), to be compared with $m+1$.

world	$m+1$	$p=60$	$p=100$	$p=200$	$p=300$
quadratic	3	3.13	3.02	3.04	3.01
degree 12	13	13.53	13.34	13.22	13.13
exponential	7	7.23	7.04	7.04	7.02
logarithmic	6	6.13	6.04	6.05	6.04
Ein-hyperplane (weight 1)	8	8.09	8.07	8.04	8.05
Ein-span	9	9.00	9.25	9.08	9.11
$\Gamma(1/4)$-locus	4	4.06	4.06	4.00	4.02

would announce itself by a forgery far cheaper than $10^{p/(m+1)}$—a coefficient vector that stops growing as p increases. Table 9 shows no such world among its seven to 300 digits, and Table 8 none among the ten to 100 digits.

Proposition 31.2 (Approximation exponents of γ)**.** (Finite certificate; script `exp13`.) *For $1 \le d \le 12$ let H_d be the height of the degree-d forgery at $p = 300$ and $w_d := 300/\log_{10} H_d$. Then*

$$(w_d)_{d=1}^{12} = 2.00,\ 3.01,\ 4.02,\ 5.03,\ 6.03,\ 7.07,\ 8.08,\ 9.10,\ 10.09,\ 11.10,\ 12.13,\ 13.13,$$

i.e. $w_d = d+1$ to within 1.1%. In Koksma's normalisation ($|\gamma - c| \le \mathrm{H}(c)^{-w_d^*-1}$) this reads $w_d^*(\gamma) = d$, which is the value for almost every real number (Sprindžuk; see [32]); γ behaves, to 300 digits and degree 12, like almost every real number. (H_d is the LLL height, an upper bound for the minimal height within the factor $\alpha^{d/2}\sqrt{d+1}$, about $10^{2.5}$ at $d = 12$; the conclusion is insensitive to this, since a world in which γ really lived would show a vector that stops growing with p.)

The two worlds read this differently and agree on its content. In B (wild), the expected value is $w_d = d+1$ for all d: γ is expected to be an S-number of type 1 in Mahler's classification, like almost all reals. In A with $[\mathbb{Q}(\gamma):\mathbb{Q}] = d_0$, Schmidt's theorem on the approximation of algebraic numbers by algebraic numbers of bounded degree [33] gives $w_d = d+1$ for $d < d_0$, while for $d \ge d_0$ the polynomial P_γ itself is a forgery of infinite precision and the finite-p ratio $p/\log_{10} H_d$ would grow without bound; the table therefore says $d_0 > 12$ or $\mathrm{H}(P_\gamma) > 10^{20.6}$ (the certified bound at 300 digits), which is weaker than the 1200-digit certificate of Part IV. Neither world is contradicted, and neither could be: the table is the shadow of A and of B alike.

32 The twist, extended

Part IV defined $\Gamma_\tau(s) = \Gamma(s)e^{\tau(1-s)}$, with $\Gamma_\tau(1) = 1$, Euler constant $\gamma + \tau$, and functional equation $\Gamma_\tau(s+1) = se^{-\tau}\Gamma_\tau(s)$. Here it acts on the incomplete Gamma function, on the mixed values of Fischler–Rivoal, and on the whole twisted system as a line.

32.1 The incomplete Gamma function

Write $\gamma(a,z) = \int_0^z t^{a-1}e^{-t}dt = z^a\sum_{n\ge0}(-z)^n/(n!(a+n))$ for the lower incomplete Gamma function (meromorphic in a, poles at $a=-n$ with residues $(-1)^n/n!$), so that $\Gamma(a,z) = \Gamma(a) - \gamma(a,z)$ is entire in a and $\Gamma(0,z) = E_1(z)$.

Proposition 32.1 (Twisted incomplete Gamma)**.** *Let $\widetilde{\Gamma}_\tau(a) := e^{-\tau a}\Gamma(a) = e^{-\tau}\Gamma_\tau(a)$ (residue 1 at $a=0$) and $\widetilde{\Gamma}_\tau(a,z) := \widetilde{\Gamma}_\tau(a) - \gamma(a,z)$. Then:*

(i) $\widetilde{\Gamma}_\tau(a,z)$ is regular at $a=0$ and $\widetilde{\Gamma}_\tau(0,z) = E_1(z) - \tau = -(\gamma+\tau) - \log z + \mathrm{Ein}(z)$: the twisted E_1 is the sheared Z_λ of Part I, and the connection formula $\mathrm{Ein}(z) = (\gamma+\tau) + \log z + \widetilde{\Gamma}_\tau(0,z)$ holds with the twisted Euler constant.

(ii) At $a=-n$, $n\ge1$, $\widetilde{\Gamma}_\tau(a,z)$ has a simple pole with residue $(-1)^n(e^{\tau n}-1)/n!$, nonzero for real $\tau\neq0$: the twist respects the Stokes structure of the incomplete Gamma function at $a=0$ only. Within the family of twists $f(a)\Gamma(a)$ with constant functional-equation factor $f(a+1)/f(a) = c$, regularity of $f\Gamma - \gamma(\cdot,z)$ at all $a\in-\mathbb{N}$ forces $f(-n) = 1$ for all n, hence $c=1$: such a twist cannot change the functional-equation constant, and if it shifts the Euler constant it does so through a 1-periodic factor, which by Bohr–Mollerup destroys log-convexity.

(iii) The residue normalisation ($\mathrm{Res}_{a=0} = 1$) and the value normalisation ($\Gamma_\tau(1) = 1$) differ by the constant e^τ; for real τ they agree only for $\tau = 0$, which is the functional equation $\Gamma(1) = \mathrm{Res}_{a=0}\,\Gamma(a)$ once more.

Proof. (i) $e^{-\tau a}\Gamma(a) = (1-\tau a + O(a^2))(a^{-1} - \gamma + O(a)) = a^{-1} - \gamma - \tau + O(a)$ and $\gamma(a,z) = a^{-1} + \log z - \mathrm{Ein}(z) + O(a)$ (from $z^a/a = a^{-1} + \log z + O(a)$ and the series with $n\ge1$); subtract. (ii) The residue of $e^{-\tau a}\Gamma(a)$ at $-n$ is $e^{\tau n}(-1)^n/n!$, that of $\gamma(a,z)$ is $(-1)^n/n!$. If f is meromorphic, regular at $-\mathbb{N}$, and $f(a)\Gamma(a)$ satisfies $f(a+1)/f(a) = c$, regularity of $f\Gamma - \gamma(\cdot,z)$ at $-n$ means $f(-n)(-1)^n/n! = (-1)^n/n!$, so $f(-n) = 1$ for all $n\ge0$ and $c = f(-n+1)/f(-n) = 1$; then f is 1-periodic with $f(1) = 1$, and the Euler constant of $f\Gamma$ is $\gamma - f'(1)$. (iii) is the definition. □

The numerical check of (i) is in script `exp13`: at $a = 10^{-24}$ the difference $\widetilde{\Gamma}_\tau(a,z) - (E_1(z) - \tau)$ is $2.8\cdot10^{-25}$ ($\tau = 0.3$, $z = 1.7$).

32.2 The mixed values of Fischler–Rivoal

Recall (Part II, Theorem 10.1(iii)) the decomposition

$$\Gamma^{(s)}(a) = (-1)^s s!\,E_{a,s+1}(-1) + e^{-1}f_{a,s+1;0}(1), \qquad E_{a,s}(z) = \sum_{n\ge0}\frac{z^n}{n!\,(n+a)^s}, \quad f_{a,s+1;0}(1)\in\mathbf{D}, \tag{32.1}$$

for rational $a>0$ and $s\ge0$; at $(a,s) = (1,1)$ it is $\gamma = \mathrm{Ein}(1) - e^{-1}\delta$.

Proposition 32.2 (The twist on mixed values)**.** *Under $\Gamma\mapsto\Gamma_\tau$, the E-side of (32.1) is unchanged and the D-side becomes*

$$f^\tau_{a,s+1;0}(1) = f_{a,s+1;0}(1) + e\big(\Gamma^{(s)}_\tau(a) - \Gamma^{(s)}(a)\big), \qquad \Gamma^{(s)}_\tau(a) = e^{\tau(1-a)}\sum_{j=0}^{s}\binom{s}{j}(-\tau)^j\,\Gamma^{(s-j)}(a). \tag{32.2}$$

At $a = 1$ the shift is e times the $\mathbb{Q}[\tau]$-combination $\sum_{j\ge1}\binom{s}{j}(-\tau)^j\Gamma^{(s-j)}(1)$; for $(a,s) = (1,1)$ it is $-e\tau$, i.e. $\delta \mapsto \delta - e\tau$, the shear. For $a \neq 1$ it involves the factor $e^{\tau(1-a)}$. Consequently, for algebraic τ and rational a, the twisted values $\Gamma_\tau^{(s)}(a)$ lie in $\mathbf{V} = \mathbf{S}\cdot\mathbf{E}$ (an element of $\mathbf{S}$ times an E-value), and the twisted mixed values $f^{\tau}_{a,s+1;0}(1)$ stay in $\mathbf{V}$: Fischler–Rivoal's containment $\mathbf{D} \subset \mathbf{V}$ (Part II, Theorem 10.1(i)) is twist-compatible. Nothing places the twisted values in $\mathbf{S}$ itself.

Proof. Leibniz' rule for $\Gamma(a)e^{\tau(1-a)}$; then solve (32.1) for f. The ring statement: $\Gamma^{(k)}(a) \in \mathbf{S}$, $e^{\tau(1-a)} \in \mathbf{E}$ for algebraic τ and rational a, $e \in \mathbf{E}$, $f \in \mathbf{D} \subset \mathbf{V}$, and $\mathbf{V}$ is a ring. □

The identity in (32.2) is verified numerically at $a = 1$ for $s = 1, 2, 3$ at 60 digits in script `exp13`. Its meaning is the one Part I assigned to the shear in the language of Stokes framings [3]: the E-functions $E_{a,s}$ are the "series side" and are fixed; the Gamma derivatives are the "Stokes side" and move. In the A-world with $\tau = \sigma(\gamma) - \gamma$, the conjugate mixed values at $a = 1$ are $\overline{\mathbb{Q}}$-combinations of the true ones, consistent with $\Gamma^{(s)}(1) \in \mathbb{Q}[\gamma, \zeta(2), \zeta(3), \dots]$; at $a \neq 1$ they pick up the E-values $e^{\tau(1-a)}$: the phantom Gamma functions have phantom values at the rationals which are true Gamma values times exponentials of algebraic numbers.

32.3 The τ-line

The twisted system is a line through the true system:

$$\tau \;\longmapsto\; \Big(\gamma+\tau,\;\; \delta - e\tau,\;\; E_1(\lambda) - \tau,\;\; \Gamma(a)e^{\tau(1-a)},\;\; \omega_{\lambda,r} \text{ with } P_r(\lambda e^{\tau};\gamma)\Big), \qquad \tau \in \mathbb{C}.$$

Each coordinate is an affine, exponential or polynomial function of τ, every connection identity holds along the whole line, and Part I's two-model theorem says that the established theorems hold along the real segment of the line permitted by the enclosures they use, and along the whole line when they use only the connection identities. The worlds of the map (Part IV, Figure 1 are statements about *where on the line a coordinate vanishes*:

- A: the γ-coordinate vanishes at some $\tau \in \overline{\mathbb{Q}}$.
- the δ-world ($\delta \in \overline{\mathbb{Q}}$): the δ-coordinate vanishes at some $\tau \in e^{-1}\overline{\mathbb{Q}}$.
- the exponential certificate world ($e^\gamma \in \overline{\mathbb{Q}}$): the γ-coordinate vanishes at some $\tau \in \log \overline{\mathbb{Q}}^\times \setminus \{0\}$.
- the Ein-hyperplane worlds: the γ-coordinate vanishes at some $\tau \in -\mathcal{A}(\Lambda)$.

The sets $\overline{\mathbb{Q}}$, $e^{-1}\overline{\mathbb{Q}}$, $\log\overline{\mathbb{Q}}^\times$, $\mathcal{A}(\Lambda)$ are countable with real points dense in $\mathbb{R}$, and they are pairwise disjoint away from 0 whenever a theorem says so: $\overline{\mathbb{Q}} \cap \log\overline{\mathbb{Q}}^\times = \{0\}$ (Hermite–Lindemann), $\overline{\mathbb{Q}} \cap e^{-1}\overline{\mathbb{Q}} = \{0\}$ (Hermite), $\overline{\mathbb{Q}} \cap \mathcal{A}(\Lambda) = \varnothing$ (Beukers); whether $e^{-1}\overline{\mathbb{Q}}$ meets $\log\overline{\mathbb{Q}}^\times$ is open (it asks whether $e^{\alpha/e}$ can be algebraic). One unconditional consequence deserves its own line.

Proposition 32.3. *There is at most one $\alpha \in \overline{\mathbb{Q}}^\times$ with $\gamma + \log\alpha \in \overline{\mathbb{Q}}$. If it exists and $\alpha = 1$, world* A *holds; if it exists and $\alpha \neq 1$, world* B *holds with the certificate $e^\gamma \in \overline{\mathbb{Q}}^\times e^{\overline{\mathbb{Q}}}$.*

Proof. Two such α, α' give $\log(\alpha/\alpha') \in \overline{\mathbb{Q}}$, hence $\alpha = \alpha'$ by Hermite–Lindemann. If $\alpha \neq 1$ and $\gamma + \log\alpha = \beta \in \overline{\mathbb{Q}}$, then $\gamma = \beta - \log\alpha$ is transcendental, since otherwise $\log\alpha$ would be algebraic. □

33 The δ-world

The world "$\delta \in \overline{\mathbb{Q}}$" is the one certificate region inside $\gamma \in \mathbf{E}$ that no proved theorem removes (Part IV, Figure 1: Rivoal's theorem removes only its intersection with A, and Fischler–Rivoal's Corollary 1 removes it under their Conjecture 2. In it, $\gamma = \mathrm{Ein}(1) - \delta e^{-1}$ is an explicit E-value with coefficients in $\mathbb{Q}(\delta)$, hence transcendental by Beukers' theorem, and the whole linear theory of γ against E-values is governed by the functional relations of the E-function $F(z) = \mathrm{Ein}(z) - \delta e^{-z}$.

Proposition 33.1 (The δ-world's certificate, forgery and phantoms)**.** *(i)* (Finite certificate; script `exp13`.) *δ satisfies no integer relation of degree ≤ 12 with height below $10^{20.6}$ (300 digits). The certifying vector is the (irreducible) minimal polynomial of a degree-12 algebraic number of height $10^{22.8}$ within $10^{-298.8}$ of δ: the δ-world has its forgeries at the same price as the others.*

(ii) In the δ-world with $d = [\mathbb{Q}(\delta) : \mathbb{Q}]$, the Galois conjugates of the system are the twists by $\tau_\sigma = (\delta - \sigma(\delta))e^{-1} \in e^{-1}\overline{\mathbb{Q}}$: the conjugate Euler constants $\gamma^{(\sigma)} = \mathrm{Ein}(1) - \sigma(\delta)e^{-1}$ are E-values, and they are the Euler constants of Γ_{τ_σ}, whose values at the integers are $n!\exp(-n(\delta - \sigma(\delta))/e)$.

(iii) The δ-world implies $\gamma \notin \overline{\mathbb{Q}}$ (Rivoal), $\gamma \in \mathbf{E}$, $\gamma \in \mathcal{A}(\{1, -1\})$, and $\mathrm{X}(W)$ of Part III fails; it is consistent with $\mathbf{E} \cap \mathbf{D} = \overline{\mathbb{Q}}$ (the collision $\delta \in \mathbf{E} \cap \mathbf{D}$ is at an algebraic point); of the conjectures of [10] it contradicts only Conjecture 2, and it contradicts Rivoal's expectation $\gamma \notin \mathbf{E}$ [16].

Proof. (i) is the computation. (ii) $\gamma^{(\sigma)} - \gamma = (\delta - \sigma(\delta))e^{-1} = \tau_\sigma$, and Part IV, Proposition 24.1. (iii) $\gamma = \mathrm{Ein}(1) - \delta e^{-1}$ has Ein-weight 1 in $V(\{1, -1\})$; the rest is Part III, Theorem 18.7 and Part II, Definition 10.2. □

The δ-world is thus the mirror image of A: there the phantoms are algebraic shifts, here they are algebraic multiples of e^{-1}; there δ is a transcendental E-value and γ algebraic, here γ is a transcendental E-value and δ algebraic; the two worlds are exchanged by the connection formula and cannot both hold (Rivoal's theorem). The A-world's shadow systems and Galois norm have their δ-world counterparts ($P_\delta(e\,\mathrm{Ein}(1)) = e^d \prod_\sigma \gamma^{(\sigma)}$, a nonzero element of $\mathbb{Q}[e\,\mathrm{Ein}(1)]$); the δ-world's Beukers theory is different in shape, since there $\mathrm{Ein}(1) - \gamma \in \overline{\mathbb{Q}}e^{-1}$ makes the family of Part IV, Theorem 23.6, dependent as soon as $-1 \in \Lambda$.

34 Questions

(Q1) Is there a world in which the forgeries are cheap? A finite basis $x_0, \ldots, x_m$ of explicit constants for which $p/\log_{10} \mathrm{H}(v)$ stays clearly above $m+1$ as p grows would be a discovery, whatever the world; the gallery is a template for looking, and the 34-element and 78-monomial windows of Part II are its largest instances so far (γ has forgeries there of the predicted cost, and no cheaper).

(Q2) The twist respects the Stokes structure of $\Gamma(a, z)$ at $a = 0$ only (Proposition 32.1(ii)). Is there a deformation of the whole incomplete Gamma family, non-constant in a, that shifts every Stokes constant consistently, and what replaces the Euler constant along it?

(Q3) The twisted Gamma values and mixed values land in $\mathbf{V} = \mathbf{S} \cdot \mathbf{E}$ (Proposition 32.2), and nothing places them in $\mathbf{S}$. Is the twist-compatibility of $\mathbf{D} \subset \mathbf{V}$ a hint about which ring the D-values "really" live in?

(Q4) Proposition 32.3 says that at most one algebraic rescaling α makes $\gamma + \log \alpha$ algebraic. Is there any statement of the same shape for the Ein-hyperplanes, i.e. at most one point of

$\bigcup_\Lambda \mathcal{A}(\Lambda)$ can equal γ? (Yes, trivially, but the interesting version asks for a bound on the height of that point in terms of Λ, which would be a Diophantine statement about γ of the kind Part III found blocked.)

35 Computations for Part V

Scripts (Python 3.11, `mpmath` 1.3, `sympy` 1.14, `fpylll` 0.6.4), archived with outputs.

- `exp13_forgeries.py`: forgeries in nine worlds at 100 digits with verification at 300 digits (Table 8); the height law at $p = 60, 100, 200, 300$ (Table 9); Wirsing exponents w_d, $d \le 12$, at 300 digits (Proposition 31.2); the δ-world certificate and forgery (Proposition 33.1); numerical checks of Propositions 32.1(i) and 32.2. All lattice reductions are re-verified exactly (unimodularity, $(0.99, 0.51)$ conditions) as in Parts I–II; irreducibility and the absence of factors in x alone are checked with `sympy`.
- `exp13b_small_forgeries.py`: the printable 30-digit forgeries of Table 7, verified at 190 digits.
- `exp13c_hyperplane_forgery.py`: the weight-one Ein-hyperplane forgeries at $p = 30$–300 and the irreducibility of the bivariate polynomials of the π- and e-rows.

Run time: about 15 minutes in total.

Conclusion

36 What the two worlds taught

The programme was to build inside A and inside B until the real world rejected something. It did not, and Lemma 15.1 says why it could not have done so cheaply: a rejection inside a world is a proof of the other world under another name. What the programme produced instead is a description of the tools.

The description has one centre. Every established arithmetic theorem about γ that we could find reaches γ through a connection formula, and is therefore blind to the one-parameter family of *twisted Gamma functions* $\Gamma_\tau(s) = \Gamma(s)e^{\tau(1-s)}$, whose Euler constants $\gamma + \tau$ fill the real line. The two-model theorem (Theorem 4.2) is the logical form of this blindness; the density obstruction (Theorem 15.2) is its extension to every arithmetic class; the tie theorem (Theorem 18.1) is its quantitative form on linear forms, where it caps the exponent of shear-invariant families at 1 and explains why Rivoal's forms sit at $\frac{1}{2}$ with $b_n/c_n \to e$; the twist (Proposition 24.1) is its function-theoretic form, in which the only things pinning $\tau = 0$ are the normalisations of the Gamma function—$\Gamma(2) = 1$, the reflection and multiplication constants, Stirling's formula, the exponential in Euler's integral—none of which enters the transcendence machinery. In the language of Fischler and Rivoal, the twist fixes the E-side and moves the D-side; in the language of the companion manuscript [3], it is the Stokes-framing shear on numbers; in the language of Fresán and Jossen, it is a $\mathbb{C}$-linear change of de Rham basis that fixes the period Ein(1) of the cycle $\sigma_1 + \sigma_2$, and the A-world is the statement that a $\overline{\mathbb{Q}}$-linear one exists which kills the period of σ_1 while, by their theorem, the motive itself does not split.

Around this centre the worlds grew. The A-world has honest E-values for all its shadow constants, a Galois theory with phantom conjugates that are Euler constants of twisted Gamma functions without an analytic definition of their own, a Galois norm, and a Beukers theory whose sentences C cannot at present prove; its Diophantine consequences—the floor, the forced

approximations along the definitions, the rigidity $\mu(\delta) = 2$—are all expected to be true, and its non-Diophantine consequences—$\delta \in \mathbf{E} \cap \mathbf{D}$, the split period matrix, the collision in the tower—are all expected to be false, each rejected by a standing conjecture evaluated at one element. The B-world is a universal negative whose only positive content is the list of certificate loci dual to the A-world's theorems; on each of them γ has a forgery at the uniform price $\mathrm{H} \approx 10^{p/(m+1)}$ and no cheaper, and the certified searches find γ on none of them. The expected truth is the wild world, in which every finite search is a search of an empty region.

Three things would change this picture, and they are the targets the programme leaves behind. A sequence of integer forms in $1, \gamma, \delta$ smaller than its own tie defect, with a proof that reaches γ through a bridge (Specification 18.6), would be the first Diophantine construction that the shear does not neutralise; the E-side produces tied forms, the D-side untied but large ones, and the two have not been combined. The point conjecture B_δ: $\delta \in \mathbf{E} \Rightarrow \delta \in \overline{\mathbb{Q}}$ (Definition 10.2, Proposition 10.3) would reject the collision and, with $\delta \notin \overline{\mathbb{Q}}$, give the middle term of the linear hierarchy of Theorem 18.7. Fullness of the period realisation for the Euler–Mascheroni motive would reject the split period matrix. Below all three sits the weakest unproved sentence above B, the linear independence of $1, e, e\gamma$ over $\overline{\mathbb{Q}}$, for which the ledger records no route at all.

We end where the session ended: not with a proof, but with a map on which the proof would have to be found, drawn to the precision the available instruments allow.

A Computational archive

All computations were performed with Python 3.11, `mpmath` 1.3, `sympy` 1.14 and `fpylll` 0.6.4 (`fplll` 5), on one core. Lattice reductions are re-verified exactly (unimodularity by Bareiss, the $(0.99, 0.51)$ reduction conditions in rational arithmetic) and the height bounds are those of Part I, Section 8, and Part II, Section 14. The only non-exact inputs are the values of the classical constants supplied by `mpmath` at 300–1660 digits; all heights, coefficients and continued-fraction data are exact integers. The archive (scripts, logs, JSON outputs, and the LaTeX sources of the five parts) accompanies this paper.

Table 10: Scripts of the archive.

script	content	used in
`exp01_identities.py`	identity certificates: $\mathrm{Ein}(1) = \gamma + \delta/e$, $\delta = eE_1(1)$, the connection formula, Rivoal's normalisation (80 digits)	Part I
`exp02_linear_forms.py`	naive Hermite–Padé forms for $(1, e^{-z}, \mathrm{Ein}\, z)$ (comparison only)	Part I
`exp03_rivoal_forms.py`	Rivoal's polynomials A_n, B_n, C_n reconstructed exactly; integer forms L_n in $1, \gamma, \delta$ and their exponents	Part I, Table 1; Parts III–IV
`exp04_delta_cf.py`	Stieltjes convergents of δ; simple continued fractions of δ and γ from 3000 digits	Part I, Table 2
`exp05_camouflage.py`	the degree-12 camouflage with full local root counts	Proposition 3.2; Part IV
`exp06_lll_certificate.py`	exact LLL certificates: no relation of degree $\le D \le 30$ below explicit heights	Part I, Table 3
`exp07_tower_identities.py`	$\omega_{\lambda,r} = e^\lambda \mathrm{Ein}_r(\lambda) - e^\lambda P_r(\lambda)$ by quadrature; g_j against the Taylor coefficients of $\Gamma(1-t)$	Theorem 11.1
`exp08_collision_probes.py`	certified probes at 1660 digits: 8 targets against 34 E-values; E–D, E–$\mathbf{S}$, D–$\mathbf{S}$ windows	Part II, Section 14
`exp09_quadratic_window.py`	γ and δ against 78 quadratic monomials in 11 core E-values	Part II, Section 14
`exp10_aworld_profiles.py`	forced approximations along the definitions (Euler–Maclaurin, Kummer, Mertens); Stieltjes forms $q_k\delta - s_k$	Table 4, Proposition 18.5
`exp11_tie_defect.py`	tie defect of Rivoal's forms; LLL-reduced shear-invariant forms and their provable bounds	Table 5

script	content	used in
`exp12_two_worlds.py`	engine ratios; the shadow world of the camouflage; the certificate zoo	Propositions 27.1, 25.2; Section 23
`exp12b_find_camouflage.py`	certified searches on c and γ at 300–1200 digits; recovery of P_c at 1080 digits	Section 23
`exp12c_camouflage_duality.py`	the roots c_p of the certificate vectors, exact re-verification, irreducibility	Proposition 23.3
`exp13_forgeries.py`	forgeries in nine worlds at 100 digits; the height law; approximation exponents w_d; the δ-world; checks of the twisted incomplete Gamma and the Leibniz shift	Part V
`exp13b_small_forgeries.py`	the printable 30-digit forgeries	Table 7
`exp13c_hyperplane_forgery.py`	weight-one Ein-hyperplane forgeries; irreducibility of the bivariate forgery polynomials	Tables 7–9

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